

DEN 590 CubeSat Project

Fall 2025 Final Report



12/19/25

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Introduction

The beginning of the Fall 2025 semester saw a descope from the initial primary mission with hosted payload(s) model into two individual missions: the *Precision Hyperspectral Agricultural Tracker Satellite* (PHAT-Sat) and the *Neutron Detector: K-State Analysis Tool* (NuKAT). There has been significant growth over the last semester in the development of both missions and the understanding/technical knowledge of the students.

The scope of the NuKAT mission includes a projected late 2027 launch of a satellite focused primarily on developing the flight heritage (NASA's TRL 7) and providing space qualification (TRL 8) for the DSC-Domino. This semester saw the inception of the NuKAT mission where the only previous work had been done taking the DSC-Domino to be a hosted payload (the mission started, for all intents and purposes, this semester). Therefore, there has been an incredible body of work done to move from an initial concept, through a Mission Design Review (MDR), and ending with a successful pre-Preliminary Design Review (pPDR). This means the design is well on its way to closing compared to where it started this semester.

PHAT-Sat is an Earth observation mission focused on agricultural applications with a projected 2028/2029 launch. This semester, an internal System Requirements Review (SSR) and external Mission Design Review (MDR) were conducted to further define the merit of the mission structure and objectives. There was a lot of effort to define university stakeholders and resources for data processing and analytics. This mission is planned to proceed to a preliminary design phase next semester.

Table 1: CubeSat Students (Fall 2025)

Name	Major(s) Minor(s)
Braden Adams	Computer Science Leadership Studies Minor
Noah Amundson	Industrial Engineering
Henry Bash	Electrical Engineering
Will Beemer	Mechanical Engineering
Michael Bennett	Electrical Engineering
Mitch Daues	Mechanical Engineering

Sophia Fenton	Mechanical Engineering
Owen Franken	Industrial Engineering Japanese Minor
Logan Hayward	Computer Science Mathematics Minor
Abigail Husted	Industrial Engineering
Colby Johnston	Mechanical and Nuclear Engineering Music Minor
Logan Mayhan	UAS Flight and Operations
Chase Newell	Mechanical Engineering Computer Science Minor
Andrew Osterhaus	Computer Science Math Minor Business Minor
Dale Petrie	Industrial Engineering
Nalen Rangarajan	Computer Science and Mathematics
Dalton Rizzo	Electrical Engineering Mathematics Minor Physics Minor
Igor Sheshukov	Mechanical Engineering
Phillip Shirkey	Mechanical Engineering Mathematics Minor
Joe Theis	Mechanical Engineering Technology Applied MFG Cert.
Bailey Walke	Mechanical Engineering
Cameron Weeks	Mechanical Engineering

Mission Overviews

NuKAT Mission

Mission Description

The downsides of current space neutron detectors drive this mission. Scintillators rely on large crystals and photomultiplier tubes (PMT) with high voltages, proportional gas detectors are pressure vessels, and planar semiconductors have low detection efficiency. Microstructured semiconductor neutron detectors (MSNDs) solve these problems and have higher detection efficiency than other semiconductors due to their non-planar geometry. Radiation Detection Technologies currently has two platforms that use MSNDs, the original Domino and the DSC-Domino. The former has been space qualified aboard the ISS as part of the NeRDI-1A mission and the latter is what NuKAT will use. The main difference is the ability of the DSC-Domino to measure fast and thermal neutrons (with background radiation correction) where the original Domino only measures thermal.

The NuKAT mission seeks to advance space-based neutron detection via MSNDs in LEO and validate the DSC-Domino as a space-qualified detector platform. Minimum mission success is achieved by building DSC-Domino flight heritage (measurement frequency of 0.0167 Hz) and producing ground-produced orbital location-correlated measurements over three months. Complete mission success is achieved by DSC-Domino space qualification (measurement frequency of 0.033 Hz) and satellite-produced orbital location-correlated measurements over one year. Additionally, NuKAT is a program pathfinder.

Mission Concept of Operations

The NuKAT mission will have an anticipated 1-month launch and initialization phase that starts from the moment it is deployed from the launch vehicle. This phase will begin with 30 minutes of sleep per regulatory rules and launch provider guidelines [1]. During this sleep, the satellite will not send any data, deploy its deployables, or otherwise perform any action other than counting down with a timer. Following the sleep step, initial startup will begin with the OBC booting and bringing ADC systems online to measure and interpret the satellite's current condition. The detumble process will then begin with ADC systems functioning to bring the satellite to a stable positional and rotational condition in the correction orientation for appendage deployment. The appendages will then be deployed, which includes the UHF antenna, solar panels, and custom ADCS magnetometer. Satellite calibration will begin for bus systems which enable us to begin transmitting and receiving health data from the satellite while inside the correct orbital window (as determined by GNSS reception and interpretation). A stand-by period will then begin as the satellite waits

for ground station commands to begin mission operations. This allows the team to establish steady communication and ensure proper satellite health.

Following these commands the mission operations phase begins which will last for at least 3 months, but nominally 1 year with the potential for even longer operations depending on stakeholder interest/support, satellite altitude from the launch provider, and satellite health. In this mode, the science will take place which involves pulling the accumulated DSC-Domino counts at least every 60-seconds (0.0167 Hz) with option for higher frequency as allowed by onboard storage and downlinking volume. While performing science operations, the satellite requires gross pointing margins (remaining within ideal nadir pointing $\pm 20^\circ$) and transmitting/receiving at least within the FOV of the K-State radio station; additional areas also included for telemetry/health (not science data) as required by FCC licensing.

There will be upset modes in case the need arises again to detumble/largely correct attitude, survive electromagnetic interference (EMI), correct for over/under voltage, and correct for over/under survival temperature. These will trigger an upset mode that will autonomously address the issue with option for a survival mode that prioritizes charging and other similar satellite-saving measures. If the event is drastic enough, the satellite can directly enter survival (i.e., critical power level). The only mode that can transition back into science is the upset mode as seen in Figure 1.

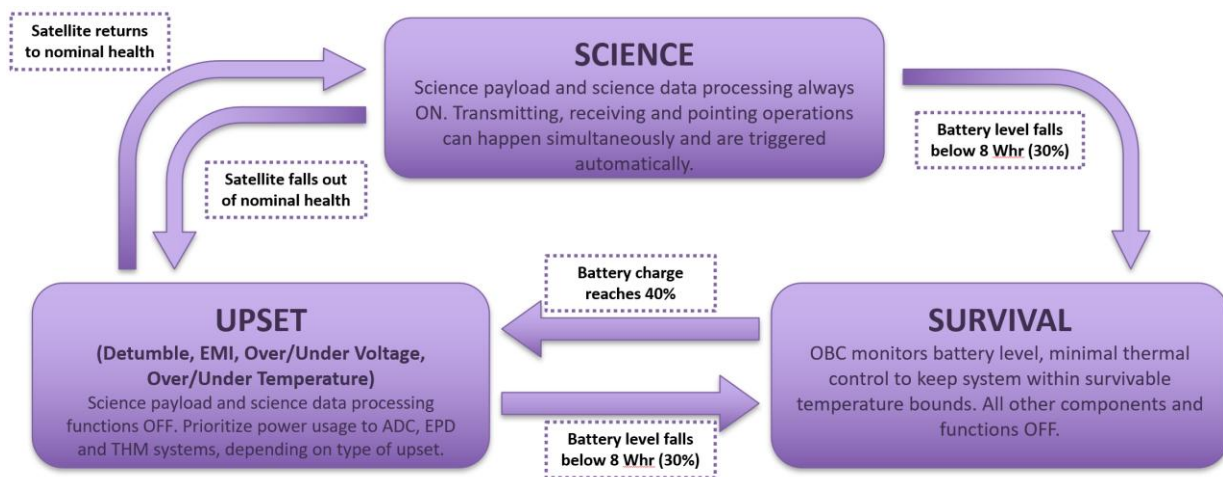


Figure 1: Mission Operations Modes

Decommissioning is the final phase of the satellite's life and consists of rotation into an increased drag orientation, final transmission, loss of communications, and finally burn up. Per FCC rule FCC 22-74, satellites must burn up within 5 years of the end of their mission. Given the satellite is in LEO and based on our current models, this will not be an issue.

The information presented in Figure 2 shows a typical “day in the life” of the satellite which consists of regularly occurring visits to the poles (shown in red and black; indicative of a polar, sun-synchronous orbit); semi-periodic visits to the South Atlantic Anomaly (SAA, shown in orange); more sparse, but still periodic passes over the K-State ground station (shown in green; approximately 4-6 visits every 24 hours with average total time per day of around 20 minutes); and regular 90-minute orbits with 60 minutes in sunlight (yellow) and 30 minutes in umbra (blue)

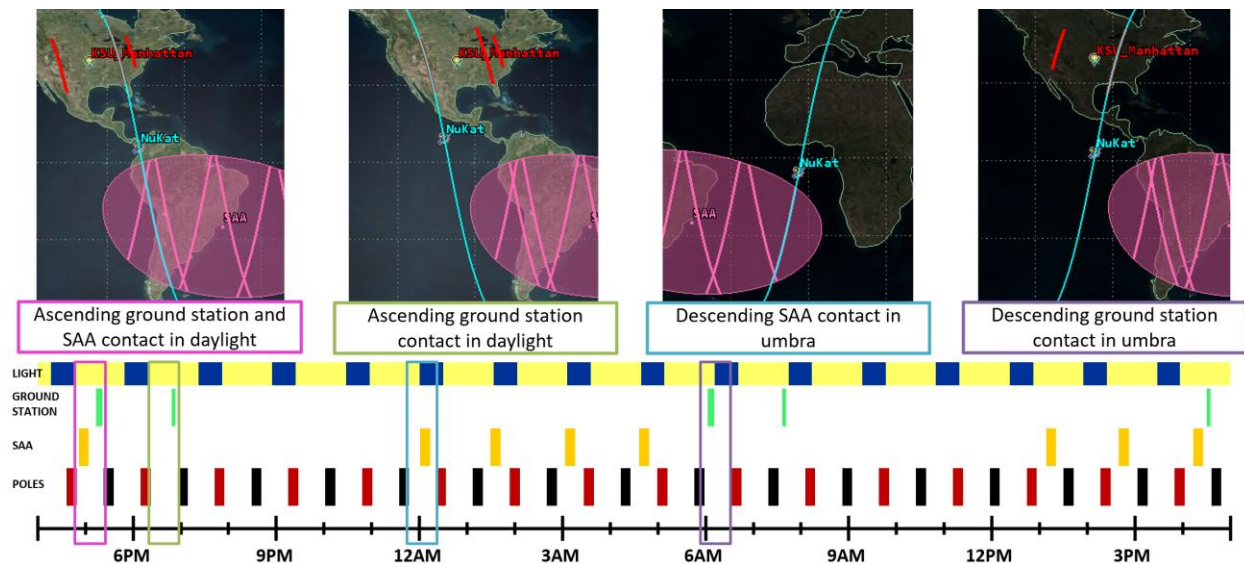


Figure 2: NuKAT "Day in the Life" with key contacts

PHAT-Sat Mission

Mission Description

PHAT-Sat (Precision Hyperspectral Agricultural Tracker Satellite) is a 3U CubeSat Earth observation mission aiming to enhance agricultural productivity and sustainability in Kansas and surrounding regions by utilizing miniaturized hyperspectral imaging and onboard data processing.

Background and Need

Agriculture plays a critical role in the Kansas and Midwest economies, contributing tens of billions of dollars annually and supporting a wide range of stakeholders. Modern agriculture faces challenges in yield production, resource efficiency, and environmental impact. Farming inputs, including irrigation, fertilizer, and pesticides, contribute significantly to operational costs. Environmental factors, such as disease, drought, and soil degradation, threaten yield production.

Precision agriculture has become an increasingly popular strategy for optimizing resource usage for farmers, with remote sensing being a key component of these methods. Current remote sensing solutions include drones and UAVs, balloons, ground-based cameras, aircraft, and satellite constellations. A competitive analysis was conducted by the PHAT-Sat team to determine the utility of the mission compared to other remote sensing precision agriculture services and was found to be the most promising solution. The creation of potential mitigation strategies was also focused on the high level risks in Phat-Sat as well, with a good majority of the highest rated risks having potential mitigation methods theorized. Of the risks, two high level red risks were at the system level, four were at the system level, and the rest were at the subsystem level or below.

Mission Objectives and Success Criteria

The mission's purpose is to design, build, test and launch a CubeSat equipped with a hyperspectral imager to monitor vegetation and soil indices, enabling proactive decisions in precision agriculture. The imager will provide improved spectral information with comparable spatial resolution to current NASA satellites, allowing PHAT-Sat's data to be used to supplement existing datasets.

Mission objectives include acquiring hyperspectral imagery of the Midwest in **20–40 spectral bands** in the visible and near-infrared range (approximately **450-1200 nm**) with a target ground resolution of **4.75m-30m**. The target mission operation life is **24 months** (6 months minimum) to capture two full growing seasons, from a 500 km sun synchronous orbit. PHAT-Sat will provide timely and useful data to stakeholders in the form of corrected wavelength data and vegetation and soil index maps for selected regions and time scales

Full mission success will be measured by successful data collection and downlink, ground validated vegetation indices, and a fully sustained active mission life of two years.

Mission Concept of Operations

Mission Phases

1. Launch & Deployment
 - PHAT-Sat will be launched to a 500km SSO by a selected launch provider.
 - Deployment from the dispenser will be followed by automatic detumbling and initial health checks.
2. Commissioning
 - Upon separation, PHAT-Sat will enter a safe mode for system boot-up, health checks, solar panel and antenna deployment, and initial communication link establishment via UHF.
 - Once stabilized, all subsystems will undergo calibration and testing.
3. Nominal Operations (Day in the Life) (Figure 3)
 - The spacecraft will enter a regular operational loop based on orbital passes:
 - o Imaging Passes: When over the region of interest, the satellite will orient nadir using ADCS and activate the hyperspectral payload.
 - o Processing: following imaging passes, data is converted from raw data to Level1B data and compressed for downlinking
 - o Downlink: When in contact with designated ground stations, science data will be downlinked via X-Band and telemetry data via UHF.
 - o Idle: prior to imaging or downlinking, satellite is pointed to correct orientation for next operation and prioritized solar charging
 - o Data Distribution/Ground Processing: Once received at K-State's ground station, PHAT-Sat mission data will be processed and validated using K-State academic entities
4. Extended Mission & Decommissioning
 - After 24 months, PHAT-Sat may continue to operate as power and system health allow. Ultimately, it will passivate and deorbit within five years of loss of communications per launch provider requirements.

Day in the Life

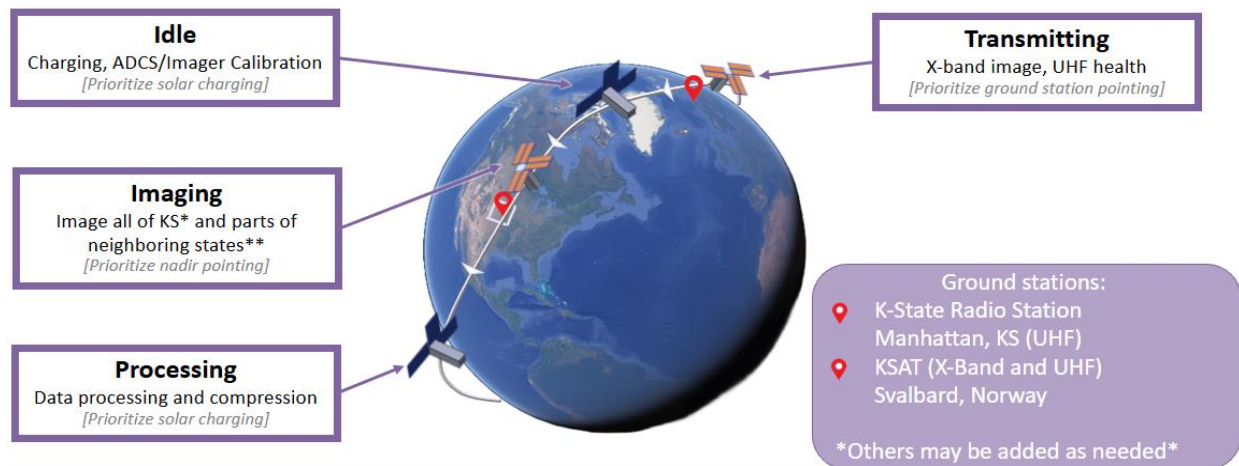


Figure 3: PHAT-Sat Day in the Life Diagram

Data Product and End Users

PHAT-Sat utilizes hyperspectral imaging to produce wavelength data products over Kansas and the Midwest. Using NASA Earth Observing System Data and Information System conventions, the mission produces Level 1B and Level 3A data products.

Level 1B data includes atmospherically, radiometrically and geometrically corrected imagery that is time-referenced and geo-referenced. This data is derived from raw image data onboard PHAT-Sat with the intent of decreasing the science data package size to downlink while still providing stakeholders flexibility of data usage for a variety of functions. Applications include producing customized vegetation indices from hyperspectral data, tracking historical trends, and space-to-ground imaging comparison.

Level 3A data consists of vegetation and soil index maps generated using K-State Precision Agriculture and Data Analytics resources. These easily interpretable spatial maps are used for resource management, identifying anomalies between passes, and triggering drone or field agent deployment to regions of interest.

End users include farming stakeholders, agricultural extension services, industry partners, and Kansas State University researchers. Data products will be distributed through established university distribution channels and websites, promoting public engagement.

Space and Launch Environment

Currently both missions are set at a 500 km altitude with an inclination of $\sim 97^\circ$, so this analysis is good for NuKAT and PHAT-Sat for the time being. From some preliminary research, the major space effect mechanisms were identified as seen in Table 2.

Table 2: Space Environment Major Mechanisms credit [2]

Mechanism	Effects	Source
Surface Charging	- Biasing of instrument readings - Power drains	- Plasma
Deep Dielectric Charging	- Biasing of instrument readings - Physically damaging electrical discharges	- High energy electrons
Structure Impacts	- Structural damage	- Orbital debris
Drag	- Torques - Orbital decay	- Thermosphere
Total Ionizing Dose	- Degradation of electronics	- Trapped electrons & protons - Solar protons
Displacement Damage Dose	- Degradation of optical components and electronics - Degradation of solar panels	- Trapped electrons & protons - Solar protons - Neutrons
Single Event Effects	- Data corruption & noise on images - System shutdowns - Electronic component damage	- Trapped electrons & protons - Solar protons - Neutrons - Galactic cosmic rays
Surface Erosion	- Degradation of components	- Particle radiation - Ultraviolet - Atomic oxygen

Using a software called OMERE 5.9 the worst-case radiation environment was roughly determined using deterministic methods. Currently the minimum thickness using the recommendation provided in [3] is to withstand twice the expected radiation. To bring the radiation to a safe internal dose is approximately 18.0 mils Aluminum. For NuKAT the exterior panel thickness is 1.5 mm (~ 60 mils) which results in an internal dose of 1.3 krad. For both missions, any unshielded components will experience a dose of ~ 417 krad. Further study of the space environment needs to be conducted on orbital debris and more in depth studies on environmental interactions with the spacecraft to ensure risks are properly mitigated. For more detailed information on the space environment refer to [4].

The launch vehicle is the SpaceX Falcon 9. The primary lateral frequency is > 10 Hz, primary axial frequency is > 25 Hz, with secondary minimum resonant frequencies > 35 Hz. Expected vibrations at 1 G are > 30 Hz, with an axial load of 11 G and a lateral load of 4-7.5 G at launch. Payload outgassing and bus depressurization occur at 4.5 kPa/s on ascent. Further power spectral density research is required to define the vibration environment.

Satellite Overviews

NuKAT Mission

Description

The NuKAT mission's payload consists of four DSC-Domino platforms, each containing four MSNDs. Specifics on these platforms can be found in the Payload Overview section. The neutron energies of interest are thermal, epithermal, and fast with adjusted detector efficiency peaks to 0.025 eV, 100 eV, and 18 MeV to capture these energy regimes.

The NuKAT satellite has underwent many redesigns. The current state consists of a 2U satellite with velocity and Earth nadir as shown in Figure 4 in the undeployed and deployed states. Going forward a trade study will be needed as it pertains to the problem with being power negative. Following the pPDR, the idea was floated to fly in a more "typical" CubeSat orientation which has the 2U stack "stacked" along Earth nadir as the moderated payload takes up less space than initially expected. This is a promising lead and should be examined further.

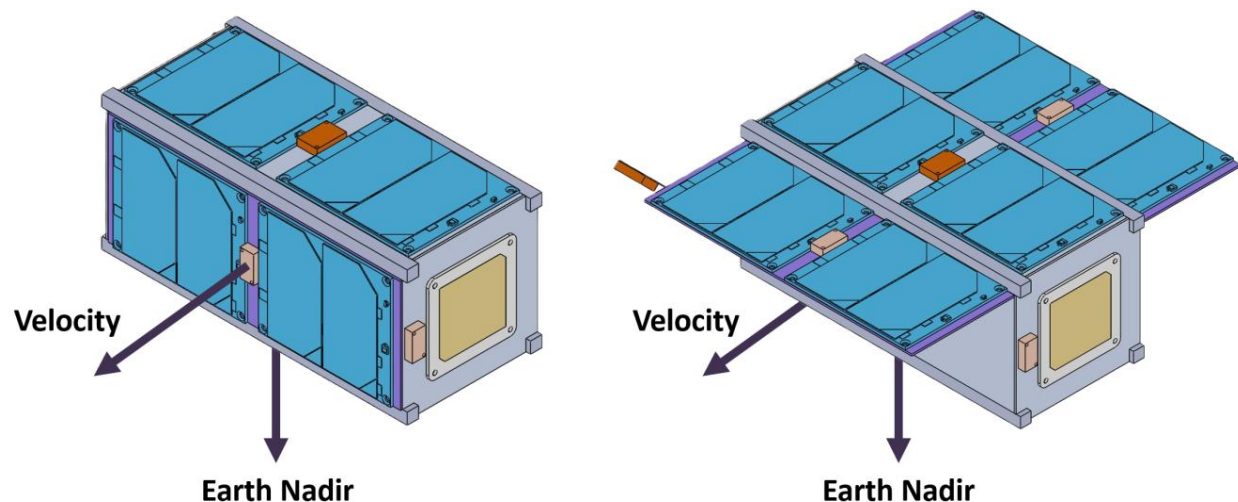


Figure 4: NuKAT Undeployed and Deployed Configurations

The current satellite consists of 6U worth of solar panel area that will need to be addressed going forward (as mentioned prior).

The driving factors of the satellite orientation were the payload orientation, solar panel configuration, and the UHF antenna placement.

The trade studies resulting in the current orientation came from the ADCS need for the magnetometer to be on the non-velocity and non-nadir face, which resulted in the velocity

vector normal to a 2U face, and the payload requirement that the unmoderated DSC-Dominos not be directly above or below the moderator which resulted in the Earth nadir vector being normal to a 2U face. These two initial constraints produced the current satellite orientation. The latter, the payload requirement, along with the realization that the moderator would not occupy an entire 1U footprint are what need to be investigated for the potential to adjust the design by rotating 90° along the velocity vector.

Isometric Satellite View

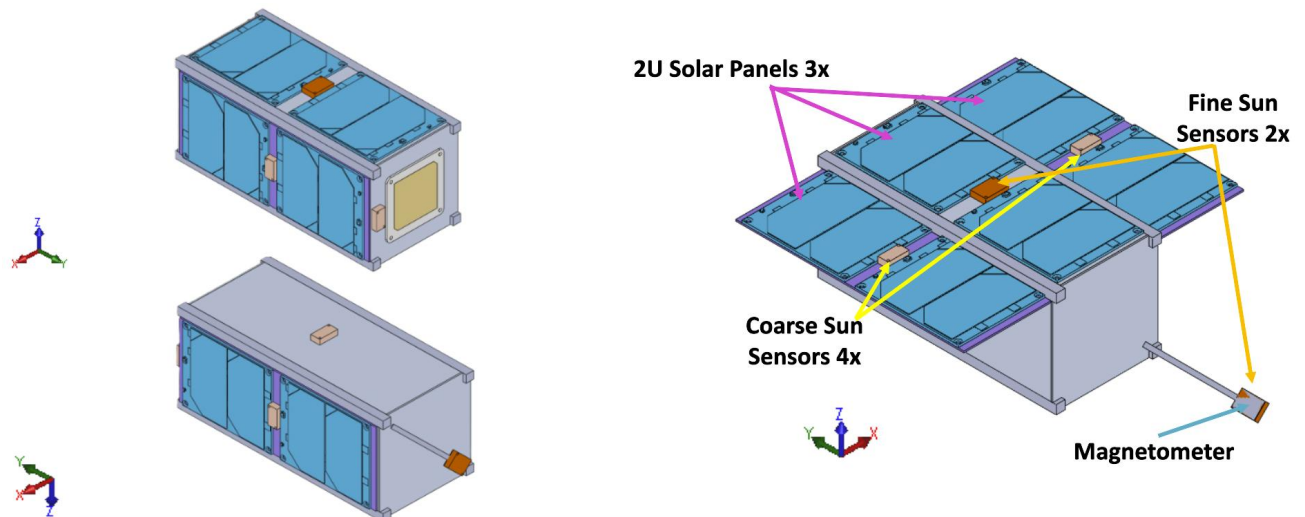


Figure 5: NuKAT Iso View (External Components)

System Block Diagram

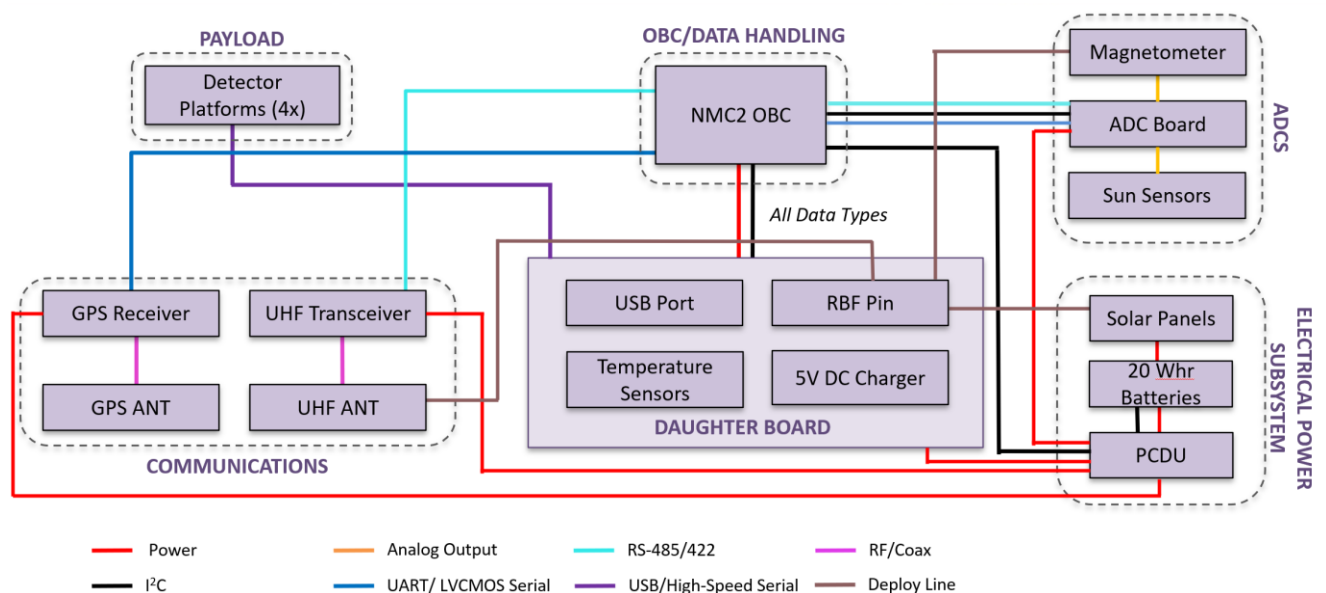


Figure 6: NuKAT Satellite System Block Diagram

A preliminary design of NuKAT electrical and communications interfaces are shown in Figure 6 with an exploded view shown later in Figure 21. All interface information was provided from subsystem component specification documents.

Satellite States and Modes

The satellite will pass through three phases, as seen in Figure 7 for “Launch & Initialization,” “Mission Operations,” and “Decommissioning.” A more in-depth graphical view of the “Mission Operations” phase can be found above in Figure 1, and the mode descriptions can be found in Table 3.

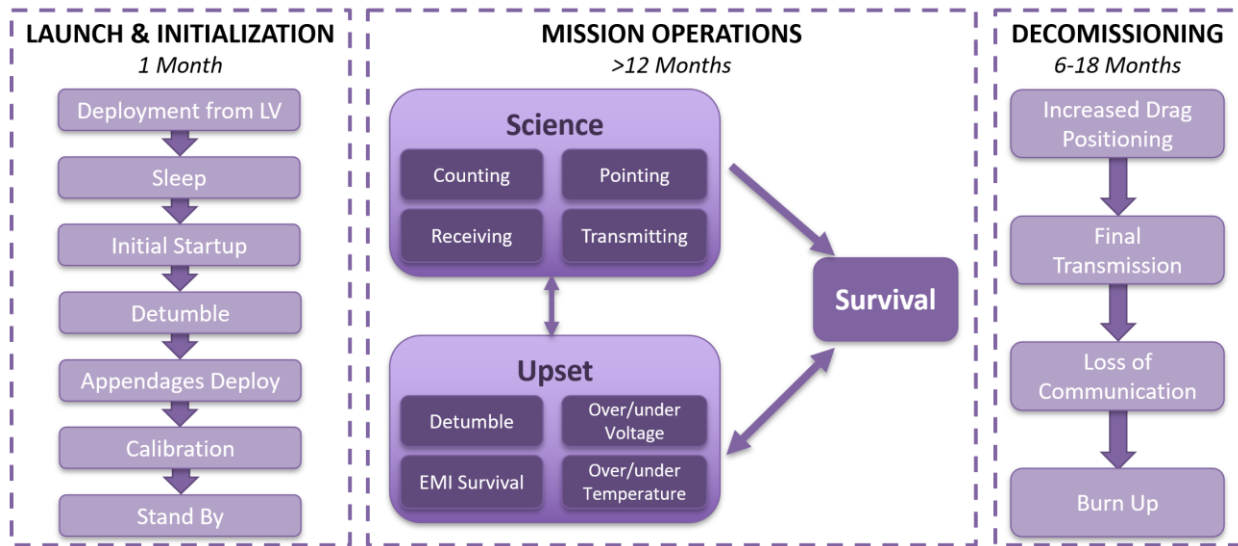


Figure 7: NuKAT System Phases, States, and Modes

Table 3: NuKAT Phases and Modes

Phase	Mode	Type	Function
Launch & Initialization	Deployment from Launch Vehicle	<i>Scheduled</i>	Ejection from the launch vehicle
	Sleep	<i>Reactionary</i>	Satellite system and transmitter powered down at least 30 minutes after deployment [1]
	Initial Startup	<i>Scheduled</i>	Following the timed sleep period, the satellite will turn on its computers and needed systems/sensors to begin detumble
	Detumble	<i>Scheduled</i>	Bring satellite into preferred, stable orientation

Mission Operations		Appendages Deploy	<i>Scheduled</i>	Extend solar panels, magnetometer, and UHF antenna
		Calibration	<i>Scheduled</i>	Desaturate torque rods and bring all other necessary systems online and perform necessary autonomous calibrations
		Stand By	<i>Scheduled</i>	Send down satellite health and telemetry data and wait for command to begin mission operations
	Science	Counting	<i>Commanded</i>	Begin pulling counts from the payload
		Pointing	<i>Reactionary</i>	Correct attitude to ensure payload, charging, and downlink needs are met
		Receiving	<i>Scheduled or Reactionary</i>	When over ground station, accept ground commands
		Transmitting	<i>Scheduled</i>	When over ground station, send science and health data
	Upset	Detumble	<i>Reactionary</i>	Return satellite to steady, science position
		Over/Under Voltage	<i>Reactionary</i>	Flag for any electrical issues, return satellite to nominal power ranges
		EMI Survival	<i>Reactionary</i>	Respond to EMI interruptions and toggle systems to reduce damage
		Over/Under Temperature	<i>Reactionary</i>	Respond to temperatures beyond acceptable ranges and return to nominal range
	Survival		<i>Reactionary</i>	Orient satellite for maximum charging and use systems only as needed for survival
Decommissioning	Increased Drag Positioning		<i>Commanded</i>	Orient satellite for maximum drag
	Final Transmission		<i>Scheduled</i>	Send final downlink
	Loss of Communication		<i>Reactionary</i>	
	Burn Up		<i>Reactionary</i>	

PHAT-Sat Mission

Satellite Description

PHAT-Sat is a 3U CubeSat spacecraft designed to support hyperspectral Earth observation operations in Low Earth Orbit. The spacecraft integrates a miniaturized hyperspectral imaging payload with a modular spacecraft bus that provides attitude determination and control, command and data handling, communications, power generation and storage, thermal regulation, and structural support. The current satellite configuration is designed to balance payload performance, data handling capability, and mission lifetime within the constraints of the CubeSat form factor.

PHAT-Sat is currently designed with 6U of solar panels but could be increased to 12U or 18U with a double-deployable design. Deployable UHF antennas for a full FOV are included, as well as a deployable magnetometer for the ADC system. An isometric view of the PHAT-Sat spacecraft in its deployed state is shown in Figure 8 and stowed state in Figure 9.

Isometric Satellite Views

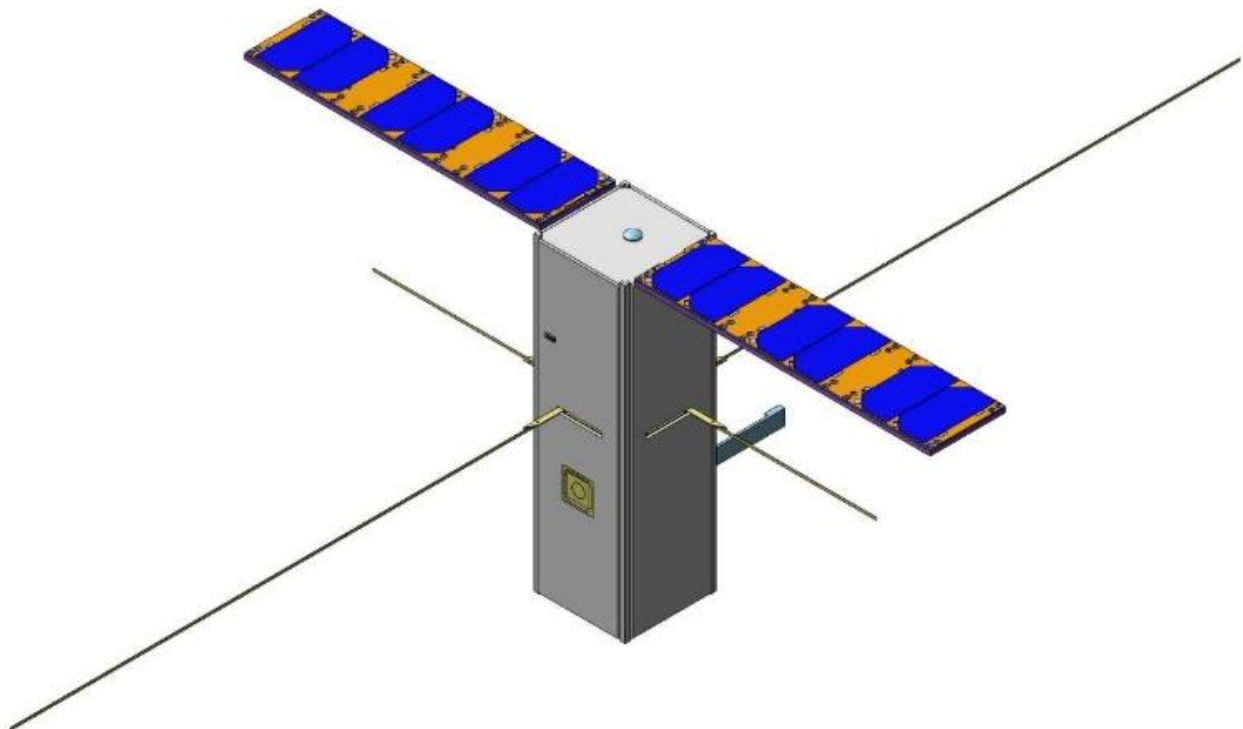


Figure 8: PHAT-Sat Isometric View (Deployed)

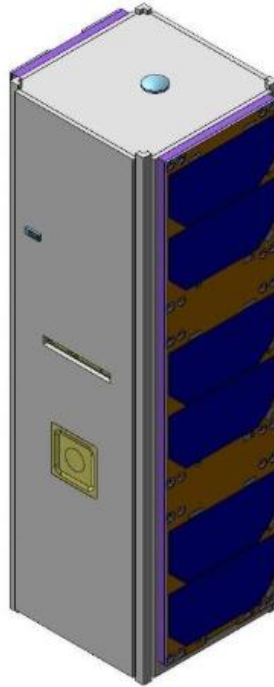


Figure 9: PHAT-Sat Isometric View (Stowed)

System Architecture

The PHAT-Sat spacecraft architecture is organized around the functional separation of the imaging payload and the spacecraft bus designed to support Earth observation. The main payload is a 1.5U hyperspectral imager operating across 20-40 bands in the visible near-infrared range (400-1200nm). The imager generates high-volume science data that drives system-level requirements for pointing accuracy, onboard processing, and downlink capability. The spacecraft bus supports payload operations through coordinated power management, precise attitude control, and autonomous command and data handling. Science data and telemetry are routed through onboard processing systems for storage and transmission to ground stations. A high-level system block diagram is provided in Figure 10. The spacecraft is designed to execute scheduled operational modes for nominal science operations, as well as planned emergency situations, as described in Table 4.

PHAT-Sat Block Diagram

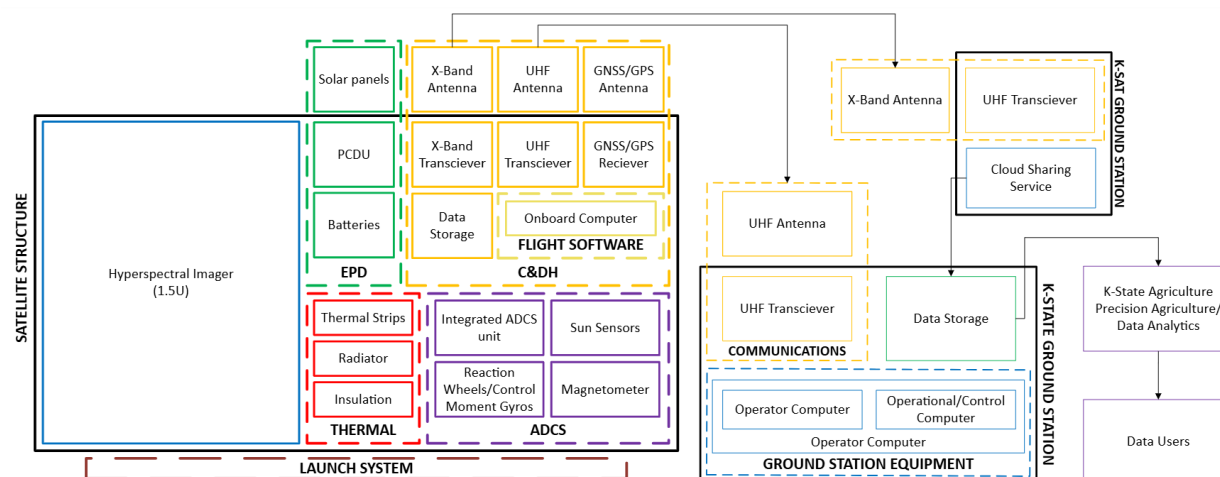


Figure 10: PHAT-Sat System Block Diagram

Modes and States

Table 4: PHAT-Sat Operational Modes

Mode	Type	Function
Deployment	Scheduled	Initialization phase immediately after ejection from launch vehicle; boot, safe startup, basic health checks, and configuration for detumble and early comm establishment
Pointing	Scheduled	Establish and maintain required attitude profile for the next planned activity (nadir for imaging, Earth/GS pointing for downlink, or sun-pointing for power generation)
Imaging	Scheduled	Activate payload and collect hyperspectral imagery over the region of interest while prioritizing nadir pointing and stability
Downlinking	Scheduled	Execute scheduled communications pass; downlink packaged science data over the high-rate link while maintaining required ground-station pointing and concurrently sending health/telemetry on the low-rate link as applicable
Idle	Scheduled	System standby between operations; prioritize solar charging, maintain time/position knowledge, and prepare for upcoming imaging/downlink events (includes low-duty housekeeping)
Raw Data Downlink	Commanded	Operator-initiated transfer of raw or minimally processed payload data for diagnostic, calibration, or validation purposes (used sparingly due to data volume)
Calibration	Scheduled	Perform planned payload and/or ADCS calibration activities, including periodic calibrations identified in the ConOps and required for long-term data quality

Survival	Reactionary	Enter a power/thermal-safe configuration that minimizes loads and preserves the spacecraft; prioritize battery protection, thermal survival, and reliable telemetry.
Upset	Reactionary	Fault response state entered after an anomaly (e.g., pointing loss, comm disruption, subsystem reset); execute recovery actions and transition to safe/survival or back to nominal when stable.
Sleep	Reactionary	Ultra-low-power state used when power margin is insufficient; non-essential subsystems off, periodic wake for health checks/telemetry as feasible
Decommission	Reactionary	End-of-life and mission termination actions following significant degradation of components and/or orbit or planned decommissioning; finalize transmissions and transition to a safe inactive configuration.

Key Design Drivers and Constraints

The PHAT-Sat spacecraft design is primarily driven by the requirements of the hyperspectral imaging payload. Payload data volume and spectral resolution create constraints for onboard processing capability, data storage, and downlink performance. Imaging quality drives requirements for attitude knowledge and pointing stability. Additional design constraints include CubeSat volume and mass limitations, available power generation in a sun-synchronous orbit, available power generation with the surface area available on a 3U CubeSat, and compatibility with standardized CubeSat launch interfaces. These factors inform system-level trades and ensure that spacecraft performance is aligned with mission objectives.

Payload Overview

Payload Descriptions

Hyperspectral Imager

The Simera HyperScape 100 is a hyperspectral imager designed specifically for remote sensing CubeSat applications. The instrument provides spatially resolved spectral measurements across the visible and near-infrared (VNIR) wavelength range, enabling vegetation monitoring and agricultural analysis. For this mission, the HyperScape 100 serves as the primary scientific payload and is responsible for generating the data required to produce regularly updated vegetation indices across the state of Kansas.

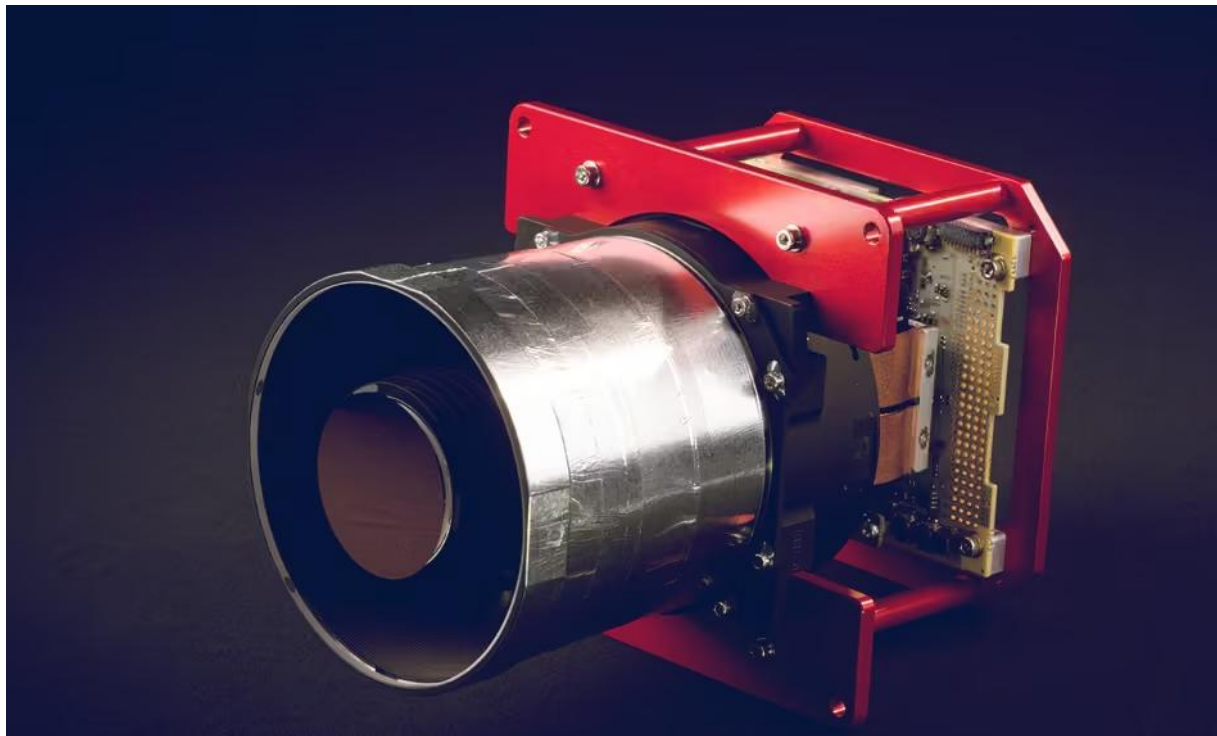


Figure : Simera HyperScape 100 hyperspectral imager

The HyperScape 100 operates as a pushbroom (line-scan) hyperspectral imager, capturing one spatial line of ground data at a time while dispersing incoming light into narrow spectral bands. As the satellite advances along its orbit, successive lines are collected and combined to form a hyperspectral data cube consisting of two spatial dimensions and one spectral dimension. This architecture enables continuous imaging over large ground areas while minimizing mechanical complexity, making it well suited for CubeSat missions.

The payload is designed to integrate within standard CubeSat volume and interface constraints, occupying approximately a 1.5U volume. The instrument has a mass of 1.26 kg and nominal power consumption during imaging operations of approximately 7 W. Science data is transferred digitally to the spacecraft avionics, with raw data rates dependent on spatial resolution, spectral binning, and imaging duty cycle, typically on the order of tens to hundreds of megabits per second prior to onboard processing or compression.

From low Earth orbit, the HyperScape 100 achieves ground sample distances of 4.75 meters, with corresponding swath width of 19.4 kilometers. This combination of spatial resolution and coverage enables efficient regional-scale monitoring and supports the mission's objective of statewide NDVI mapping within the desired revisit period.

Spectrally, the HyperScape 100 provides fine spectral resolution through narrowband filtering, with spectral sampling on the order of 1 nm and a full width at half maximum (FWHM) bandwidth specified as 3.5% of the central wavelength. With the spectral ranges for our applications that gives us a spectral resolution of approximately 23nm. The instrument's VNIR spectral coverage includes the red and near-infrared bands required for NDVI computation, allowing detection of subtle reflectance variations associated with vegetation health.

The HyperScape 100 was selected due to its combination of spectral performance, compact form factor, and CubeSat-optimized design. Its relatively high technology readiness level reduces mission risk while enabling the collection of scientifically valuable data products. Overall, the payload provides the spatial and spectral capabilities required to meet the mission's agricultural monitoring objectives while remaining compatible with the power, volume, and data constraints of a 3U CubeSat platform.

Neutron Detector

The Dry Storage Cannister (DSC) Domino is a neutron detection platform designed and produced by Radiation Detection Technologies (RDT) as a follow-up to the original Domino. Both technologies utilize four microstructured semiconductor neutron detectors (MSNDs) but the original Domino is only equipped to measure thermal neutrons. Thus, to get efficiency peaks in other energy ranges multiple Domino platforms are needed in varying thicknesses of moderator as was done on the NeRDI-1A mission. The new DSC-Domino, however, is designed to measure two energy ranges, thermal and fast neutrons, which are done using different MSND trench geometry and backfill. The specifics on the geometry are

excluded here to prevent any unintended disclosure of intellectual property, but the backfill is ^6LiF for thermal neutrons and Parylene N for fast neutrons.

While the NeRDI-1A mission space qualified the original Domino, the newer DSC-Domino does not have any flight heritage and possesses a few distinguishing capabilities that the original lacked. For example, the DSC's integrated driver board greatly reduced the size of the separated original Domino + driver board. Because of these differences, and the benefits of an MSND compared to the competition (scintillators, proportional gas detectors, and planar semiconductors, as discussed much earlier), the DSC-Domino is a clear choice for this mission for the value it can provide for data collection and the value to RDT for raising their product's TRL.

The DSC-Domino has a USB-C interface port where all data and power is routed through. The overall dimensions are $27.17 \times 66.96 \times 5.78$ mm. The layout of a DSC-Domino is shown in Figure 11 and the location of the MSNDs shown on the left is general and just meant to provide a visual representation of the backfill (i.e., the specific location of, for example, the Parylene MSND may be different from the one shown).

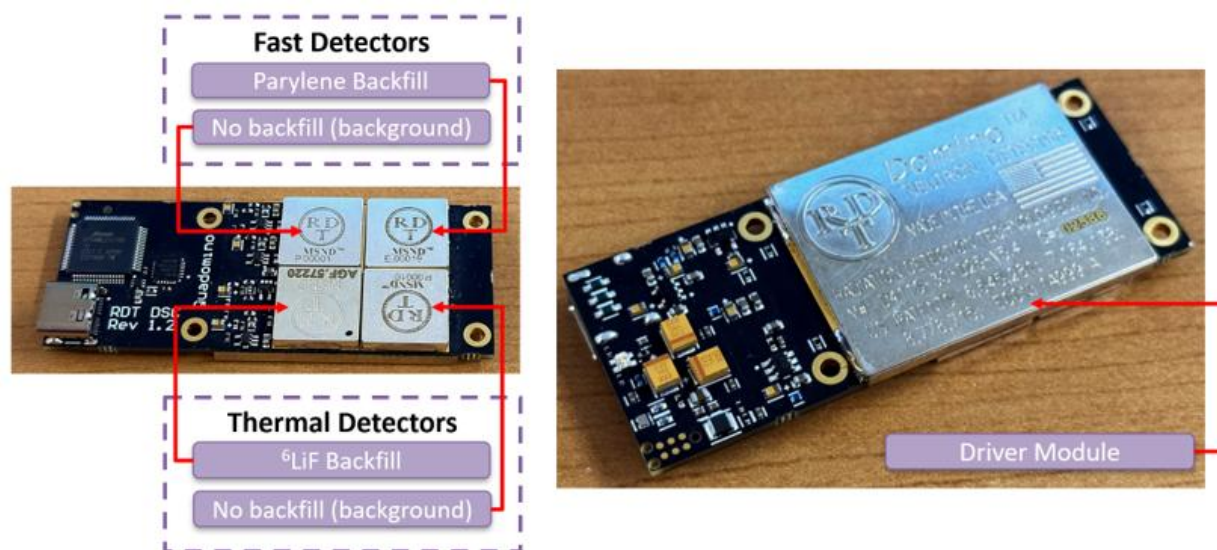


Figure 11: DSC-Domino Components (Left Image: top face with the MSNDs, Right Image: back face with the driver board)

For this mission, there will be Domino platforms in moderated and unmoderated configurations to allow us to shift the ^6LiF MSND's detection efficiency peak into the epithermal range. The chosen HDPE (C_2H_4) moderator "filters out" the low energy neutrons primarily through absorption while the epithermal and fast neutrons are still able to reach the detectors, albeit with reduced energy, due to scattering while en route through the moderator. This reduced energy, however, is why the ^6LiF backfilled detector can have its detection efficiency shifted using moderators.

MCNP 6.2 simulations were performed using an isotropic, uniformly distributed neutron energy source (from 0.025 eV to 20 MeV with an explicit set of source energy distribution bins) outside of an RPP-shaped moderator. An F4 tally was adjusted using a tally multiplier of the form $(-0.648, M, Z)$ where the 0.648 value comes from RDT and their evaluation of the efficiency of the detector, the “M” value relates to the specific material number for the backfill (assumed to be dry air for the non-backfilled MSNDs), and the “Z” value is the specific neutron interaction we are relying on for a detector response (105, or $X(n, t)Y$ for ${}^6\text{LiF}$ and 103, or $X(n, p)Y$ for Parylene)). There was also a special treatment “FT4 SCX” tag onto each F4 tally that leveraged the source’s explicit distribution bins to report the tallies according to the particle’s **initial** energy instead of its energy when it interacted with the detector. These steps resulted in the desired moderator thickness of 10-mm on each side of the detector platform. An example of the modeled moderated and unmoderated systems are shown in Figure 12 with the resulting unmoderated and moderated relative ${}^6\text{LiF}$ platform detection efficiencies shown in Figure 13. Because of the relatively low scattering cross sections of carbon and hydrogen in fast energy regimes, the high-fidelity efficiency spectrum shown in Figure 14 shows negligible change between the moderated and unmoderated configurations.

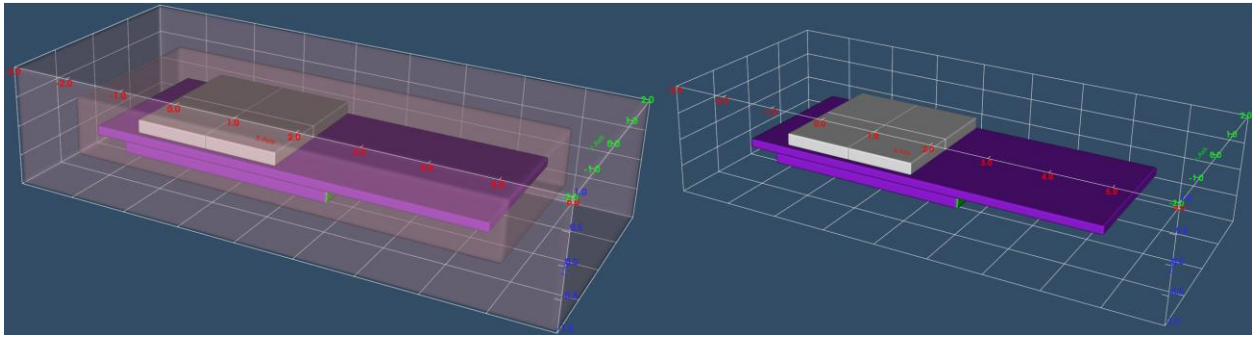


Figure 12: MCNP Modeled DSC-Domino (Left Image: with moderator, Right Image: without moderator)

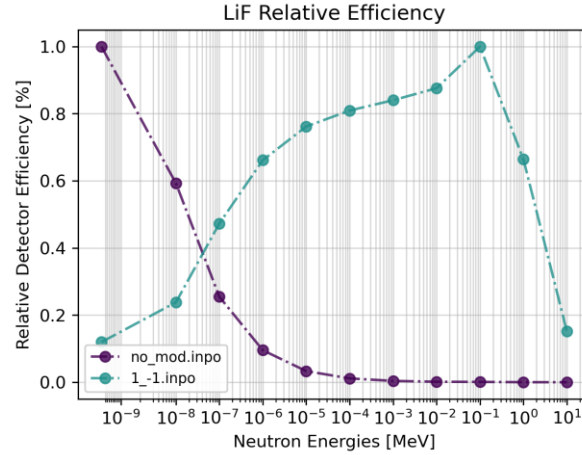


Figure 13: ${}^6\text{LiF}$ MSND Relative Detection Efficiency

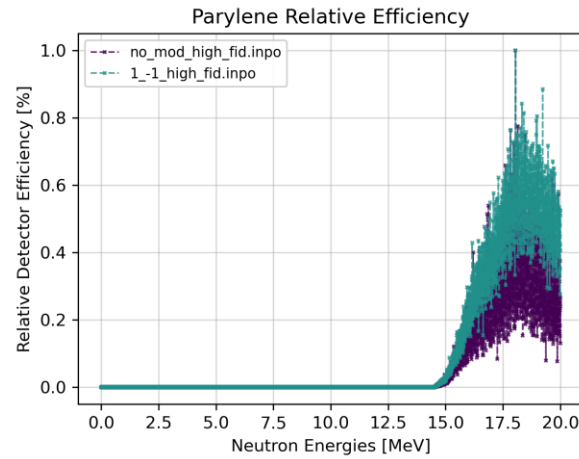


Figure 14: Parylene MSND Relative Detection Efficiency

Therefore, to achieve energy bins in the thermal, epithermal, and fast neutron energy regimes, a 10-mm thick HDPE moderator shell shall be used around the detector platform to achieve peak efficiencies at 0.025 eV, 100 eV, and 18 MeV. Further work should be performed to determine more accurate detector responses/efficiencies as well as background rejection in environments with particles representing all those the satellite will encounter in space (GCRs, gammas, other charged particles, etc.). Additionally, more accurate geometries should be pursued once the full satellite model is complete.

Experiment Plan

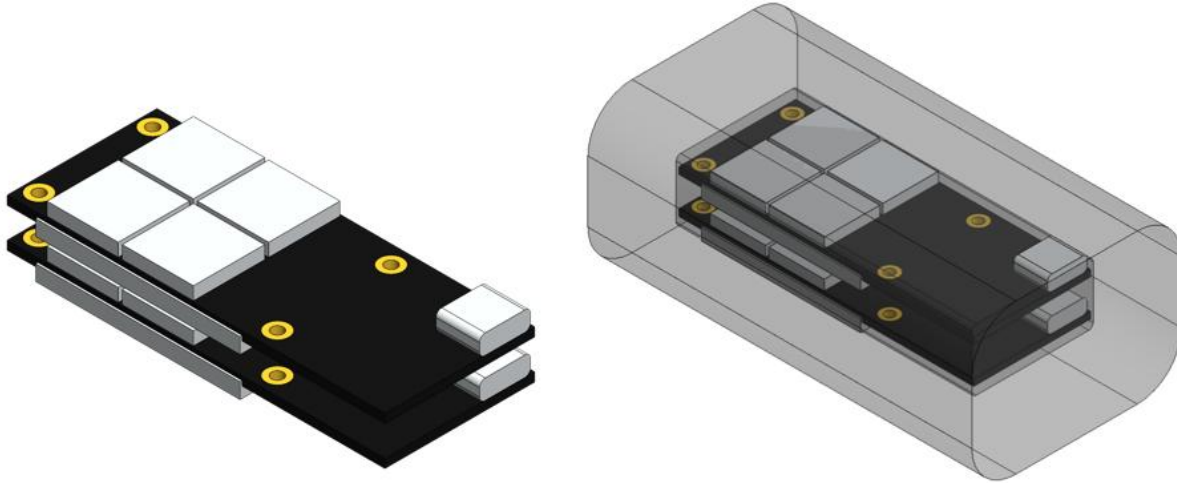


Figure 15: DSC-Domino Unmoderated and Moderated Configurations (CAD)

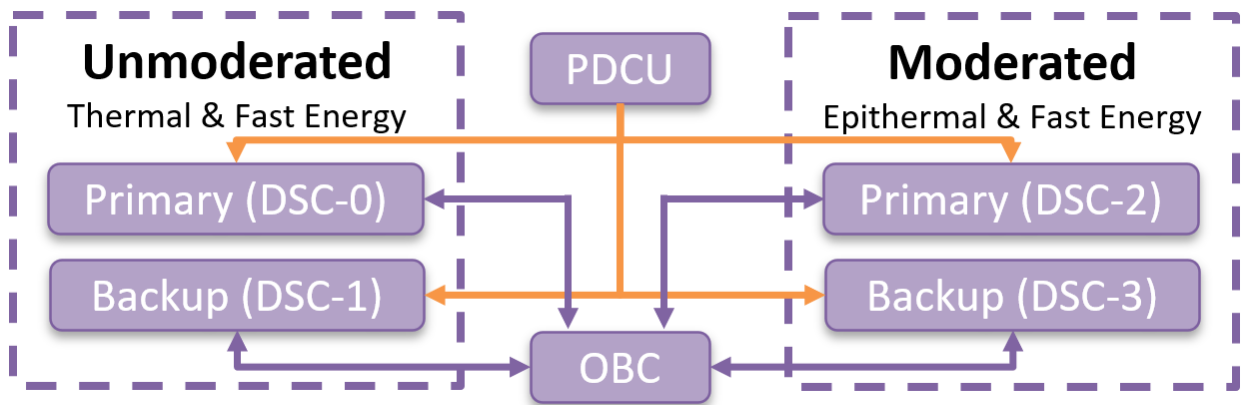


Figure 16: NuKAT Payload System Block Diagram

The NuKAT mission is interested, as mentioned above, in the thermal, epithermal, and fast neutrons in LEO. While this is a lower radiation flux environment compared to higher orbits due to Earth's magnetic field and slight atmosphere, there is still a significant neutron flux. Especially around the polar regions (from the Van Allen radiation belt) and in the South Atlantic Anomaly (SAA), there are much higher local fluxes for charged particles and neutrons. So, the neutron energy profile of these regions is of particular interest. Up through the pPDR, the experimental plan was to gather counts at a set frequency (determined by C&DH data limits) throughout the orbit with a minimum frequency of 0.0167 Hz or pulling the counts once every minute. A reviewer on the pPDR panel presented the idea of throttling down our frequency in the regions of less importance which would then enable a higher sampling rate in the poles and SAA—this is something that will be investigated next semester. At this point, the ideal sampling rate (given data downlink constraints) is pulling counts every 30 seconds (or a frequency of 0.0333 Hz)

which we hope will be even more frequent as the mission progresses and data rates are set by the FCC.

To contextualize this minimum and ideal measurement frequency, the average velocity is around 7.8 km/s. When the payload is accumulating counts with pulls every 30 and 60 seconds, the resulting measurement reflects the total neutron counts over an arc distance of $x = \frac{v}{f} = \frac{7.8}{0.0333} \approx 234$ km and $\frac{7.8}{0.0167} \approx 468$ km, respectively. When extrapolated to the average 90-minute orbit of LEO, there are around 90 to 180 different measurements per orbit. While this is a coarse look at LEO, it still provides enough fidelity to correlate counts to certain regions. Additionally, after extended mission life, interpolation will enable us to develop a finer view of the environment because of the overlapping passes.

The satellite will have an ongoing detection mission throughout the mission operations phase with the only exception being when the health/stability of the satellite and its systems require (e.g., in a critical battery state). The satellite payload will consist of four DSC-Domino platforms, two encased in HDPE and two unmoderated as shown in Figure 15. Within each configuration, there will be a primary and a backup platform which will allow measurement validation by comparing the measurements between the similar detectors which is displayed in Figure 16. If the measurements that are expected to be similar fall outside of a certain percentage window from each other (e.g., 5% over the averaged values from the past 20 counts), then a flag will be raised that alerts the ground station to the deviation which allows us to proceed using only the detector/platform that is still in a nominal operating condition. From a concept of operations standpoint, it may be valuable to periodically switch which detector is treated as primary and backup, thus adjusting which platform's data is primarily downlinked. For example, a monthly rotation would have DSC-0 and DSC-2 as the primary platforms for 30 days before switching to treating DSC-1 and DSC-3 as the primary platforms for the next 30 days.

Now, focusing on the data package, each detector platform has four MSNDs which each return a 32-bit unsigned integer when counts are pulled. Since there are four platforms, we have 16 MSNDs total which, when a full raw data package is examined, results in $4 \text{ platforms} \times 4 \text{ MSNDs/platform} \times 32 \text{ bits/MSND} = \boxed{512 \text{ bits/measurement}}$ or **128 bits/DSC-Domino**. While this is data that the OBC must handle, the amount for downlink can be greatly reduced by processing the data. Firstly, since there is a redundant platform in each moderated/unmoderated configuration, only the data from one in each case must be downlinked. This reduces the number of reported MSNDs from 16 to 8 and halves the bits per measurement to **256 bits** ($2 \text{ platforms} \times 4 \text{ MSNDs/platform} \times 32 \text{ bits/MSND} = \boxed{256 \text{ bits/measurement}}$). Secondly, each DSC-Domino only has two

MSNDs that are specifically geared toward measuring neutrons with the other two intended to allow the subtraction of background radiation's influence on neutron measurements. By subtracting the background counts from the neutron measurements, there are only two numbers to report from each platform and we have $2 \text{ platforms} \times 2 \text{ MSNDs/platform} \times 32 \text{ bits/MSND} = \boxed{128 \text{ bits/measurement}}$. So, our data load can be significantly reduced such that the data the ground station receives is four unsigned 32-bit integers relating to the thermal, epithermal, and fast energy spectrum counts. There are two downlinked fast counts (one from moderated and one from unmoderated detectors) which will allow us to perform rough validation on the ground that compares these two values without needing to downlink the whole science data profile from the satellite. That said, periodic raw data should be downlinked periodically (at least for the primary platforms) to verify proper science collection.

This data package will also be tagged with the orbital location where it was recorded using GNSS which allows us to determine if we are in specific regions of interest and to aid in stakeholder research/classroom examples that require knowledge of the positional data for measurements. Additionally, this will allow us to validate data with prior LEO neutron detection missions such as NeRDI-1A and other similar missions (see referenced documents) [5] [6] [7] [8] [9]. This location tagging is also important to produce visual results such as heat maps and topological maps. Examples of these types of data products are shown in Figure 17 and Figure 18.

To effectively produce this data, there are some non-payload requirements that dictate satellite operation and layout. To begin, the unmoderated platforms must not be placed above or below the moderator to reduce/avoid the influence of the moderator on the neutron spectrum we will be measuring. Additionally, to ensure at least 90% of the expected neutron flux is impinging on the main detector faces, the platforms (and thus satellite) must be kept within 20° of the nominal nadir vector using ADC systems. This is a simple inverse cosine relation ($\cos(0.9) \approx 25.8$ degrees), so if the attitude of the satellite is known and transmitted as part of the data package, we can back out the expected neutron flux without the need for fine pointing control. The detector platforms must also be kept within their operating thermal limits of -20°C to 55°C .

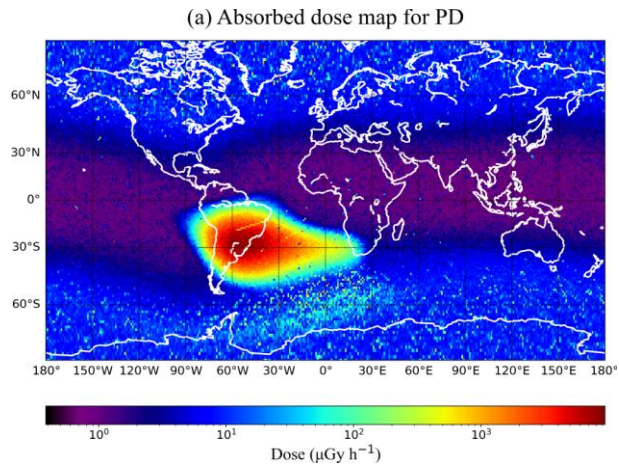


Figure 17: Example Radiation Heat Map: *Journal of Astronomy and Space Sciences* | LEO Radiation Dosimetry, Uk-Won Nam et al.

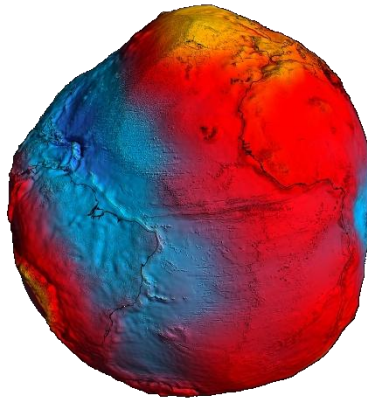


Figure 18: Example 3D Topological Map: *National Geographic* | New Gravity Map Reveals Lumpy Earth

Ground Overview

NuKat

Ground System Description

Nukat's ground system is like that of many other types of ground systems. As shown by the block diagram under the NuKAT, the ground system for the mission consists of the KSU radio station, the satellite in space, and the web application that communicates with end users. More specifically, the second block diagram under NuKAT shows the satellite in space once again, but also shows the antenna, rotator, amplifier, transceiver, and radio station computer on the ground. The ground station computer connects to the rotator, transceiver, and amplifier which controls these tools and their functions. All telemetry, payload data, and commands go through the KSU radio station as the frequency in which these operate under UHF bands. UHF is a lower frequency on the spectrum than other frequencies such as S-band or X-band.

Ground System Block Diagram

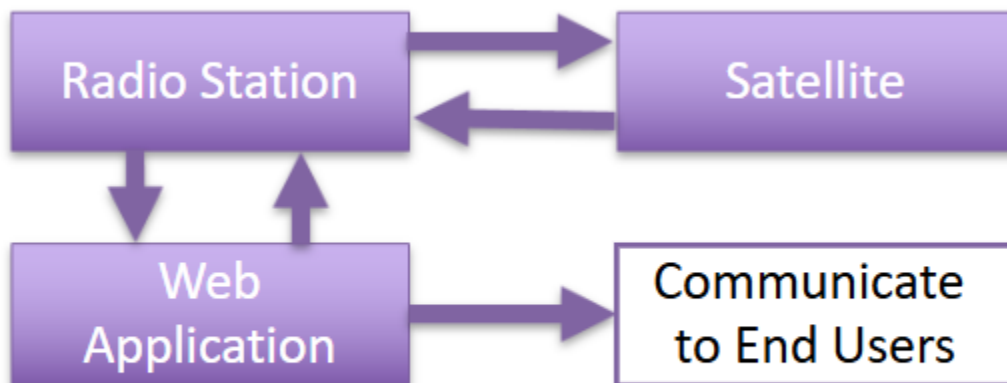


Figure 19: NuKAT Ground Station Block Diagram (part 1)

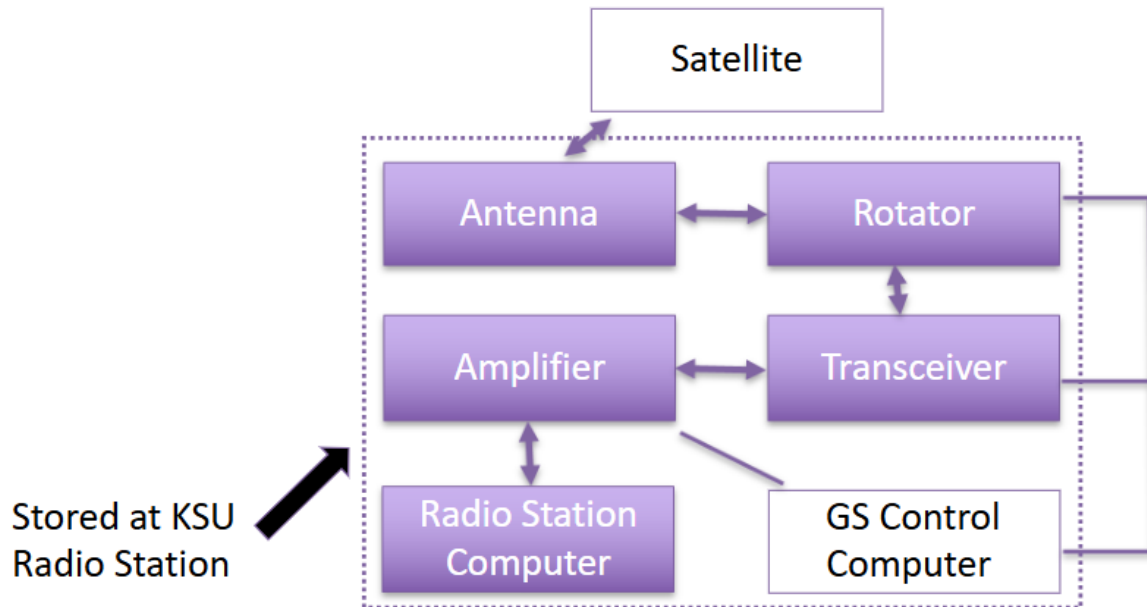


Figure 20: NuKAT Ground Station Block Diagram (part 2)

Ground Software Description

The main purpose of the ground station software will be to handle telemetry data from satellite, display the status of the satellite, send commands up to the satellite, and distribute data to end users. To meet these demands, we will use the free open-source software COSMOS as its foundation. COSMOS has most of the functionality that the ground station will need already built into it. It can interface with ground stations across multiple modalities, create and show displays, and has strong automation tools. Additionally, any extra functionality can be added through importing and integrating pre-built packages from Python or Ruby libraries. An Influx database will act as the main storage point for the neutron counts that are collected. It is a time-series database that is built for storing and aggregating data across different time windows. This allows for easy collation and dissemination of mission data to end-users, which is the primary goal of the ground station software. Testing for the ground station software can be done by creating simulated data of neutron counts and passing it into COSMOS through a TCP / UDP packet simulator.

Ground Data Description

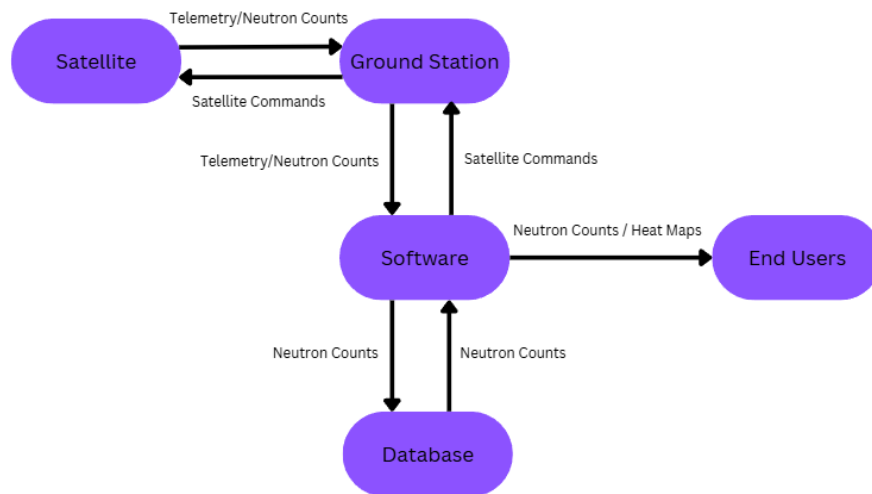


Figure xx: NuKat Data Flow Diagram

The NuKAT mission will collect neutron counts using the DSC-Domino from Radiation Detection Technologies Inc. These counts will be downlinked through the ground station, which will be the Kansas State Radio Station, to the software. The software will store the neutron counts in an Influx database and will process and deliver to end users either on demand or at a set time interval. Data delivered to end users can come in the form of raw counts or as heat maps.

Ground Support Equipment Description

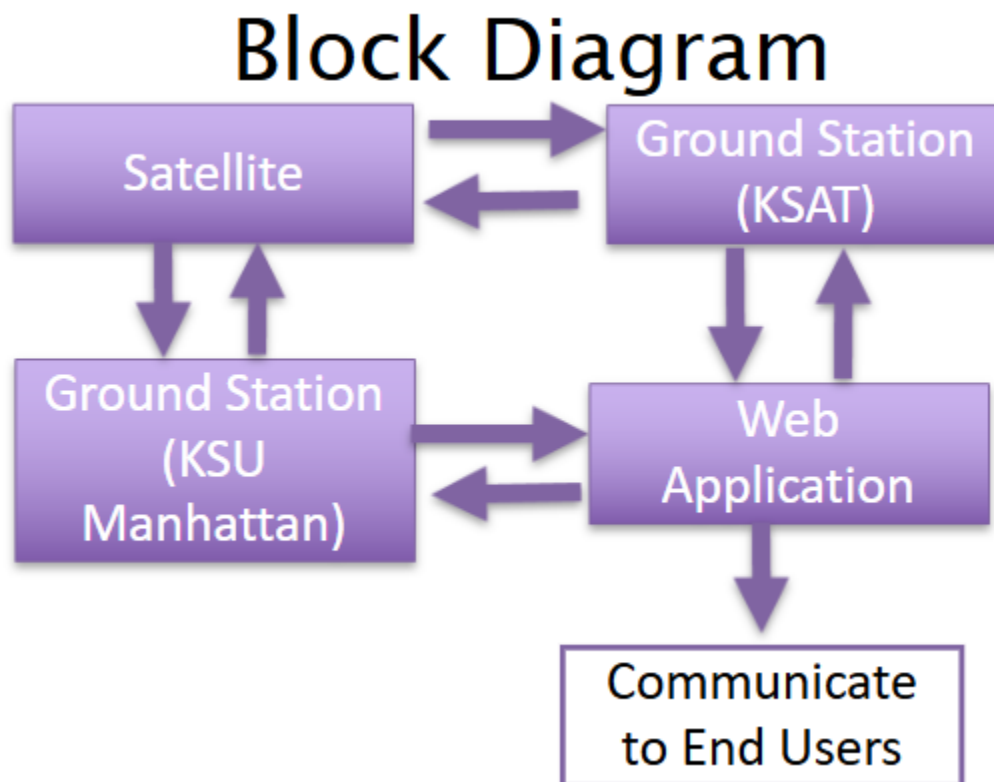
The NuKat ground support equipment was selected to provide the tools and hardware necessary to support all assembly, integration and testing. Electrical equipment focuses on subsystem interfacing, signal emulation, and data acquisition. This calls for standard lab instruments for basic verification and troubleshooting. Additionally, software driven data acquisition devices are required to emulate signals and capture data. We can simulate environment inputs to evaluate responses. Mechanical support requires tools for assembly and integration. Precision measurement instruments, mass scales, and surface roughness gauges. Tools required to verify assembly. Together, electrical and mechanical ground support equipment capabilities provide for efficient integration and testing.

PHAT-Sat

Ground System Description

Phat-Sats ground system is more complex than the NuKAT mission since Phat-Sats payload data can only be sent down through X-band frequencies unlike NuKAT which can be sent down via UHF bands. All the telemetry, tracking, and commands will go through the KSU radio station operating in UHF bands. A second ground station will be needed to operate through X-band frequencies to relay payload data to the ground. We are currently working with KSAT in Norway to hopefully setup a ground station to hand the payload data. The operations at the KSU radio station will remain like the NuKAT. Both missions will require licensing for operation such as Federal Communications Commission (FCC), International Amateur Radio Union (IARU), Federal Aviation Administration (FAA), and International Telecommunication Union.

Ground System Block Diagram



Ground Software Description

The PHAT-Sat mission will use the same software structure as the NuKAT mission except for the Influx database, which will be replaced with Amazon Simple Storage Service (S3)

buckets, and it will send the imaging data received from satellite to the Kansas State Agricultural department for processing. Once the data is done being processed, it will be distributed to the web service All Things Kansas, which will act as the access point for end users. PHAT-SAT testing can be done in a similar manner to NuKat using simulated data to test functionality, automation, and storage capabilities.

Ground Data Description

Currently, PHAT-SAT will collect imaging data over the United States and will compress this along with time and location data to downlink via X-Band to the KSAT ground station. It is important to note that this is based off current data budget expectations, and if these calculations change, we will be forced to calculate vegetation and soil indices onboard and downlink those instead. Commands and health information will be transmitted via UHF through the Kansas State radio station. The data that is downlinked will be stored in Amazon Simple Storage Service (S3) buckets, which will be distributed to K-State Precision Ag/Data Analytics who will process our data into vegetation and soil indices and then into maps. These maps will be sent to the web service All Things Kansas, which will act as the distribution point for end users.

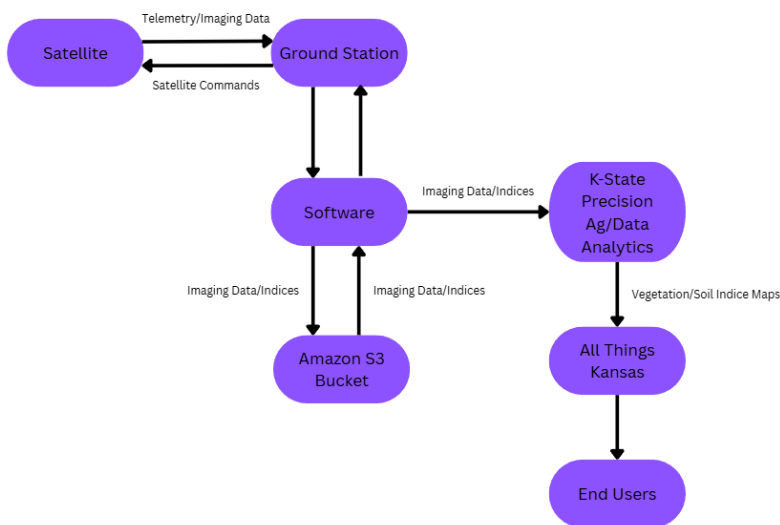


Figure xx: PHAT-SAT Data Flow Diagram

Ground Support Equipment Description

The PHAT-Sat ground support equipment promotes assembly, integration, and testing. Mechanical support capabilities include controlled handling and fixturing to prevent damage during integration, alignment verification of structural components, and dimensional inspection. Mechanical equipment also supports inspection before and after

environmental testing to assess changes. Electrical equipment allows for satellite interfacing and data logging. Software driven hardware allows for signal emulation, allowing us to simulate space environment inputs and gage responses. These ground support capabilities allow for proper assembly integration and testing.

Bus Overview

NuKAT

The NuKat spacecraft bus provides the core functionality required to support continuous neutron detector payload operations and survivability of launch and on-orbit conditions. The bus is responsible for maintaining satellite health, supplying power, managing science and telemetry data, providing coarse attitude stabilization, ensuring structural integrity, and regulating the thermal environment from launch to decommissioning. The bus architecture is designed to prioritize payload functions, reliability, simplicity, manufacturability, and operational success.

The dual E-Stack design was chosen for the selected satellite flight orientation (Mission Description), payload orientation requirements (Neutron Detector), external interface constraints, and heat dissipation considerations. External interface requirements specify that inhibit pins and external charging ports need to be accessible on a long face that is not occupied by solar panels. Board spacing within E-Stacks promotes passive thermal dissipation through conduction to the spacecraft structure and radiation across internal surfaces. This approach limits localized heat buildup and reduces the risk of overheating avionics during on-orbit operations. The 1U Avionics E-Stack allowed for external interface ports on the daughterboard to exist on the negative z-face, while the 1U Payload E-Stack correctly positioned the detectors in line with Earth nadir. Both E-Stack are secured with fasteners to the chassis and are removable through the y-faces and support integration and testing, as demonstrated in Figure 21.

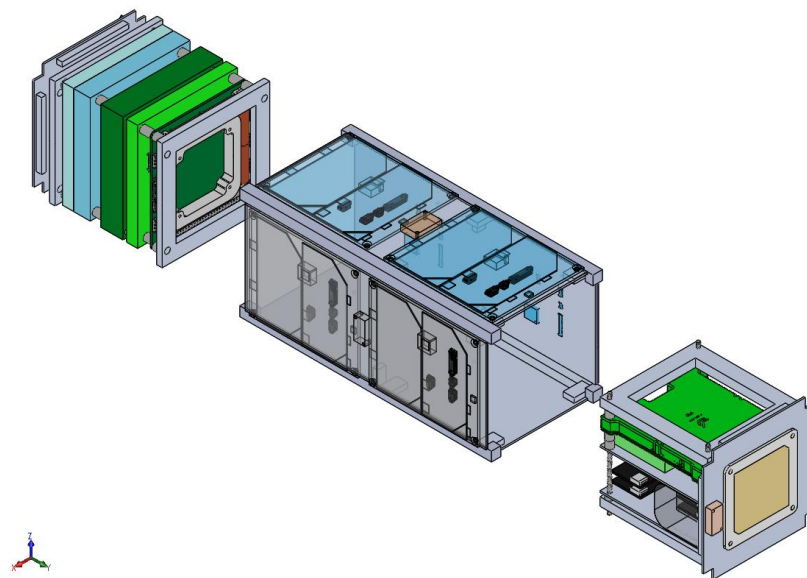


Figure 21: NuKat E-Stack Exploded View

The Avionics E-Stack houses the onboard computer, daughter board, power distribution and control unit, battery, and ADC board for centralized data handling, command, and power distribution in a compact volume. Designing these components within the same stack minimizes harnessing length and electromagnetic interference during operations. The Payload E-Stack consists of the payload bay, GNSS receiver, and UHF transceiver, as well as extra allocated space for cabling connections to the Avionics E-Stack. A breakdown of the internal components within each E-Stack is provided in Figure 22.

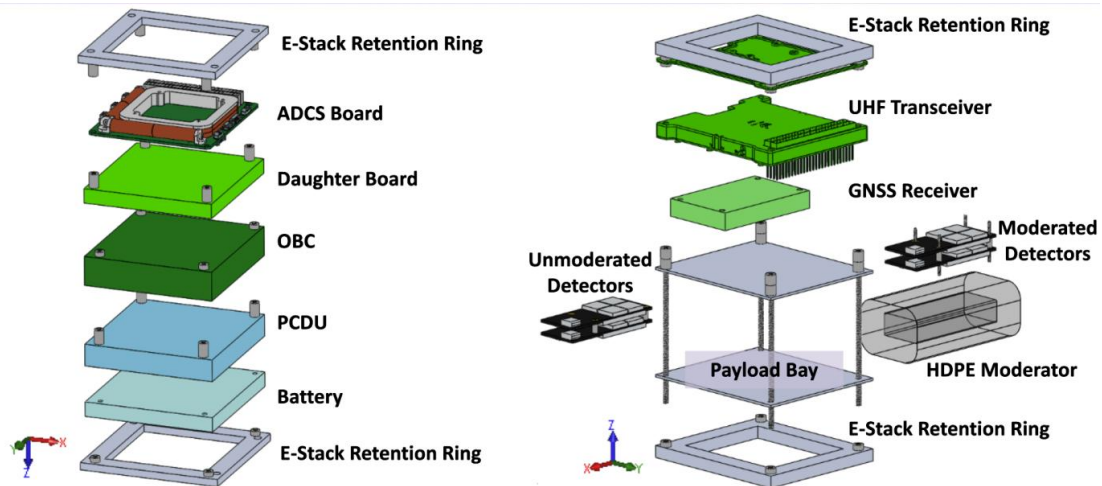


Figure 22: NuKat Avionic E-Stack and Payload E-Stack Exploded View

The GNSS/GPS antenna is placed on the positive y-face and the magnetometer arm deploys from the negative y-face to minimize magnetic interference from the solar panels. Each x-face stores a deployable solar panel and will become exposed post-deployment. The negative z-face will host the UHF antenna (TBD) to allow for payload operations to continue during downlinking. Sun sensors are located on each face for full FOV coverage for attitude determination. The external components for the undeployed and deployed configurations are shown in Figure 23.

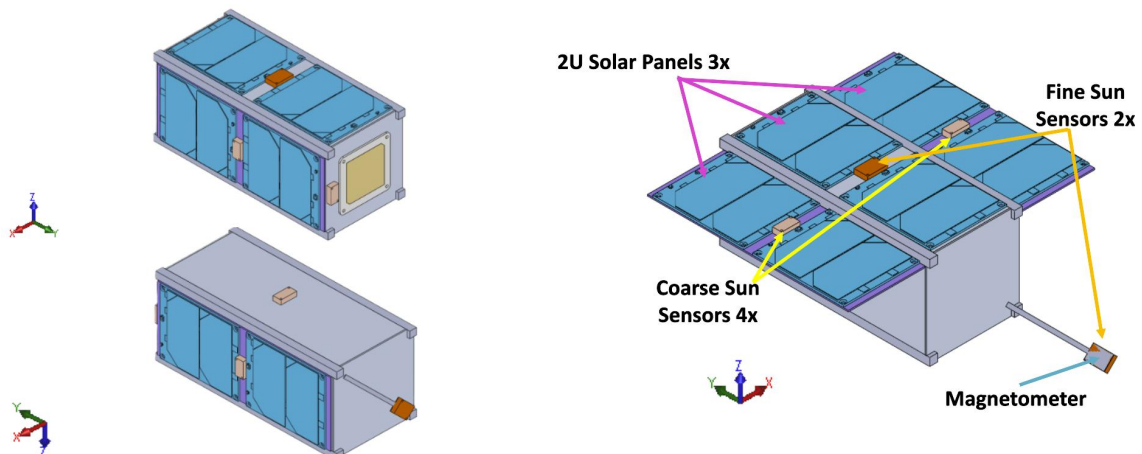


Figure 23: NuKat External Components

Electrical, data, and RF interfaces within the bus are implemented using coaxial cabling, standard harnessing, and PC104 stacking connectors are currently designed for the selected components. A high-level cabling and connection diagram is shown in Figure 24.

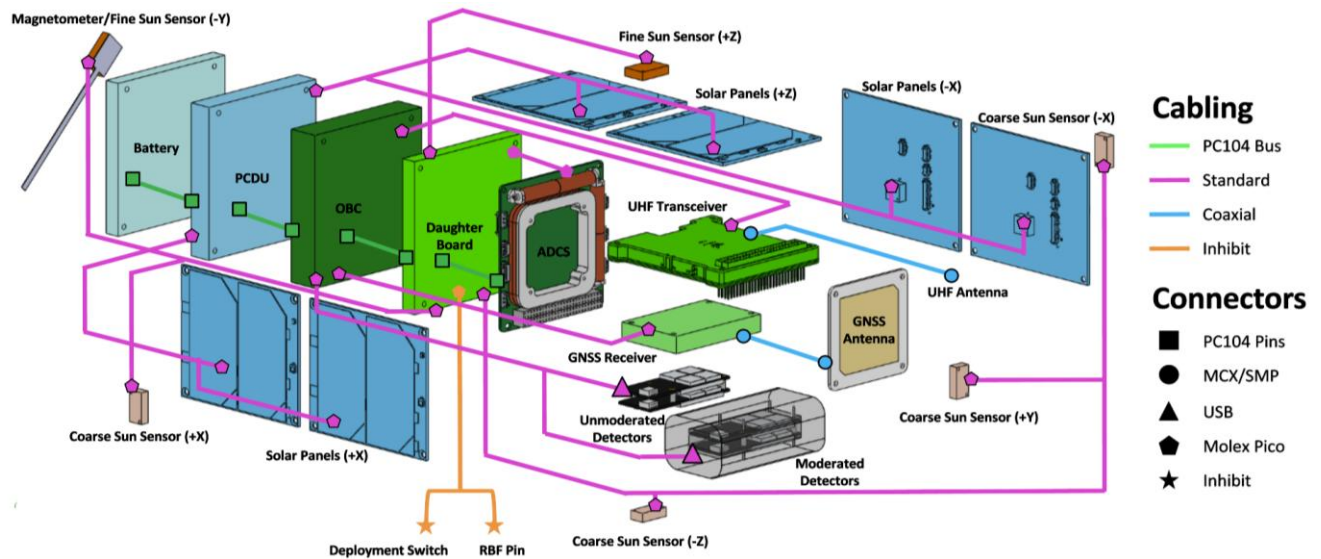


Figure 24: NuKat Cabling and Connection Diagram

Attitude Determination and Control

The Attitude Determination & Control (ADC) requirements of NuKAT are 1) to allow Nadir awareness and pointing so our payload can collect scientific data, 2) to allow for Sun pointing so our satellite can charge more effectively under normal and upset conditions, 3) to allow for ground station pointing so our satellite can complete uplink and downlink of commands and scientific data, 4) to recover from upsets during deployment and mission operations, and 5) to allow for de-orbit less than 5 years after end-of-mission by pointing different surface areas at the incoming atmosphere (controlling drag). To do this, the subsystem shall include:

- Hardware: a magnetometer, fine and coarse sun sensors, magnetic torque rods, possibly a subsystem computer, integration with a GPS/GNSS receiver
- Software: Nadir awareness, Sun awareness, angular velocity estimation, Nadir pointing mode, Sun pointing mode, Ground Station pointing mode, Detumble safe pointing mode, Temperature regulation (roisserie) safe pointing mode, calibration mode
- Interfaces: CDH (position & time IN; telemetry OUT), FSW (mode, commands IN; health and status OUT)

ADC must also be mechanically, electrically, and thermally integrated with the bus.

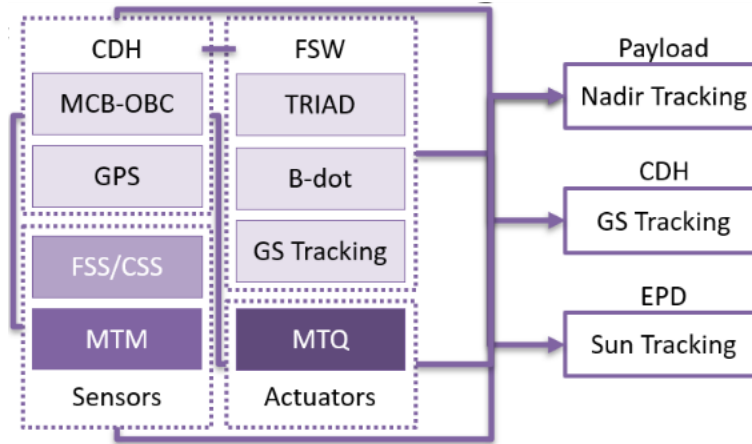


Figure 28: ADC Subsystem Block Diagram for NuKAT.

All sensors and actuators are routed through an ADCS computer, purchased along with the other components. While not required, redundancy is improved by using flight-proven computers integrated with system components by the provider.

Subsystem	Mode	Component	Target	Axis to Align and Frame of Axis	Accuracy (+,-)	Precision (+,-)	FOV (+,-)	Function
PLD	Science (Nadir Pointing)	Neutron Detector	Earth Normal	(+,-) Z Axis of Spacecraft to be aligned with Nadir of Earth	5 deg	5 deg	20 deg	Point for science mode; align to collect minimum 90% of expected particles
CDH	Up/Downlink (GS Pointing)	Comms Antenna	Ground Station	(+,-) X or Y Axis of Spacecraft to be aligned with GS, which one based on structure	UNKNOWN	UNKNOWN	UNKNOWN	Point for up/downlink, align to send minimum X% of packets
	Science, Calibration (Sun + Nadir)	GPS Receiver	GPS Satellites	Aligned with Equator during operation	5 deg	5 deg	30 deg	Position Determination
ADC	Science, Up/Downlink, Calibration	Magnetometer	Earth Normal	No alignment required, will calibrate on orbit to find relative direction of Earth Normal vector	5 deg	5 deg	360 deg	Attitude Determination
	Safe Mode (Sun Pointing)	Coarse Sun Sensor	Sun	Aligned with Sun vector during operation	10 deg	10 deg	45 deg	Attitude Determination, Sun Awareness
	Science, Up/Downlink, Calibration, Safe Mode	Fine Sun Sensor	Sun	Aligned with Sun vector during operation	10 deg	10 deg	45 deg	Attitude Determination
	ALL	Magnetorquer	N/A	Aligned internally, all 3 components must be orthogonal	5 deg	5 deg	N/A	Attitude Control; limit sensor resolution
EPD	Safe Mode	Solar Panels	Sun	(+) Z Axis of Spacecraft to be aligned with Sun	10 deg	10 deg	20 deg	Sun tracking to charge

Figure 29: ADC Pointing Budget for NuKAT.

The pointing budget includes requirements for three primary pointing modes, as well as calibration and safe mode. Ground station mode is not fully defined, because the antenna package (and thus FOV) is not known.

While the subsystem requirements are defined, the preferred supplier is not. It is imperative that the next team make an informed decision on provider selection next year, in order to close all integration requirements with all other sub-teams. This will ensure parts are ordered with the appropriate lead time.

Communications and Data Handling

The NuKAT Communications and Data Handling (CDH) subsystem serves as the primary onboard processing and communications system. It is being designed to support UHF

communication for ground station uplink and payload/telemetry downlink, with GNSS for spacecraft positioning and metadata generation.

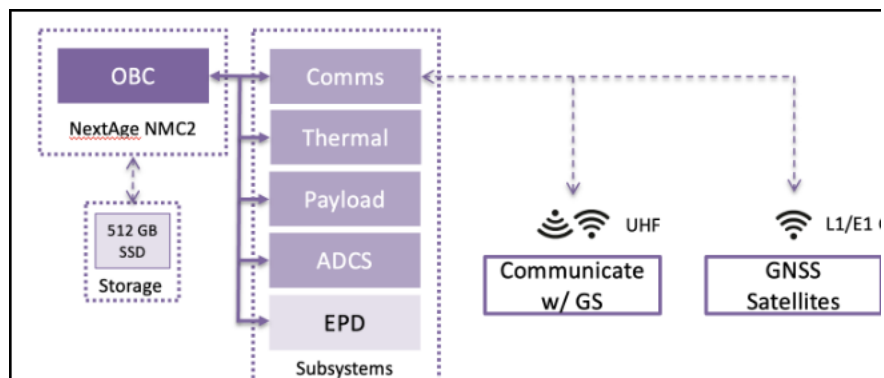


Figure 32: Block Diagram for NuKAT CDH Subsystem

UHF operations have been planned to operate in the 435-438 MHz range per IARU standards, with the possibility of pursuing experimental licensing. A preliminary link budget was developed using specifications for the EnduroSat UHF Transceiver II and estimated values for the K-State ground station (W0QQQ) and antenna (TBD) due to ongoing design changes and limited vendor responses. Current antenna considerations include the EnduroSat 2U UHF Deployable Antenna, KairoSpace UHF Patch Antenna, and a custom-designed antenna. A preliminary link budget using key values (Free Space Path Loss based on altitude, elevation angle) and estimates for ground station parameters is included below.

UPLINK		DOWNLINK	
Ground Station EIRP (rough estimated value)	25 dBW	Satellite Transmit Power	0 dBW
Free Space Path Loss	-176.5 dB	Antenna Gain (at half-power beam edge)	-3 dB
Antenna Gain (at half-power beam edge)	-3 dB	Free Space Path Loss	-176.5 dB
Received Power	-154.5 dBW	Ground Station Gain (estimated)	40 dB
Required Eb/N0 (GMSK, 10 ⁻⁵ BER)	10.6 dB	Received Power	-139.5
System Noise Temperature (at Satellite, estimated)	300K	Required Eb/N0 (GMSK, 10 ⁻⁵ BER)	10.6 dB
Channel Bandwidth	25 kHz	System Noise Temperature (at Ground, estimated)	700K
Bitrate	4800 bps	Channel Bandwidth	25 kHz
k * T_sys	-203.9 dB/Hz	Bitrate	4800 bps
Calculated Eb/N0	12.51 dB	k * T_sys	-200.2 dB/Hz
Link Margin	1.89 dB	Calculated Eb/N0	23.9 dB
		Link Margin	13.3 dB

Figure 33: Preliminary Link Budget for NuKAT UHF Communications

GNSS capabilities are required to associate payload data with positional information for verification and mission deliverables. The current baseline design utilizes the EnduroSat GNSS L1/E1 Passive Patch Antenna with NovAtel OEM719 Receiver, selected for flight

heritage and familiarity with the PhoenixSat mission. Trade studies and research on at least 3-5 options are planned before final selection.

The on-board computer (OBC) for NuKAT is the NextAge NMC2, selected for radiation-tolerant design, error detection and correction features, and redundancy capabilities. NextAge has indicated intent to prove the processor and engineering support at no cost to establish flight heritage and establish university relations; though this agreement must be formally documented. The NMC2 offers 400+ GPIO pins, as well as PC-104 interfacing for adding on daughter boards for breakouts, transceivers, ADC boards, etc. Note that this processor does not, to our current knowledge, natively support USB-C, which is required for DSC-Domino interfacing. A potential solution is to have four host microcontrollers to act as driver boards. This architecture must be resolved early in Spring 2026.

NuKAT Data Calculations

Orbital analysis indicates an average of 23 minutes of daily downlink time. This corresponds to maximum daily downlink capacities of 13,248,000 bits (1,656 KB) at 9600 bps and 6,624,000 bits (828 KB) at 4800 bps. Using an estimated 296 bits per sample for DSC-Domino data, data includes four detector counts, temperature, latitude, longitude, altitude, UNIX timestamp, and earth normal angle with 3.8x margin was applied to account for data framing, modulation and forward error correcting. Note that data compression should be explored moving forward. Based on these assumptions, the following table was developed to select an appropriate sampling period for CONOPs.

Frequency (Hz)	Sample Period (s)	1 Domino (KB / day * 3.8 mrg)	4800 bps Compliant?	9600 bps Compliant?	2 Dominos (KB / day * 3.8 mrg)	4800 bps Compliant?	9600 bps Compliant?
0.017	60	202.46	Yes	Yes	404.93	Yes	Yes
0.033	30	404.93	Yes	Yes	809.86	Yes	Yes
0.05	20	607.39	Yes	Yes	1,214.78	No	Yes
0.1	10	1,214.78	No	Yes	2,429.57	No	No
0.2	5	2,429.57	No	No	4,859.14	No	No
0.5	2	6,073.92	No	No	12,147.84	No	No
1	1	12,147.84	No	No	24,295.68	No	No
2	0.5	24,295.68	No	No	48,591.36	No	No
5	0.2	60,739.20	No	No	121,478.40	No	No
10	0.1	121,478.40	No	No	242,956.80	No	No

Figure 34: Payload Data Calculations for NuKat

Electrical Power & Distribution

During the Fall of 2025, the EPD team efforts included preliminary block-diagram based designs for powering the satellite bus, analysis of various solar array solutions, establishing mission requirements, and the formulation of the power budget.

The block diagram below outlines the interconnect structure of the EPD subsystem, showing simple relationships between hardware. Given that differences between NuKAT and PHAT-Sat will be in size and power customers, the EPD block diagram remains consistent across multiple vehicle configurations. The Pico-EPS PCDU is the central hub for power conditioning and distribution for the subsystem. Interfacing between the solar arrays which are our vehicle power source, the batteries which are our power storage, and the voltage buses providing power to customer subsystems downstream. The Pico-EPS was chosen as this hardware met vehicle size constraints and provided the correct output buses needed for subsystem power. The Pico-EPS also includes built in battery storage with optional modular upgrades to capacity. The STM32 MCU is a reliable and proven heritage MCU and will be used to supplement voltage and current measurements to the Flight Software in addition to the measurements provided by the Pico-EPS. These measurements will focus on the solar array inputs, the battery, and each voltage bus. The MCU measurements will offer valuable developmental flight Information t

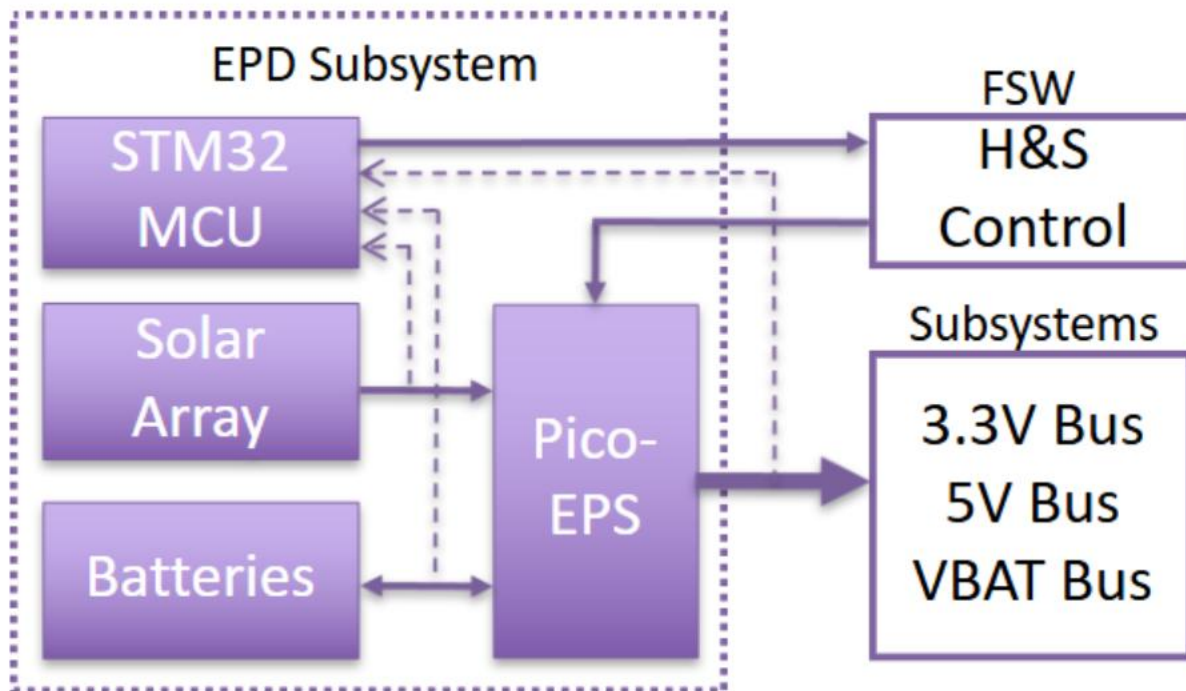


Figure 36: Block Diagram for General EPD Subsystem

Insights on Related Standards

The commercial off the shelf hardware we've selected comply with the following testing standards: The EnduroSat 1U solar panel complies with the European Cooperation for Space Standardization ESA Standard ECSS E ST20-08C, photovoltaic assemblies and components and ECSS-E-ST-10-03C, testing for random vibrations, radiation, and thermal vacuum testing. The General environmental verification standard for Goddard Space Flight Center GSFC-STD-7000A demonstrates minimum workmanship quality and provides confidence that components, like solar panels, will not fail due to the mechanical and thermal stresses of the mission. These rigorous requirements for the qualification and testing of photovoltaic cells and bypass diodes ensure their reliable electrical performance and survival against the extreme radiation and thermal cycling of space. Additionally, the hardware was manufactured in international organization for standardization ISO 14644 class 6 compliant clean room, which allows a maximum of 10^6 particles per cubic meter at sizes of $0.1\mu\text{m}$ or larger. The Pico-EPS PCDU has a NASA Technology Readiness Level of eight, meaning that the hardware is flight qualified through test and demonstration.

Preliminary Power Budget

An understanding of how the available power should be allocated across the bus is vital for beginning design work. To reach such an understanding, the textbook, *A Guide to CubeSat Mission and Bus Design*, was consulted. It provided an initial power budget from New SMAD's design as an example. The payload's power consumption had been tested this semester and is negligible compared to the other subsystem loads, so the example had to be adjusted for NuKAT's bus. We started with the percentages from the example, then dropped the payload from 40% to 5%. The ratio of the remaining percentages to the other 60% was calculated and applied to the newly freed 35%. The results can be seen in the figure below, as well as some simple parameters.

Subsystem	% Allocated to Subsystem	Maximum Simultaneous Power Allocated to Subsystem (Watts)	Minimum Simultaneous Power Consumption in Idle Mode (Watts)	% of Known Operating Power Consumption
Payload	5%	0.6	0.25	3%
ADCS	10%	1.2	1.50	17%
C&DH	55%	6.5	5.30	59%
Thermal	10%	1.2	0.00	0%
Power	20%	2.4	2.00	22%
Structure	0%	0.0	0.00	0%
Total Consumption		11.9	9.05	
Max Generation			13.9	
Balance	100%	2.0	4.85	

Figure 37: Preliminary Percentages of Power Budget for NuKAT

These allocations are considered the maximum power each subsystem can consume while all subsystems are on. This budget does not restrict the subsystems to a maximum power consumption at any moment, but rather the maximum power consumption when all electronics are powered simultaneously. Thus, the “Minimum Simultaneous Power...” column is the known values for the electronics that have been chosen so far and shows whether the minimum consumption from each system is acceptable when all subsystems are operating. A safety margin of 2 Watts is applied, which accounts for the degradation of the solar panels, the inevitable inefficiencies in all subsystems, and slight deviations from datasheet values. The number for power generation is the notable input, as it may change depending on how the mission’s scope changes. Based on calculations discussed later, the maximum generation during orbit is 13.9 Watts, which occurs when the array is completely normal to the sun.

Communications & Data Handling (C&DH) has the highest allocation, as the On-Board Computer will always need power at different rates, and transmitting data is periodically cumbersome on power systems. The EPD subsystem itself will always be managing power generated by solar panels, monitoring the battery, and maintaining the voltage rails, which introduces a constant demand for a portion of power generated. The ADCS and Thermal subsystems gained an equal allocation, as both typically have a lower minimal power consumption, but do require periodical rises in power during certain operating modes. The Thermal team did not decide their electronics this semester, which is shown as a zero in the “Minimum Simultaneous Power...” column. EPD expects the ADCS subsystem will have a continual power demand due to the preference of knowing exact positioning during the science operation of detecting neutrons. Currently, the bus has a lower overall power consumption during simultaneous operations than was originally allocated, which allows some portion of power generated to recharge the vehicle’s battery.

Solar Array Configurations

An important realization occurred when a portion of the entire mission team was debating the solar array configuration. While a flat surface loaded with solar cells pointed at the sun is the most obvious way to maximize power, there is another method that is comparable. If two faces are loaded with panels, then the trigonometric nature of the angle of incident for the solar irradiance allows for a configuration with negligible difference from the flat surface facing the sun. The cosine factor involved in the transformation of solar irradiance to power generated is not a strictly proportional constant. The equation below is the power generated for any instant of time, where η is efficiency, “A” is the total area of photovoltaic cells, “I” naught is solar irradiance (which can be taken as the solar constant of 1367 W/m^2 at Low-Earth Orbit). The $f(T)$ is a fraction of one depending on the temperature of the solar cells, which will fluctuate throughout each orbit. The $f(t)$ is a fraction of one depending on the degradation of the photovoltaic cells over time.

$$\text{Instantaneous Power} = P = \eta \cdot A \cdot I_o \cdot \cos(\theta) \cdot f(T) \cdot f(t)$$

Figure 38: Instantaneous Power Generated by Solar Array

In the case of a CubeSat having two faces with solar panels, the power generated has a maximum that is 141% of a single face’s maximum generation when that single face is normal with the sun. In the figure below, the corner between two-faces is aligned with the sun, such that the angle of incident on both faces is 45-degrees. Using this angle for the cosine and multiplying the instantaneous power by two (since there are two faces that are equal) introduces the 141% increase. This will change if the faces are not equal, but in any configuration with two faces, the shown relation between the sun and solar array results in maximum power generation. The same argument can be made for a three-faced solar array, where the optimal angle of incident is 54.7-degrees for all three faces and the maximum power generated is a 173% increase from a single face normal to the sun.

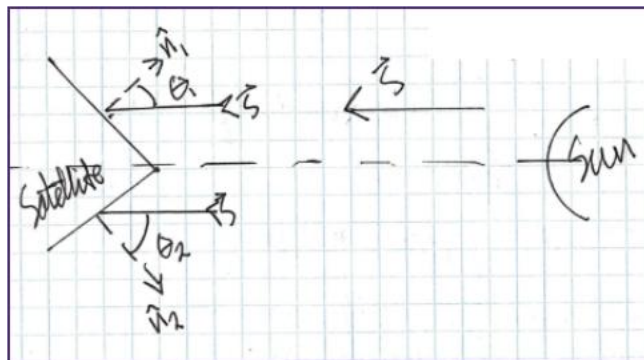


Figure 39: Optimal Angle of Incidence for Two-Faced Solar Array

Part of the decision for implementing a deployable solar array was supported by calculations of power generated by affixed solar panels on multiple faces of the vehicle. These calculations can be seen in the appendices included at the end of this report.

Degradation Factor

Another important aspect of calculating the power generated by a solar array is the degradation of the photovoltaic cells over time, referred to as the Degradation Factor. Some manufacturers include the Beginning-of-Life (BOL) and End-of-Life (EOL) values for their equipment's electrical characteristics. As seen in the equation below, the factor can be estimated by taking the ratio of the BOL and EOL values, which gives a percentage of the power that is lost due to the degradation of the photovoltaic cells and other materials in the solar panel. The power budget described later uses this factor when calculating the power generated, so there is another safety margin introduced and the EPD design assumes degradation from onset of the mission.

$$\text{Degradation Factor} = \frac{P_{EoL}}{P_{BoL}}$$

Figure 40: Equation for Simple Degradation Factor

In the current design, the EnduroSat solar panel uses the Azur Space photovoltaic cell, which is also used in the solar panels produces by DHV Technology. Using the values provided in the DHV Technology brochure, which are included in the table below, a degradation factor for each operating temperature can be calculated. In the power budget, 93.33% is applied to the instantaneous power equation in place of $f(t)$.

Operating Temperature (Celsius)	Beginning-of-Life Power (W)	End-of-Life Power (W)	Degradation Factor
-50	2.48	2.41	97.18%
28	2.07	1.98	95.65%
125	1.5	1.4	93.33%

Table 5: DHV 1U Solar Panel Values

Instantaneous Power Budget

In the most convenient manner, *A Guide to CubeSat Mission and Bus Design* provided a template for a power budget that is extensive and acts as a strong basis for EPD's technical feasibility. This template was modified into a workbook that aligned with NuKAT's concept of operations subsystems. Due to the instantaneous nature of the workbook, there is a multitude of rows where every row is another time step. The time step, along with other

inputs for the “Typical Mode Sequence” spreadsheets in the workbook, can be seen in the table of input cells shown below.

Settings (Inputs)	
Parameter	Value
Time Step (sec)	20
Max Charge (Wh)	20
Min Charge (Wh)	0
Min Safe Charge (Wh)	10
Initial Charge (Wh)	20
Orbit Period (hrs)	1.572

Table 6: Input Cells for Instantaneous Power Budget

The operational modes for the mission were implemented throughout this workbook, as they decided the instantaneous power consumption for each time step. The power consumption of every electronic during each operational mode is a vital set of information that informs the instantaneous power consumption. The table for inputs from this set of information is shown below and is a part of the larger Power Budget workbook.

		Power Usage by Mode (W)								
Parts		Science				Upset				Safe
Subsystem	Component	Counting	Pointing	Receiving	Transmitting	Detumble	EMI Survival	Irregular Voltage	Irregular Temperature	Safe
Payload	Neutrino Detector	0.25								
C&DH	OBC	3.0	medium	medium	medium					
C&DH	GNSS Antenna	0	0	0	3.5					
C&DH	GNSS Receiver	1.3	1.3	1.3	1.3					
C&DH	UHF Transceiver	1	2.8	7.3	10					
C&DH	UHF Antenna	0	0	0	3.5					
ADCS	ADCS Main Board	1.5	5	1.5	1.5					
ADCS	Coarse Sun Sensors (6x)									
ADCS	Deployable Magnetometer									
EPD	Control & Distribution Unit	2	2	2	2	2	2	2	2	2
Thermal										
Average Power During Mode		9.05	11.1	12.1	21.8	2	2	2	2	2

Table 7: Portion of “Power by Mode” spreadsheet in Power Budget workbook

Another important set of information related to the “Power by Mode” spreadsheet is the on/off status per mode for each electronic. This is used as a verification elsewhere in the workbook that the instantaneous power consumption is correct. Graphs are also set to show the status of each electronic throughout an entire orbit. The table for these inputs is shown below and is part of the larger Power Budget workbook.

Subsystem	Component	Counting	Pointing	Receiving	Transmitting	Detumble	EMI Survival	Irregular Voltage	Irregular Temperature	Safe
Payload	Neutrino Detector	☒	☐	☐	☐	☐	☐	☐	☐	☐
C&DH	OBC	☒	☐	☐	☐	☐	☐	☐	☐	☐
C&DH	GNSS Antenna	☒	☐	☐	☐	☐	☐	☐	☐	☐
C&DH	GNSS Receiver	☒	☐	☐	☐	☐	☐	☐	☐	☐
C&DH	UHF Transceiver	☒	☐	☐	☐	☐	☐	☐	☐	☐
C&DH	UHF Antenna	☒	☐	☐	☐	☐	☐	☐	☐	☐
ADCS	ADCS Main Board	☒	☐	☐	☐	☐	☐	☐	☐	☐
ADCS	Coarse Sun Sensors (6x)	☒	☐	☐	☐	☐	☐	☐	☐	☐
ADCS	Deployable Magnetometer	☒	☐	☐	☐	☐	☐	☐	☐	☐
EPD	Control & Distribution Unit	☒	☒	☐	☐	☐	☐	☐	☐	☐
Thermal		☒	☒	☒	☒	☒	☒	☒	☒	☒
EPD		☐	☐	☐	☐	☐	☐	☐	☐	☐
Parts On During Mode		11	2	1	1	1	1	1	1	1

Table 8: Portion of "Subsystem (On/Off)" spreadsheet in Power Budget workbook.

The previous two tables are not complete due to the lack of component-specific design and details for each mode of operation. This information will become vital as the design team analyzes the power balance throughout orbits and over the entire mission. The instantaneous power balance can be generated as long as each electronics on/off status and power consumption is known for each mode of operation.

Baseline Instantaneous Power Balance

During the Fall of 2025, only one mode was considered, which is the counting mode. This is considered the idle or minimal power consumption state, where it is preferable to always be detecting neutrons. This allowed for the last set of data to be used for baseline power balance analysis. The set of instant power generation values dependent on time was calculated using orbital information from the STK simulations, which provided a set of data for both power generation and effective area of the solar panels. For more control over the instantaneous power generated and the implementation of the aforementioned safety margins, only the effective area data was used. This subset of data has an interval or time-step of 20 seconds, which is highly accurate time-step for the power budget.

This subset is proportional to the photovoltaic area, with the constant being the cosine of the incident angle, and the photovoltaic area being taken from the solar panel datasheet. Using the equation for instantaneous power, the calculated degradation factor, the known panel efficiency, and the irradiance of 1367 W/m^2 (solar constant), the instant at every time-step can be calculated. This calculation occurred outside the power budget workbook, as the STK data had to be organized, the degradation factor had to be deduced from known datasheets, and the efficiency gathered. The resulting set of instant power generation was then organized for inputting into the workbook. A portion of the raw and calculated data can be seen in the table below.

Time (UTCG)	Effective Area (m ²)	Simulated Power (W)	Calculated Power (W)	Solar Intensity
4:30:00 PM	0.0124	5.2060	4.7366	1
4:30:20 PM	0.0131	5.5220	5.0241	1
4:30:40 PM	0.0139	5.8290	5.3039	1
4:31:00 PM	0.0146	6.1440	5.5906	1
4:31:20 PM	0.0153	6.4500	5.8692	1
4:31:40 PM	0.0161	6.7520	6.1437	1
4:32:00 PM	0.0168	7.0560	6.4204	1

Table 9: Portion of Calculations Based on STK Simulation Data.

Power Balance for NuKAT

The configuration shown earlier in the report was analyzed through the Power Budget using modified STK simulation data for twenty-five hours of orbit. Two separate sets of data were generated by the simulation, where one set has the vehicle permanently Earth nadir, and the other has the solar array normal to, or pointing at, the sun permanently. After modifying each set of simulation data using the instantaneous power equation, then inputting the data into the Power Budget workbook for the two orientations, the following results were observed. The power consumption of 9.05 Watts, which is found in the figure for a portion of “Power by Mode” is applied in every result.

Earth Nadir Orientation

In the figure below, the power balance over nearly four orbits can be observed. The average power generated over those orbits was 8.48 Watts and the maximum power generated is 13.9 Watts. There is a positive power balance for about 20 minutes in each orbit.

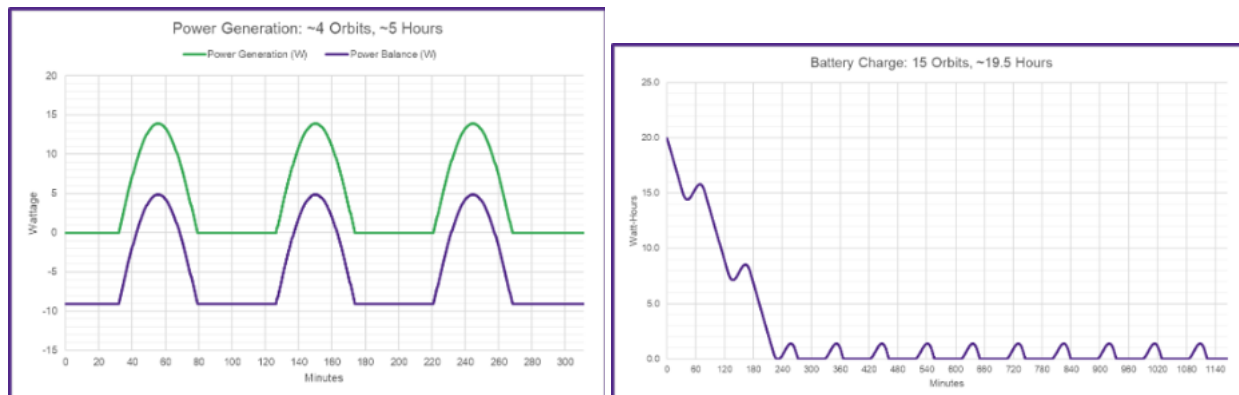


Figure 41: Power Balance & Battery Charge Graphs for Earth Nadir Orientation

Next, the State-of-Charge (SoC) is observed as calculated by the Power Budget. The battery reaches zero charge after 227 minutes in orbit, as seen in the graph above.

Sun Normal Orientation

These results are useful for knowing how the maximum power generation effects the power balance, as well as understanding the impact of any mode that points the array directly at

the sun. The maximum generation of 13.9 Watts remains, but there is much more time spent with a positive power balance, as seen in the graph below.

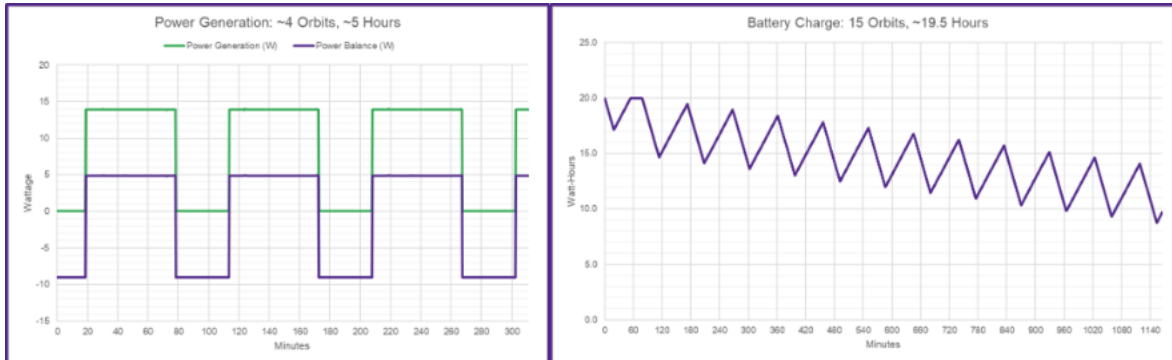


Figure 42: Power Balance Battery Charge Graphs for Sun Normal Orientation

In this orientation, the battery does not reach zero charge immediately, since the power balance is positive more often. The battery did not reach zero charge during the twenty-five hours of data provided by the simulation. Since there is a downward trend in the battery's charge, we can safely assume that the battery will reach zero charge sometime past the known twenty-five hours. It can be seen above that the battery's charge does reach 50% SoC at around 980 minutes.

Larger Solar Array Results

In an attempt to provide a solution to the deficient power generation, simulation data for a 10U-sized solar array was generated. The results, shown in the graphs below, reveal a maximum generation of 23 Watts and an average generation of 14.5 Watts. A positive power balance is much more frequent at nearly 60 minutes per orbit. The battery does not immediately reach zero charge, but it does occur nearly 600 minutes into the mission.

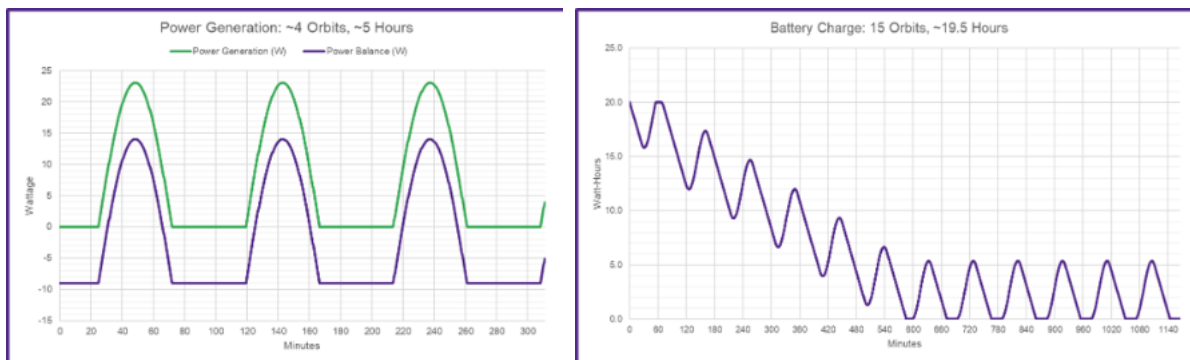


Figure 43: Power Balance & Battery Charge Graphs for 10U Solar Array in Earth Nadir Orientation

The permanently sun-normal orientation was also observed. It shows the battery charge to be stable, with only a 27% Depth-of-Discharge or 73% SoC, which allows NuKAT to survive during continuous and minimal power consumption. This observation shows the

importance of the “Pointing” mode of operation and that it may be necessary at different times in any orbit.

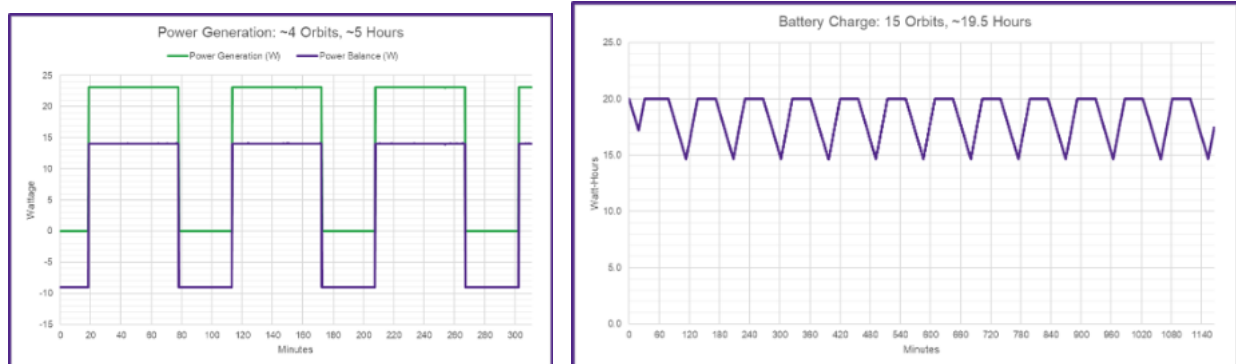


Figure 44: Power Balance & Battery Charge Graphs for 10U Solar Array in Sun Normal Orientation

Overall, the current design for EPD is not technically feasible and the ability to produce design is not yet proven for NuKAT. To balance the power budget, either power consumption must decrease, or power generation must increase. A decrease in consumption becomes a challenge for the bus design and each subsystem. An increase in power generation may change the overall structure of NuKAT, which will require consideration at the mission level. For a better-defined power budget, more details regarding electronics at every mode of operation is necessary.

Thermal

The thermal subsystem will use a combination of passive and active regulation methods to ensure the satellite maintains a safe, stable range of temperatures. Additionally, temperature measurement devices such as thermistors will be placed at critical locations to monitor the condition of the satellite.

The temperature range NuKAT should be kept within is based on the operating temperatures of the individual components within the satellite. Currently, for NuKAT, that range is between -20 and 50 degrees Celsius. This range may narrow if a more restrictive component is added to the satellite.

Measurement

To monitor the temperature of the satellite, thermistors will be used at certain locations in the satellite along with any built-in temperature sensors from individual components (i.e. Pico-epds, DSC-Domino). The basic outline is to provide to the on-board computer measurements for the following regions:

- Each Exterior Face (6)

- Comms (1)
- Battery (1)
- Payload (2)
- OBC (1)

This currently totals 11 measurement devices; however, that amount is flexible to change based on how the needs of the satellite evolve. Below is a table summarizing potential devices that can be used in NuKAT.

Device	Collins 0118MM	TE 44900	Variohm 311P18
Op Temps (C)	-269-400	-55-90	-55-90
Flight Qualified	☑	☑	☑
Interchangeability	±1.0%	±0.1 C	±1.0 C

Table 10: Potential Options for Temperature Measurement

Each of the shown options operate well within our target temperature range, but further trade studies should be completed before choosing which option best meets the needs of the satellite.

Radiation

Rough calculations have been made to begin understanding the thermal cycling of the satellite in a space environment. These consider two extremes of operations: the Hot case characterized by the maximum incident radiation and minimum outgoing radiation, and the Cold case characterized by the opposite. These calculations also do not account for real properties of the materials on a satellite and currently assumes the 2U satellite as a black body with a long face pointing normal to the sun. The results of these calculations are shown below in Table 11.

Radiation IN			Radiation OUT		
	Hot	Cold		Hot	Cold
qsun [W]	27.22	0	q" [W/m ²]	271.27	544.20
qeir [W]	5.32	4.28	q [W]	27.13	54.42
qalb [W]	9.53	0			

Table 11: Calculations for Incident and Outgoing Radiation of a 2U black body

Keep in mind that these numbers make generous assumptions and before final decisions are made, much more detailed calculations and simulations should be performed. However, they do provide valuable guidance for future choices related to surface coatings and potential, dedicated radiative surfaces.

Heaters

We have a basic understanding of how the satellite interacts with space via radiation, and to prevent systems from going under operating temperatures, heaters will likely be needed to provide additional heat. Unfortunately, further research is needed before making a decision on the type and size of said heaters. However, they will likely be flexible, resistance-based heaters.

Structure

There was a lot of progress made on the development of the structure for NuKAT this semester, as we determined an orientation of the overall satellite and made decisions on specific components so we could model the NuKat satellite. Within given requirements, we designed a structure that could secure each component oriented as needed. Our current configuration is a horizontal orientation (Y long axis) with X axis being velocity and Z being Nadir, as well having 4U of deployable solar panels and 2U fixed.

The overall structure of the NuKAT satellite must fit within the outer dimensions of 120mm x 120mm x 227mm, with the outside of the four rails being 100mm x 100mm x 226mm. The rails extend past the Y faces of NuKAT by at least 3mm and have a width of 8.5mm from the paneling. The Z faces and X faces extend past the rails while fitting within the requirements given by standard dispenser for NanoRacks.

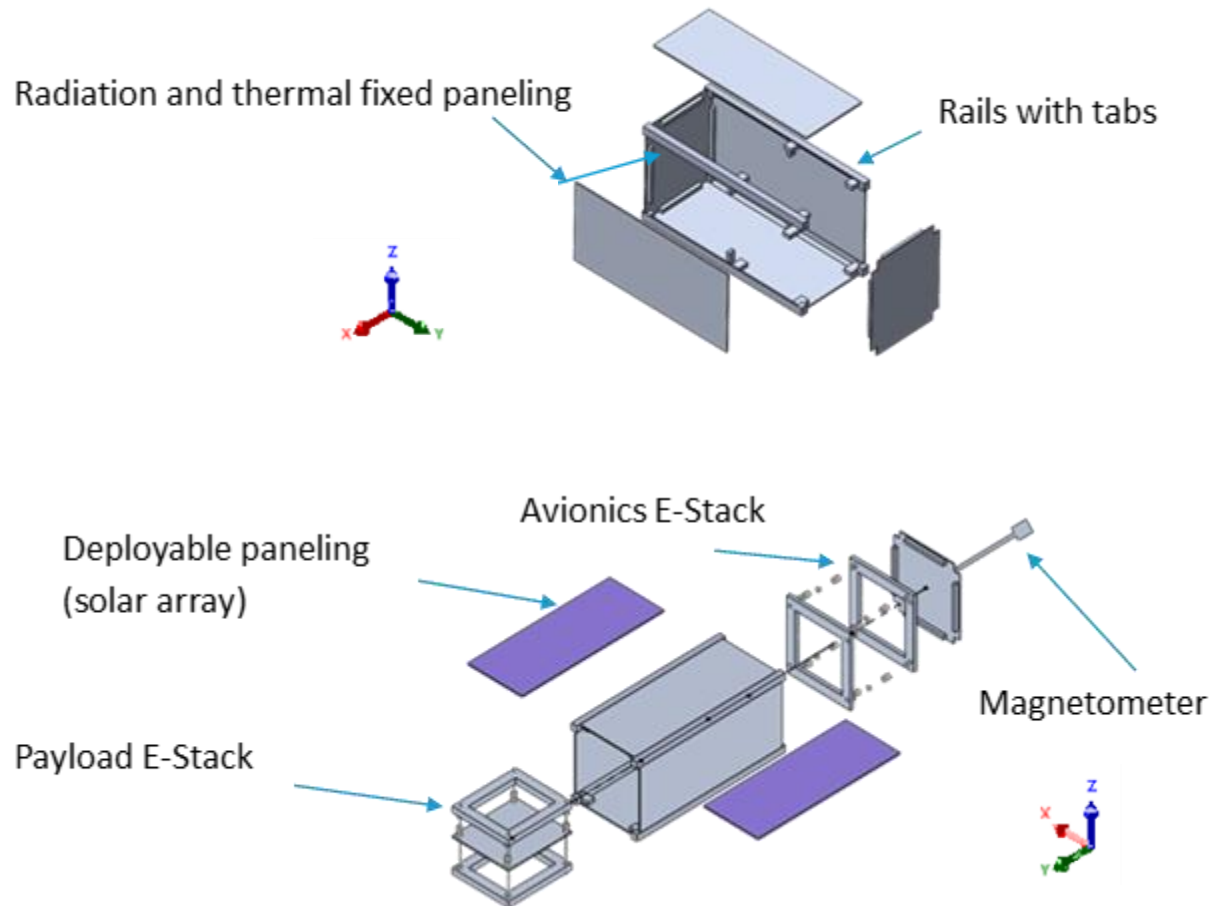
The NuKAT structure consists primarily of the aluminum chassis that will be manufactured in house with K-State TDI. It will be constructed of 6061 aluminum and include rails that will interface with the launch dispenser. The rails will also be made of 6061 aluminum through the K-State TDI and each rail will be 8.5mm x 8.5 mm x 226mm. The chassis will have tabs attached to the internal edges of the rails to assist with internal mounting as well as mounting components from the external.

Within the chassis, there will be four retainers to hold two “stacks” that we have assembled to secure the individual components. These retainers will also be manufactured at K-State TDI out of 6061 aluminum. The four retainers will be fastened to the internal tabs and rods will be inserted through the individual boards and retainers. The Payload E-Stack will be normal to the Z faces while the Avionics E-Stack will be normal to the Y faces and secure to the Y- face.

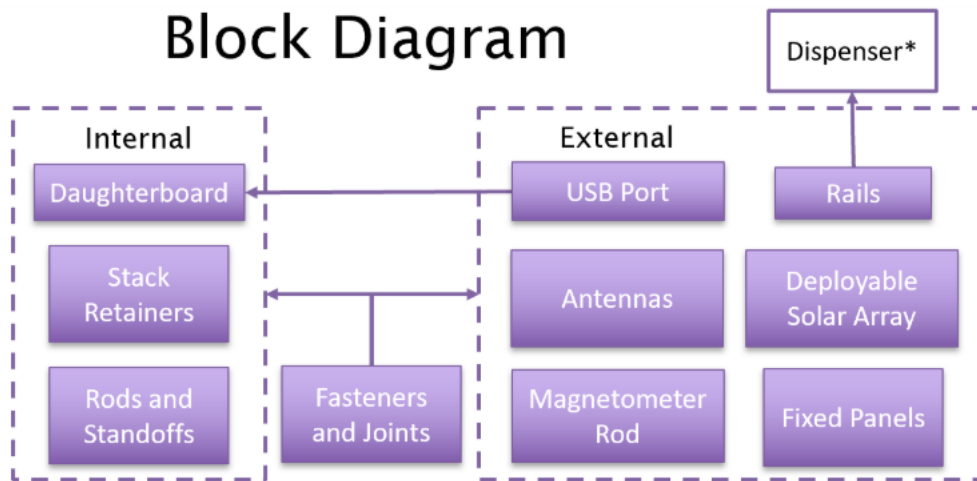
On the external faces, there will be deployable solar cells, the panels themselves will be a custom build from K-State TDI to attach the cells to, from the X- and X+ faces, as well as a deployable magnetometer from the Y- face and likely a deployable UHF antenna from the Z- face. All 6 faces will have 1.5mm thick 6061 aluminum for radiation shielding and thermal conduction. There will also be fixed components secured by bolts externally such

as solar panels on the Z+ face, solar sensors on all 6 faces, and a GNSS antenna on the Y+ face.

The fasteners and bolts that have been selected will include M1-M3 bolts of 18-8 stainless steel as well as 4-40 rods and standoffs for the stacks.



Block Diagram



*Dispenser provided by launch provider

Launch Interface

NuKAT will use the SpaceX Falcon 9 rocket as a launch vehicle. Launch environment concerns are described earlier. The interface shall be a rail-type dispenser from the launch provider.

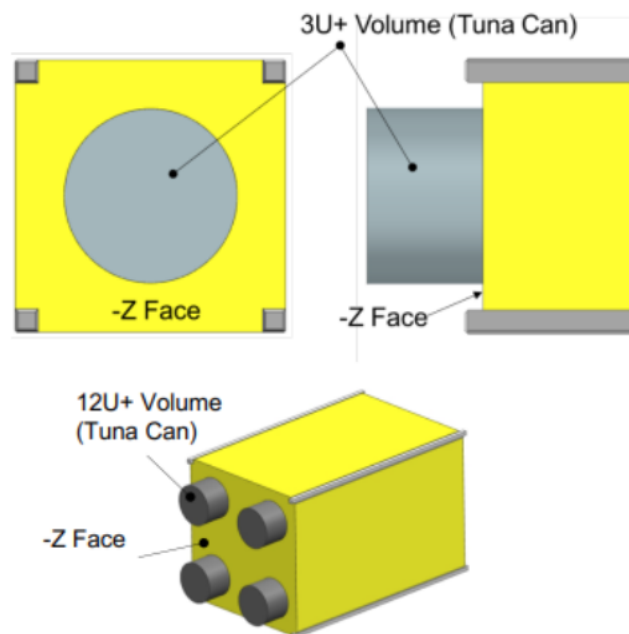


Figure 45: Diagram of the volume inside a standard cube-sat dispenser. Each of four 'tuna cans' represents a 3U rail section.

NuKAT is a 2U mission, which means that it will fit in one of the four 3U sections to this typical dispenser. If it is at the front of the dispenser, NuKAT will be allowed to use the 'tuna can' volume. This is offered on request, but the team currently plans not to use it.

The dispenser will be included in the launch provider's quote. This makes vehicle integration easier, because the provider company pushes their requirements onto our team, and knows the dispenser our system is being put into. This same company will complete the pre-loaded and loaded vibration testing required by the launch vehicle company. Currently, the favored launch provider is Arrow Science and Technology.

PHAT-Sat

The PHAT-Sat spacecraft bus provides the core spacecraft functions required to support hyperspectral Earth observation operations throughout the mission lifetime. While the hyperspectral payload defines the scientific objectives of PHAT-Sat, the bus is responsible for maintaining spacecraft health, supplying and distributing electrical power, managing command and data handling, providing attitude determination and control, regulating the thermal environment, and ensuring structural integrity from launch through decommissioning. The bus architecture is defined to enable reliable payload operation within the mass, volume, and power constraints of the CubeSat form factor.

PHAT-Sat's mission imposes system-level demands on the bus due to the pointing stability, data handling, and power requirements associated with hyperspectral imaging. The bus must support stable nadir pointing during imaging opportunities, maintain accurate time and attitude knowledge, and provide sufficient onboard data handling and storage capacity to accommodate payload data prior to downlink. Power generation and energy storage functions must support intermittent high-power payload operation while maintaining power-positive behavior across the orbit. These requirements define the functional role of the bus without prescribing a specific hardware implementation.

The spacecraft bus coordinates command, telemetry, and science data management to enable reliable end-to-end mission operations. It must support scheduled imaging activities, autonomous transitions between operational modes, and fault detection and recovery to protect both the spacecraft and payload during off-nominal conditions. Communications support is required to maintain spacecraft commanding and health monitoring while enabling periodic downlink of science data products.

Structurally and thermally, the bus provides a stable platform for payload integration, launch survivability, and on-orbit operations. Thermal regulation must accommodate variations in orbital environment and payload duty cycle while maintaining acceptable operating temperatures for onboard components. Collectively, these bus functions provide the foundational infrastructure that enables PHAT-Sat to successfully collect and deliver hyperspectral data in support of precision agriculture objectives.

No further overall satellite bus design was conducted this semester. The following images are from the May 2025 PHAT-Sat Feasibility Review.

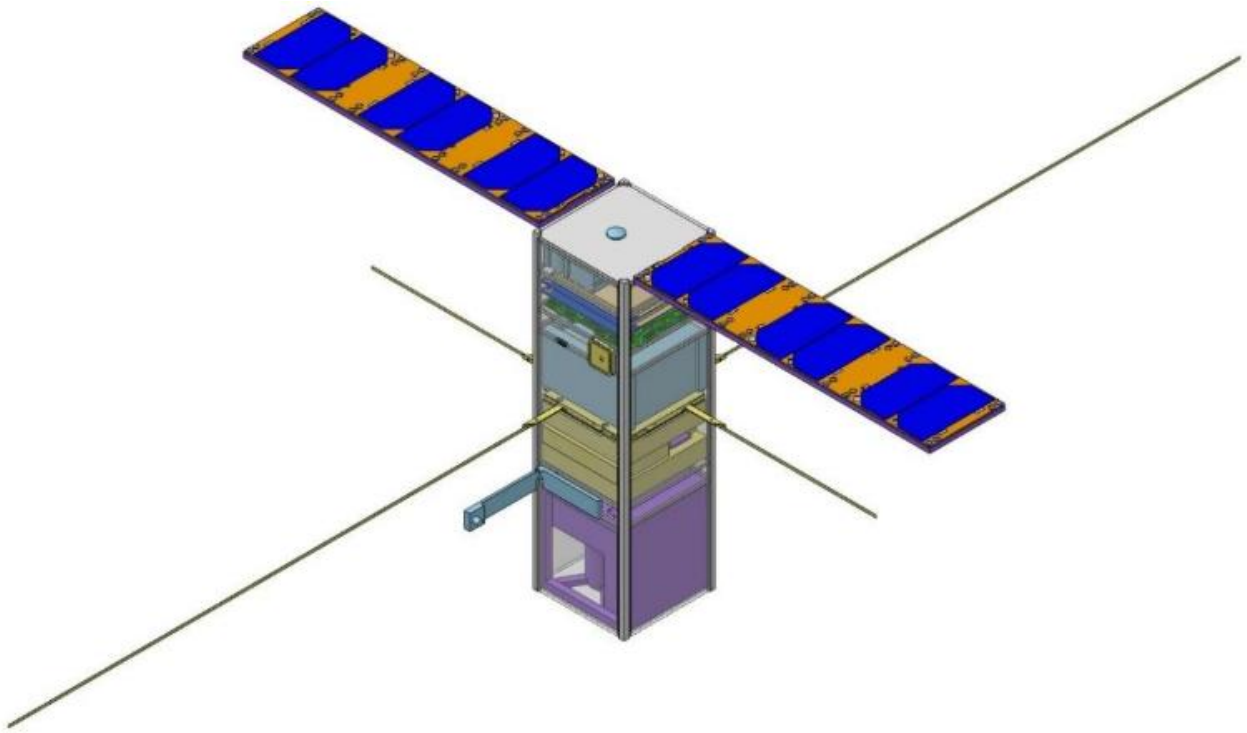


Figure 25: PHAT-Sat Isometric View (Deployed, Transparent Paneling)



Figure 26: PHAT-Sat Isometric View (Stowed, Transparent Paneling)

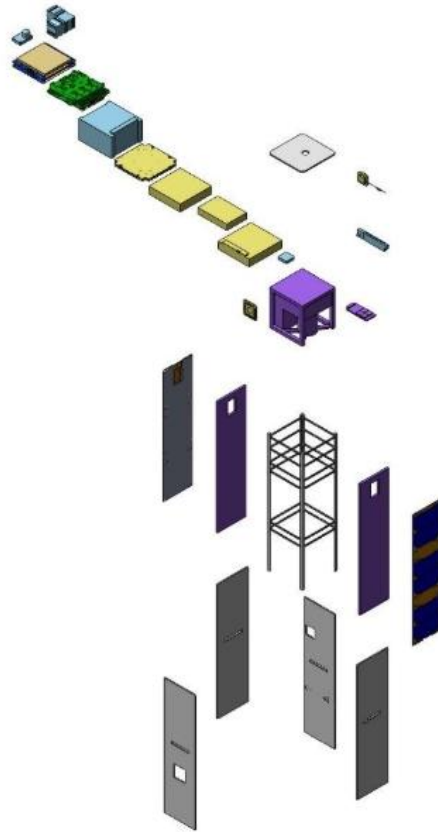


Figure 27: PHAT-Sat Component Exploded Isometric View

Attitude Determination and Control

The ADC requirements of PHAT-Sat are 1) to allow ground target identification and tracking so our payload can collect scientific data, 2) to allow for Sun pointing so our satellite can charge more effectively under normal and upset conditions, 3) to allow for Sun avoidance on certain faces so our sensors will not be damaged, 4) to allow for ground station pointing so our satellite can complete uplink and downlink of commands and scientific data, 5) to recover from upsets during deployment and mission operations, and 6) to allow for de-orbit less than 5 years after end-of-mission by pointing different surface areas at the incoming atmosphere (controlling drag). To do this, the subsystem shall include:

- Hardware: magnetometer, fine and coarse sun sensors, magnetic torque rods, reaction wheels or a control moment gyroscope, a rate gyroscope, possibly a star tracker, possibly a subsystem computer, integration with a GPS/GNSS receiver
- Software: Ground area spatial and angular awareness, Sun awareness, angular velocity estimation, Ground area scanning mode, Sun pointing mode, Ground Station pointing mode, Detumble safe pointing mode, Temperature regulation (rotisserie) safe pointing mode, calibration mode

- Interfaces: CDH (position & time IN; telemetry OUT), FSW (mode, commands IN; health and status OUT)

ADC must also be mechanically, electrically, and thermally integrated with the bus.

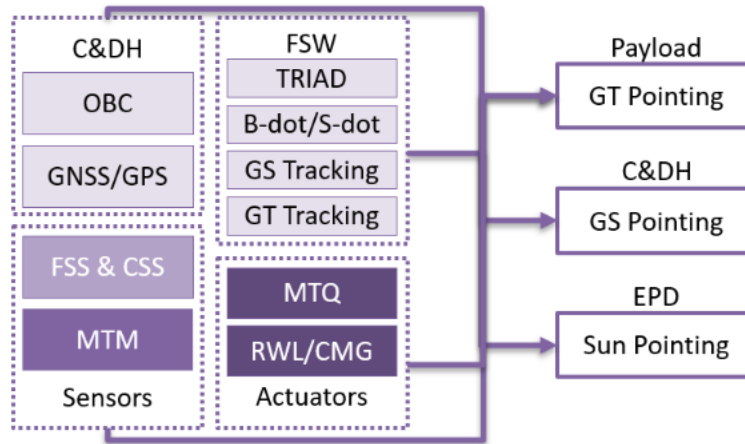


Figure 30: ADC Subsystem Block Diagram for PHAT-Sat.

In addition to all of the requirements of NuKAT, PHAT-Sat also requires a special ground target tracking algorithm and high-precision actuators. It is possible that the system will require a rate gyroscope, a control moment gyroscope, a star tracker, and/or an earth horizon sensor. This depends on the imager resolution calculations, which must be immediately verified next semester.

Subsystem	Mode	Component	Target	Axis to Align and Frame of Axis	Accuracy (+,-)	Precision (+,-)	FOV (+,-)	Function
PLD	Science (GT Pointing)	Camera	North America, Earth	(+) Z Axis of Spacecraft to be aligned with Nadir of Earth	UNKNOWN	UNKNOWN	UNKNOWN	Point for science mode, align to collect minimum [x boundary condition]
CDH	Up/Downlink (GS Pointing)	Antenna	Ground Station	(+,-) X or Y Axis of Spacecraft to be aligned with Nadir of Earth, which one based on structure	UNKNOWN	UNKNOWN	UNKNOWN	Point for up/downlink, align to send minimum X% of packets
	Science, Calibration (Sun + GT)	GPS/GNSS Receiver	GPS Satellites	(+,-) X or Y Axis of Spacecraft to be aligned with Nadir of Earth, which one based on structure	5 deg	5 deg	30 deg	Position Determination
ADC	Science, Up/Downlink, Calibration	Magnetometer	Earth Normal	No alignment required, will calibrate on orbit to find relative direction of Earth Normal vector	5 deg	5 deg	360 deg	Attitude Determination
	ALL	Inertial Measurement Unit (ACC, GYRO)	N/A	No alignment required, will calibrate on orbit to find angular velocity	UNKNOWN	UNKNOWN	N/A	Velocity Determination
	Science	Star Tracker	Space (cold side)	Aligned with cold space during operation	UNKNOWN	UNKNOWN	UNKNOWN	Attitude Determination
	Safe Mode (Sun Pointing)	Coarse Sun Sensor	Sun	Aligned with Sun vector during operation	10 deg	10 deg	45 deg	Attitude Determination, Sun Awareness
	ALL	Fine Sun Sensor	Sun	Aligned with Sun vector during operation	10 deg	10 deg	45 deg	Attitude Determination
	ALL	Reaction Wheel	N/A	Aligned internally, all components orthogonal	science mode	science mode	N/A	Attitude Control
EPD	ALL	Magnetorquer	N/A	Aligned internally, all components orthogonal	5 deg	5 deg	N/A	Attitude Control
	Safe Mode	Solar Panels	Sun	(-) Z Axis of Spacecraft to be aligned with Sun	45 deg	45 deg	20 deg	Sun tracking to charge

Figure 31: ADC Pointing Budget for PHAT-Sat.

The preliminary pointing budget includes requirements for three pointing modes, as well as calibration and safe mode. Primary mission requirements for sensors and actuators are not defined, because imager calculations are not complete. Ground station mode is not fully defined, because the antenna package (and thus FOV) is not known.

It is imperative that the next team make an informed decision on component and provider selection next year, in order to close all integration requirements with all other sub-teams. This will ensure parts are ordered with the appropriate lead time.

Communications and Data Handling

The PHAT-Sat CDH team efforts this semester included mission definition and review of work done in the prior semester, revisiting link capabilities of UHF, S-Band, and X-Band through detailed data calculations, and updating requirements to reflect changes and new research. X-Band for payload downlink has been identified as the lowest frequency capable of handling payload downlink, with UHF handling ground station uplink and telemetry downlink operations. This communications system will need to add GNSS capabilities for precise satellite tracking and pointing. Link budgets will need to be generated. Additionally, due to the large data size of images, the data handling system will need to be capable of processing and storing large data quantities with redundancy. For this, the NextAge NMC2 processor is desirable, though trade studies are required to move forward with OBC selection.

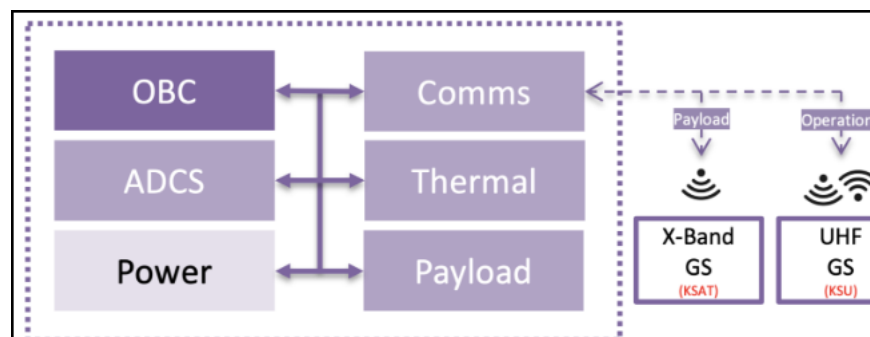


Figure 35: Block Diagram for PHAT-Sat CDH Subsystem

PHAT-Sat Payload Data Calculations

The data calculations for PHAT-Sat have used 50% margins as a conservative estimate and are as follows:

$(18600 \text{ Mb image} / (6:1 \text{ compression} * 0.5) / (100 \text{ Mbps} * 0.5) / (60 \text{ seconds})) \rightarrow 2.1 \text{ min}$
downlink time per image.

Simera Hyperscape100 with 4.75 km swath $\rightarrow$ 837 images to cover region of interest $\rightarrow$
Estimated 850 images taken due to inconsistencies in orbit from launch provider.

$(850 \text{ images} * 2 \text{ for redundancy}) * 2300 \text{ MB} \rightarrow 3.9 \text{ TB}$ of storage required.

Given 6-12 min. of downlink per pass, 3-5 images are estimated to downlink each pass.

Estimated 170-284 passes to downlink 850 images to 1 ground station.

Electrical Power & Distribution

The PHAT-Sat flight software will remain largely similar to NuKat with a few changes due to a difference in payload data. The payload data will include images taken by the hyperspectral camera, as well as the altitude, location, and date and time that an image was taken at. We will follow a similar FSW structure as NuKat utilizing C, C++, Python scripts, TriCortex, and the cFS framework. In addition, due to the large quantity of raw image data we will need to downlink, we will implement an image compression algorithm to downlink this data more efficiently. While we have not selected a specific algorithm, some promising ones include Embedded Zerotree Wavelet (EZW) or JPEG2000. The current plan is to aim for visually lossless compression. This means that pre-processing and post-processing differences are imperceptible to humans. Lastly, our software will follow multiple sets of industry standards to ensure that our code is safe and reliable.

Thermal

The thermal system for PHAT-Sat has not made much development outside of general ideas. Given the larger size, different flight orientation, and larger deployable area, PHAT-Sat presents a similar yet unique set of challenges to solve. The fundamentals used for NuKAT will apply to PHAT-Sat as well, but specific considerations will need to be made to account for the vast differences between the missions.

Structure

There was not much progress on the development of the structures for PHAT-Sat but we did present more requirements for PHAT-Sat structure based on information gained from NuKat. These requirements were presented for PHAT-Sat System Requirement Review. The PHAT-Sat structure should be simpler in some senses, as we are flying a more standard orientation and dimension of CubeSat, thus most of our components will be standard and easy to purchase off the shelf such as 3U deployable solar panels, a 1U UHF that means our needs, and a vertically oriented (Z is long) 3U chassis itself with racks that are all normal to the Z axis.

Launch Interface

PHAT-Sat will use the same launch vehicle, launch dispenser, and launch provider as NuKAT. PHAT-Sat is a 3U mission, which means that it will use one full rail slot (including 'tuna can').

Management Overview

Project management

This semester, the project management work primarily focused on three aspects: branding, outreach, and the progression towards creation of the Wildcat Space Program.

For branding, the main work was the design and launch of a new logo for the NuKat mission. Six logo options were created and were proposed to the class. Everyone had a chance to vote on their favorites as well as provide feedback to improve the logo design. This process produced a finalized logo for our NuKat mission which was revealed during the NuKat Pre-Preliminary Design Review.

For outreach, our team had the privilege of participating in two events during the semester. Celebrate Ag Day is hosted each year by Kansas State University and this year, our satellite design team presented a booth in “Ag Ville” in Bramlage Coliseum. Our PHAT-Sat mission aligned closely with the agriculture focus of the event, and it was a great opportunity to share our mission with the community, potential stakeholders, and potential donors. The second event we participated in was the K-State Research and Extension Conference where we held a breakout session for attendees to come and learn about our PHAT-Sat mission. Our leadership team presented our mission, primarily focusing on background, science objectives, and the value we can bring to the agriculture industry in Kansas. We also we able to get great feedback and ideas from experts in the industry.

Finally, Wildcat Space Program was an idea dreamt up by our leadership team during the semester. While not fully established at this point, progress was made towards student organization approval and creation. Purpose and objectives were established for the organization including branding, marketing, promoting, fundraising, customer-facing, traveling in support of space-related objectives for the university. A constitution was created and submitted for approval. This organizational status will also provide our program the ability to get support on branding/marketing as well as access to fundraising avenues like Crowdfunding and Day of Giving. Next semester, Wildcat Space Program will be fully formed and operating.

NuKAT

Risk Management

For NuKAT, risks were identified based on the system requirements. Each requirement correlated to one risk. From those risks, they were ranked both on their likelihood to occur and the consequences that would occur if said risk were to occur. These rankings were

both on a scale from one to five, with five representing the greatest likelihood or risk for their respective categories.

After this, the likelihood and consequences were multiplied to create an overall risk score. These risk scores were calculated and put into three categories of low, medium, and high risks. In total, over 275 risks were identified, with 40% of those as low risks, 50% of those as medium risks, and 9% categorized as high risks. Potential risks were also marked as mitigate if action was planned to reduce the risk, accept if the current risk level were to be accepted, or unknown if the choice was to watch the risk for future updates. The full layout of risk results is shown below in Table 2

Table 5: NuKAT 5x5 Matrix Number Breakdown of RVM

Likelihood	5			1	1	
	4			3	5	8
	3	2		12	12	10
	2		9	26	31	28
	1	13	21	35	34	28
		1	2	3	4	5
		Consequence				

For the risks, priority to documentation and potential mitigation strategies was given to the highest level red risks. Of these high level risks, two were on the mission level, three were on the system level, and the rest were in the subsystem or below level risks.

Competitive Assessment

NuKAT already has a primary corporate mission (to space-qualify a radiation detector), so the competitive assessment is not necessary.

Mission Definition Review Feedback

On October 15, 2025, the team presented a Mission Definition Review for the NuKat mission. Six reviewers provided feedback for our team including Dr. Raju Dandu, Dr. Amir Bahadori, Professor Paul Snider, Glen Fountain, Dave Copeland, and Chris Hersman.

Dr. Dandu completed the team on a great presentation. He recommended that moving forward, we spell out the subsystem names in the slides instead of using acronyms.

Dr. Bahadori recommended looking further into the mitigation of the red risk, “Radiation Environment”, based on the suggested mitigation being more of a characterization of the risk. He noted that looking into alternative orbits could be a feasible way to mitigate radiation environment issues.

Professor Snider was complementary of the time management and Q&A section of the presentation. He recommended hemi coverage with Earth pointing, or omni coverage without Earth pointing for the antenna field of view. He also wondered about the feasibility of data transmission 100% of the time for world-wide amateur radio contact. Finally, he recommended further research into interaction between the flight computer and the EPD and ADC systems, as well as what oversees the operational modes.

Glen Fountain wanted to see the details built out like budgets and SWAP, while recommending a 25% margin for these details. Glen provided in-depth feedback which can be found in the Appendix section under “Glen Fountain NuKat MDR Comments.”

Dave Copeland was complementary of the team and provided feedback primarily focused on the comms system, while also agreeing our 100k budget is feasible. He also questioned the need for ADC system, but understood the necessity of the pathfinder nature of the mission for PHAT-Sat. He suggested a low-gain antenna, and looking into guidelines from the International Amateur Radio Union for frequency for amateur radio community. Finally, he recommended prioritizing a five-year decay, as well as developing minimum mission success criteria.

Chris Hersman was looking for another level of detail, and more info payload needs like how much power and data.

NuKAT Preliminary Design Review Feedback and Action Items

At the time of writing this report, the team is still awaiting formal feedback from most of our review panel. However, one reviewer, Zach Metzinger, provided feedback regarding our radio communication. He directed our team towards 47 CFR 97.113(b) for regulations regarding broadcasting our data and also provided information for Part 5 licensing. We look forward to gathering this formal feedback before next semester.

In addition the following are comments from reviewers:

"Great job on the PDR!" - Zach Metzinger

"You all really knocked it out of the park (and right into orbit). Great job and have a wonderful holiday break." -Jeff Sheldon

"It was a very thorough and impressive presentation" -Taylor Ochs

"I want to share with you how impressed I am with the quality of your presentation and your growth as students this semester. Well done!" - Paul Snider

"Well organized, managed, and presented." - Dr. Raju Dandu

"This group has come far in its collaborative efforts. Very proud of the work you've all invested, the challenges you've crossed, and the barriers you've identified." - Jeff Sheldon

We also received the following from feedback from Chris Hersman during our dry run prior to the final pPDR presentation.

The duration of the PDR is insufficient for a comprehensive review, with dialog and interaction with the review panel. If there is a possibility of having more time for the review, it would be beneficial.

Add slide numbers to allow for referencing comments.

Compare PDR Entrance & Success Criteria to NASA 7123.1 (Just for ideas, in case there is anything that you determine is worth adding to your criteria.)

Maps of science data were mentioned. Who make these maps? What do they look like?

Reverse the order of the stakeholders to put the primary on top, tertiary on the bottom.

Show component callouts on Bus overview.

Show orientations (Nadir, Ram) on all pictures of the Bus.

On Day-In-The-Life chart it shows times when there is overlap between SAA and K-state. I don't think that is possible.

What is EMI Safe? There 2 different blocks labelled “Safe.” Can you explain the difference between these two safe states? (one should be Survival)

Interfaces are missing from block diagram. (was mentioned that they would be added)

If the RF requires pointing to the ground station, does that mean there’s no science during downlink contacts? (determine if the link closes without pointing to ground station)

What are the headings for the 3 columns showing Red/Yellow/Green with numbers from 1 to 5? Use consistent table format for all sections.

Is there a “Software” Subsystem in the ADC? What tools are to be used? What processors are in the system?

Power system shows a microcontroller, but no mention of microcontroller flight software.

Does ADC use the CDH computer?

Is Sun Synchronous really necessary? This may be more restrictive than necessary.

I don’t understand some of the notes indicating non-compliance.

Launch environments are specified in g’s. Are these g’s RMS?

Is there an instrument calibration activity?

Check power budget to confirm payload

Solar array sizing? Degradation over mission life?

Solar array power calculations incorporating solar angle as a function of orbit?

C&DH Quantify ranges for low/med/high, or add numbers.

Communications and Data Handling

Subsystem	Component	Throughput Estimation	CPU Utilization	Non-volatile Memory	Volatile Memory	Update Frequency	Timing Accuracy
Payload	DSC Domino	High	Medium	High	Medium	Medium/High	Low
Structures and Mechanisms	Deployment Mechanisms	--	Low	--	--	--	Strict
Thermal Control	Temperature Sensors	Medium	Low	--	--	--	Low
	Radiators	--	--	--	--	--	Low
Power	PCDU	Medium	Low/Medium	Medium	Low	High	Medium
Telemetry and Control	UHF Transmitter	High	Medium	High	--	--	--
	UHF Receiver	Medium	Low	Low	--	--	--
	GNSS Receiver	Medium	Low	Low	--	--	--
Command and Data Handling	On-Board Computer	High	High	High	High	High	Strict
Attitude Determination and Control	Sun Sensors	Medium	High	Low	--	--	--
	ADC Board	Low	Low	Low	Medium	Medium	Strict
	Magnetometer	Low	Low	Low	Low	Low	Low

Complete list of software subsystem requirements other than CDH were missing

How much does the NASA core Flight System cost?

Structures: Mass budget seem to be inconsistent 1417.47 g vs. 2.x kg?

Thermal indicated -40C to 85C was the requirement, but initial presentations showed -20 to 50C.
(inconsistent?)

How much storage is needed in the ground station (3 years of data)?

Ground computer interface to control the rotator is not shown.

Plans for tests with the ground station and the satellite before launch?

Will you have a copy of the satellite computer during development of the ground station software?

Ground system testing not shown in the AI&T flow. What GSE is needed?

Is there an environmental test over temperature planned?

Is there a vibration test planned?

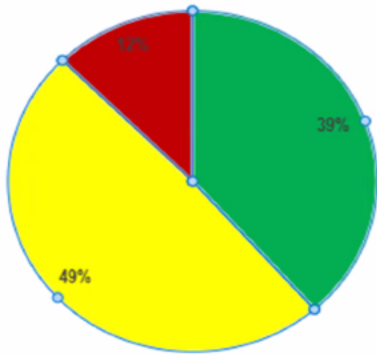
If they are outsourced, have they been costed?

I didn't see any computers in the ground support equipment, such as necessary to send command and receive telemetry.

32 red risks seems like a lot, but the details of all these red risks was not provided, so there's no way to evaluate these risks.

Risk Assessment Strategy

Risk Ratings



■ Green ■ Yellow ■ Red

	Frequency	Percent
Green	99	38.52%
Yellow	126	49.03%
Red	32	12.45%
Total	257	100.00%

PDR Board Feedback

- What mission and design definition content is missing?
- What mission and design definition needs more development?
- What mission and design definition trade studies are needed?
- Feedback on path to compliance/noncompliances

NuKAT Proposal Feedback and Action Items

section	scoring criteria	person(s) responsible	targeted completion date	AT status (score)	PS status (score)	CL status (score)	notes
General				3	3	0	
	Follows Formatting Guidelines			3	3	0	
	Includes all Required Sections			3	3	0	
	Meets Submission Deadline			3	3	0	
Technical Competence				3	3	0	
	Submission details the applicant's plan, resources, and know-how to deliver their payload within a 12- to 24-month timeline			3	2	0	
	Submission details applicant's ability to navigate the appropriate licensing work			3	3	0	
	Meets Technical Requirements			4	4	0	
	Complies with Do No Harm Assessment			3	4	0	
	Submission details payload's scientific objectives in space			3	4	0	
Benefits				3	1	0	
	Technology that Firefly May Benefit From				0	0	Does this apply? If so, it needs to be stated as with the words similar to "this technology will translate back to Firefly..."
	Technology that Others May Benefit From			4	4	0	
	Charitable Organization			2	0	0	See note in document
	Plans for MentorshipTechnical				0	0	I'm not sure I can evaluate this one.
TOTAL				9	7	0	

Table xx: Red Team Review Feedback and Action Items

Our team submitted a rough draft of our NuKat proposal to our red team reviewers. They used the criteria given by FireFly for their Dream program and graded us on a nine-point scale. Table xx shows that we were given seven out of nine points, with the two missing points coming from our submission details and their charitable donation requirements.

One focus of the comments was on word choice. The reviewers gave many suggestions for better phrasing and word choice, pointing out areas where they were unsure of the meaning of sentences and how we might reword them. Much of this helped us reduce the size of our proposal, as the first draft was over the page limit. Furthermore, the rephrasing created a more consistent tonal flow to our proposal.

Finally, the reviewers focused on helping us improve the use of visuals in our proposal. They suggested switching our organization's visual as well as our budget visual. The previous organization graphic focused on subsystems and team breakdown, while the new one is broken down by campus and major. Our new budget graphic uses a bar chart instead of a pie chart and breaks down the costs into four categories instead of every subsystem.

Verification and Validation Planning

Planning verification methods for each aspect of the mission is, of course, paramount to the success of a CubeSat. Contained within the NuKat RVM are 298 specifically defined

mission requirements that require a simple and direct method for confirming or denying their compliance status. Our method of accomplishing this was simple and effective.

To classify each RVM requirement, we chose to arrange them all into 4 separate categories: Analyze, Inspect, Test, and Demonstrate (AITD). The analysis class refers to requirements that may need to be evaluated in a simulated environment. This could include FEA in SolidWorks Simulation or other software, analysis using MATLAB, various electrical circuit simulation software, and the like. Inspection refers to taking a direct measurement from the CubeSat, be it physical or electrical. This includes overall dimensions, weights, fit checks, power requirements, checks for shorts, surface hardness, and other relevant factors. The test category requires that conditions be altered to match the satellite's operating conditions as closely as possible to test hardware, or a simulated test be conducted on the satellite. This includes various flight software tests that confirm its ability to perform the desired tasks, as well as many others. Demonstrate, arguably the simplest verification type, is a simple pass-or-fail condition. Does the Integration and Testing facility provide the necessary power to operate all testing hardware? Is the neutron detector mounted as specified? Etc.

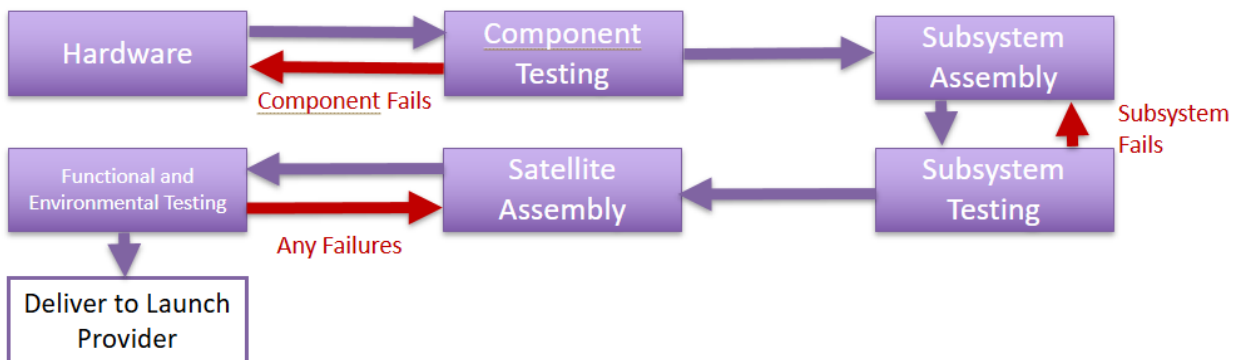
It is also worth noting that each requirement is not limited to only one verification type. For example, when selecting an LNA for the ground station transceiver, an appropriate classification is A&T. Analysis can be conducted on the expected receive signal strength to determine the required amplification, and a physical test can then be performed to verify that the received signal is boosted to acceptable levels.

Integration and Test Planning

The NuKat integration and test plan is designed to verify that the CubeSat meets all requirements necessary to ensure mission success. The plan covers system level behavior, electrical and mechanical requirements, software performance, and environmental tolerances. The end goal is to have an operational satellite that is validated and prepared to be turned over to the launch provider.

The plan for NuKat follows a bottom up approach, beginning at the component level, moving through subsystem integration, then ending with full satellite testing. Each stage includes testing to verify function and detect isolated issues early in the integration process. This serves to increase efficiency with troubleshooting prior to advancing to

higher levels of assembly.



This flow diagram better illustrates the approach. The process begins once hardware has been selected and received. Individual components are first verified to detect faulty hardware prior to integration. When hardware passes verification, they are assembled into subsystems. Electrical, mechanical, and functional testing is carried out on subsystems to ensure correct operation. After subsystems pass verification, the satellite is assembled. Functional and environmental testing can be carried out at this stage. Hardware is used to interface with the flight software to simulate space environment inputs and responses are evaluated to ensure readiness. If at any point in the process errors or unexpected outcomes should occur, troubleshooting procedures are carried out to identify and resolve the issue. This approach ensures that testing errors can be corrected early, avoiding issues compounding into problems later in the integration process.

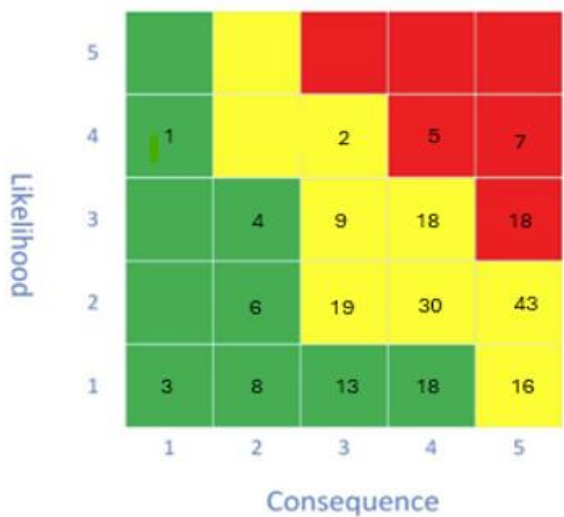
Most testing procedures can be performed at the dedicated campus AIT facility, currently designated at Seaton 0004. This includes all electrical verification processes as well as some environmental testing such as sunlight simulation, RF communication verification, and ADCS control verification. Certain environmental testing procedures will be too technically intensive to self perform, these processes will be outsourced to third party testing services to ensure reliable results. Processes currently include vibration and shock testing needed to simulate the launch environment, and thermal vacuum testing to replication the orbit environment.

PHAT-Sat

Risk Management

For Phat-Sat, the same methods were used to identify and rank the potential risks based on their likelihood and potential consequences. Over 230 risks were identified for Phat-Sat, with 21% of those as low level risks, 61% as medium level risks, and 13% of those categorized as high level risks. The full layout is shown below in Table 3

Table 6: Phat-Sat 5x5 Matrix Number Breakdown of RVM



The creation of potential mitigation strategies was also focused on the high level risks in Phat-Sat as well, with a good majority of the highest rated risks having potential mitigation methods theorized. Of the risks, two high level red risks were at the system level, four were at the system level, and the rest were at the subsystem level or below.

Competitive Assessment

To compare PHAT-Sat against other realistic ways of collecting land and crop data, we evaluated several commonly used options. These included drones, commercial satellite imagery, crewed aircraft services, balloon platforms, and people walking fields. The goal was to see how PHAT-Sat performs relative to existing solutions and whether it offers a real advantage for large-scale monitoring.

We scored each option using the same set of criteria: cost, revisit time, time to scan one acre, capabilities, max lifetime, key issues, and key advantages. These criteria were weighted based on how important they are for this mission. Speed and coverage were weighted the highest since PHAT-Sat's main strength is scanning very large areas quickly. Capabilities and cost were also emphasized because they affect how useful the data is and whether the system is practical. Lifetime and limitations were considered but given less weight.

Each option was rated on a 1 to 5 scale using vendor specifications, public information, and reasonable assumptions about real-world use. The scores were combined with the weights to get a final score for each option. The full scoring and weighting comparison is shown directly below in Table 5.

Table 7: Phat-Sat Competitive Analysis

Criteria	Weight	Phat-Sat		Drone (DJI Matrice 350RTK)		Other Satellite (PlanetScope)		People		Balloon Option		Aircraft	
		Score	Score with weight	Score	Score with weight	Score	Score with weight	Score	Score with weight	Score	Score with weight	Score	Score with weight
Cost	15%	3	0.45	4	0.6	1	0.15	5	0.75	4	0.6	3	0.45
Revisit Time	10%	2	0.2	2	0.2	5	0.5	1	0.1	1	0.1	2	0.2
Time to scan 1 Arce (Speed)	30%	5	1.5	3	0.9	5	1.5	1	0.3	4	1.2	3	0.9
Capabilities	20%	5	1	5	1	3	0.6	4	0.8	3	0.6	3	0.6
Max Lifetime	10%	4	0.4	4	0.4	4	0.4	5	0.5	2	0.2	5	0.5
Key Issues	10%	3	0.3	4	0.4	4	0.4	3	0.3	2	0.2	2	0.2
Key Advantages	5%	5	0.25	5	0.25	4	0.2	3	0.15	1	0.05	5	0.25
Total Score			4.1		3.75		3.75		2.9		2.95		3.1

PHAT-Sat received the highest overall score. This is mainly due to its ability to cover thousands of acres almost instantly while providing more spectral information than most alternatives. Other options perform well in specific areas, but they become slower, more expensive, or harder to scale as coverage increases. PHAT-Sat fills the gap between high-detail local sensing and expensive large-scale commercial systems, making it the best overall fit for the mission goals.

Mission Definition Review Feedback

On November 12, 2025, the team presented a Mission Definition Review for the PHAT-Sat mission. Four reviewers provided feedback for our team including Professor Paul Snider, Chris Hersman, Glen Fountain, and Dr. Tim Steffensmeier.

Professor Snider shared compliments about the presentation, especially noting the Q&A section, and had a few recommendations about timing in the presentation. He also noted that the orbit minimum, maximum, and average needs to be more clearly communicated, and acknowledged the technical challenges of an imaging mission like technical aspects and cost.

Glen Fountain acknowledged that our team covered a large amount in the presentation and recommended that placeholders be placed in the budget for unknown costs. He also had a few questions about instrument sensitivity, signal to noise sensitivity, and time to dwell on a point without smear. He also suggested a gyro for pointing accuracy.

Chris Hersman pointed out that the mission will require a high FOV for downlink, especially for off-NADIR imaging. He suggested that it would help to have a little more concept for a timeline in an orbit and questioned how many passes would be needed to image the entire state, as well as downlink that data. Chris also expressed concern surrounding the estimated revisit time.

Finally, Dr. Steffensmeier focused feedback on our agriculture focus as well as end-user access to collected imagery. Tim asked what our targeted crop was for imaging and recommending looking into a potential partnership with All Things Kansas as a potential data delivery method.

Verification and Validation Planning

The process for assessing the PHAT-Sat RVM was the same as Nukat, so there is nothing new of note in that regard. However, PHAT-Sat has only 225 requirements as opposed to the 298 present in the NuKat RVM.

Integration and Test Planning

The PHAT-Sat integration and test plan follows largely the same methodology as the NuKat integration and test plan. The components and subsystems will be validated through the same bottom-up approach, detailed in the flow diagram above.

PHAT-Sat has stricter requirements for the assembly, integration, and testing of many of its components: in particular, the precision hyperspectral imager is expected to have dust/cleanliness requirements. These specifics will be better solidified as level 4 (component) and 5 (part) requirements are completed for the PHAT-Sat Requirements Verification Matrix. The AIT facility will be flexible and designate controlled areas such that these specifications are met.

Acronyms

ADC(S) – Attitude and Direction Control (Subsystem)
AHB – Advanced High-Performance Bus
AITD – Analyze, Inspect, Test, & Demonstrate
BOL – Beginning-of-Life
bps – bit(s) per second
C&DH – Communications and Data Handling
cFS – core Flight System
DSC – Dry Storage Cannister
EOL – End-of-Life
EZW – Embedded Zerotree Wavelet
FCC – Federal Communications Commission
FEA – Finite Element Analysis
FOV – Field of View
GB – Gigabytes
GCR – Galactic Cosmic Ray
GPS – Global Positioning System
GNSS – Global Navigation Satellite System
KB – Kilobyte(s)
LEO – Low Earth Orbit
LNA – Low Noise Amplifier
Mb – Megabit(s)
Mbps – Megabit(s) per second
MB – Megabyte(s)
MDR – Mission Definition Review
Mrg - Margin
MSND – Microstructured Semiconductor Neutron Detector
NASA – National Aeronautics and Space Administration
OBC – Onboard Computer
PDR – Preliminary Design Review
PI – Principal Investigator
pPDR – pre-PDR
PMT – Photomultiplier Tube
PV - Photovoltaic
RDT – Radiation Detection Technologies

RVM – Requirements Verification Matrix
SAA – South Atlantic Anomaly
SOC – State-of-Charge
SSD – Solid-State Drive
STK – Systems Tool Kit
STM – Science Traceability Matrix
TB – Terabyte(s)
TBD – To Be Determined
TCP – Transmission Control Protocol
TDI – Technology Development Institute
TRL – Technology Readiness Level
UDP – User Datagram Protocol
UHF – Ultra High Frequency

References

Johnstone, Alicia. *CubeSat Design Specification (1U – 12U) REV 14.1 CP-CDS-R14.1*, February 2022. https://www.nasa.gov/wp-content/uploads/2018/01/cubesatdesignspecificationrev14_12022-02-09.pdf

Appendices

Requirements Verification Matrix

Linked here are the RVMs for both the PHAT-Sat and NuKat Missions.

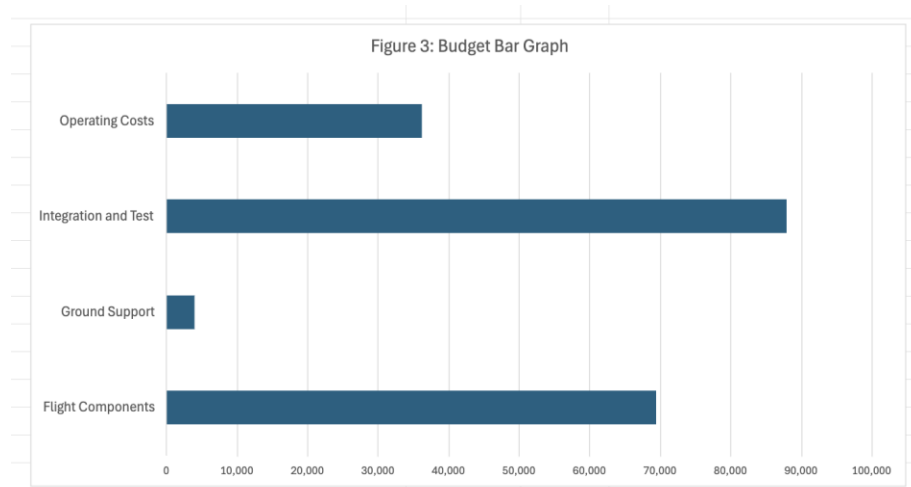
[NuKat RVM_Fall 2025 Final Report_12-17-25.xlsx](#)

[PHAT-Sat RVM_Fall 2025 Final Report_12-17-25.xlsx](#)

Master Equipment List spreadsheets

Budget spreadsheets

NuKAT Budget Bar Graph Representation



NuKAT Subsystem Budget Breakdown Spreadsheet

1			
	Subsystem	Budget	Contents
2	Payload	5,000	Core hardware, electronics, mounting, and prototype units required to build the instrument. Processing, software, calibration, and testing tools needed to verify performance. Ground data handling and operational support
3	EPDS	11,300	Power generation, storage, regulation, and distribution hardware to supply the spacecraft and payload. Control electronics, interfaces and protection systems ensuring stable, safe, and reliable power delivery.
4	ACDS	10,000	Sensors, actuators, and hardware for orientation control. Electronics and software for attitude estimation and stabilization. Calibration, testing and ground support for pointing accuracy.
5	Launch	76,000	Covers integration, processing and fees required to get the satellite approved for launch. Includes the launch service itself and associated safety and handling requirements. Ensures the spacecraft can be delivered to orbit reliably and on schedule
6			

Subsystem	Budget	Contents
C&DH	30,716	Hardware and processors that manage spacecraft commands and data flow. Software and interfaces that control subsystems and store mission data. Testing and ground support to ensure reliable communication and data management.
OBC	0	Central processor and electronics that control spacecraft functions. Software and interfaces for executing commands and managing subsystem data. Testing and support to ensure reliable, fault-tolerant operation in orbit.
Structure	4,000	Primary frame and panels that hold and protect all spacecraft components. Mechanical interfaces for mounting subsystems and maintaining rigidity. Testing and verification to ensure strength, alignment, and launch survivability.
Subsystem	Budget	Contents
H&C	1,500	Wiring and connectors that distribute power and data between subsystems. Mechanical routing and mounting to keep cables secure and organized. Testing to ensure reliable signal integrity and power delivery.
Thermal	1,000	Insulation, coatings and hardware that regulate spacecraft temperature. Sensors and control elements to monitor thermal conditions. Testing to ensure components stay within safe operating limits.
Flight Software	6,000	Code that controls spacecraft functions, autonomy and subsystem coordination. Interfaces for command execution, data handling, and fault management. Testing and simulation to ensure reliable performance in orbit.

Subsystem	Budget	Contents	Subsystem	Budget
Integration	6,000	Assembly and installation of all subsystems into the full spacecraft. Alignment, fit checks, and interface verification across components. Testing to confirm the spacecraft functions correctly as a unified system.	Spares	2,400
Testing	6,000	Functional and performance tests to verify subsystem operation. Environmental tests (vibration, thermal, vacuum) to ensure launch and space readiness. Tools, equipment and analysis needed to validate mission requirements.	Backup Fund	33,789
Ground Station	4,028	Antennas, radios, and equipment for communicating with the spacecraft. Software and interaces for commanding, data downlink and health monitoring. Setup, testing, and support to ensure reliable mission operations.	Total (Hardwarees)	105,705
			Grand Total	197,733

PHAT-Sat Budget Tracking Methodology

Total Cost PHAT-SAT									
Mission		\$0.00							
Satellite		\$0.00							
Launch		\$0.00							
Ground		\$0.00							
Total		\$0.00							
Subteam	Item	Company	QTY	Total Cost	Lead Time	Ordered	Receipt	Purchahsed By	Notes
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
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						FALSE	FALSE		
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						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		

PHAT-Sat Comms Budget – Finalized

Comms	Sub Category	Item	AH Guess L	AH Guess H	Item	Price
75,000	Radio Hardware	Transceiver	3,000	10,000	UHF Transceiver	7,900
	Radio Hardware	Power Amplifier	1,000	5,000	UHF Antenna	5,500
	Radio Hardware	Software Defined Radio	2,000	8,000	X-Band Transmitter	34,300
	Radio Hardware	Beacon Transmitter	500	1,500	X-Band Antenna	10,000
	Antenna System	Deployable Antennas	2,000	6,000	GNSS Receiver Antenna	15,000
	Antenna System	S-Band Patch Antenna	1,000	3,000		
	Antenna System	Antenna Deployment Mechanism	500	1,500		
	Antenna System	Antenna Coax & RF Harnessing	200	800		
	Supporting Electronics	RF Switch / Diplexer	200	1000		
	Supporting Electronics	Low-Noise Amplifier	500	1,500		
	Supporting Electronics	Grounding & EMI Filtering	100	400		
	Supporting Electronics	Power Conditioning Board	200	1,000		
	Onboard Data Handling	Interface				
		OBC Interface Board (if not integrated)	300	800		
	Onboard Data Handling	Interface				
		Software Integration / Telemetry Packets (low if work done in house)	500	200		
	Ground Segment Costs	Ground Station Access Fees	5,000	20,000		
	Licensing and Testing	FCC/ITU Frequency Licensing	1,000	5,000		
	Licensing and Testing	EMC/EMI Testing	3,000	5,000		
	Licensing and Testing	Thermal-Vac/Vibration Tests	5,000	15,000		
			LOW	HIGH	Real	
		GRAND TOTAL	26,000	85,700		72700

Schedule spreadsheets

Week by week Timeline for Spring Semester 2026 - NuKAT

Time Period	Task	In Charge
Dec 1-7	- Finalize CAD file - Draft SubSystem Breakdown - Created PDR slide deck/practice	- Sophia - All teams - All teams
Dec 8-19	- PDR week! - Red Team Report Draft - CAD file done?	All teams/Red team
Jan 20-27	- Preliminary Manufacturing Plan - Integration Drawings (Struct, PwrSys, Comms, Thermal, Payload) - Interface agreements - Update CAD file with agreements	- TBD - All teams (Bailey) - All subsystems - Anyone with new part information pertaining to CAD
Jan 27-31	Feasibility Analysis	- All systems - Power, thermal, PM

	2. Calculations for loads, thermal limits, power budgets 3. Update Risk Matrix with new CAD model info 4. CAD Refinement: internal layout, brackets and mounts	3. Risk Analysis Team 4. Structures, with the corresponding subsystems
Feb 2-7	1. Prelim integration plan done 2. Design Validation Planning 3. Component Selection list V1	1. Subsystems listed with integration drawings 2. Management 3. All subsystems ordering parts
Feb 10-14	1. Deep Dive into Structural Analysis 2. Power Modeling (Solar, batteries , power flow) 3. CAD updates from Analysis Results	1. Structural 2. Power Systems 3. Structural
Feb 24-28	1. Component Selection V2 2. Supplier Research + BOM Draft 3. Risk Matrix 4. Drafting Critical Design Report Sections	1. Misc Subsystems 2. Biz Ops 3. Risk Analysis Team 4. Management
Mar 3-7	1. Subsystem Design Review #2 2. Interface requirements update 3. CAD + simulation integration	1. All 2. All 3. Sophia x whoever does the simulation
Mar 10-14	1. Finalize subsystem designs 2. Prototype simulations 3. Check manufacturability and Assembly Feasibility	1. All systems 2. All systems 3. Management in collaboration with subsystems

Mar 17-21	Spring Break- catch up time	Whoever needs it
Mar 24-28	1. Full system integration pass 2. Update CAD for final assembly 3. Draft system test procedures	1. All 2. Sophia 3. Management

Mar 31-Apr 4	1. Begin CDR slide deck 2. Perform internal design review	1. All 2. All
Apr 7-11	1. Prep for CDR 2. Final performance checks	1. All 2. Mngmt
Apr 14-18	1. Final CAD release 2. Final risk matrix update 3. Simulated integration walkthru	1. Sophia 2. Risk team 3. All involved previously
Apr 21-25	1. CDR 2. Document Feedback	1. All 2. All
Apr 28-May 2	1. Post-CDR revisions 2. Prep final documentation package 3. Assign summer work?	1. All 2. All 3. Management

Payload-Specific Deliverables (Required Milestones for launch in May 2027) - NuKAT

Event	Date (L-xx format)	Date (Month format)
Payload MICD Due	L-6 Months	Nov 2026
Payload test reports due	L-3 Months	Feb 2027
Payload License Received	L-2 Months	Mar 2027
Payload delivery for integration	L-45 days	Late Mar 2027

PHAT-Sat Schedule Tracking

Milestones	Assigned Semester	Start Date	Semester Completed	P.I. Sign Off	Professor Sign Off	Scheduler Sign Off	Has Everyone Signed Off?
System Engineering Phase	Spring 2025	N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Mission Formulation/Merit review		N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Requirements Review		N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Conceptual Design/Feasibility Review		N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Design showcase		N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Final Report for Semester Data Transfer		N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Preliminary Design Proposal Phase	Fall 2025	August.25	Fall 2025			Abigail Husted	NO
Previous Semester Report Review		August.25	Fall 2025			Abigail Husted	NO
Schedule Review		October.25	Fall 2025			Abigail Husted	NO
Risk Reevaluation		December.25	Fall 2025			Abigail Husted	NO
Preliminary Design/Pre-PDR Review		December.25	Fall 2025			Abigail Husted	NO
Software Development		December.25	Fall 2025			Abigail Husted	NO
Final Report for Semester Data Transfer		December.25	Fall 2025			Abigail Husted	NO
Detailed Design Phase	Spring 2026						NO
Previous Semester Report Review							NO
Preliminary/Critical Design and Reviews							NO
Mid Semester Progress Review							NO
Stakeholder/Budget Review							NO
Mission Advisory Board Progress Review							NO
Licensing Evaluation	Fall 2026 - Spring 2028						NO
Schedule Review							NO
Risk Reevaluation							NO
Design showcase							NO
Final Report for Semester Data Transfer							NO
AIT Phase							NO
Previous Semester Report	Fall 2026 - Spring 2028						NO
Build							NO
Test							NO
Mid Semester Progress Review							NO
Stakeholder/Budget Review							NO
Mission Advisory Board Progress Review							NO
Licensing Evaluation	Fall 2028						NO
Schedule Review							NO
Risk Reevaluation							NO
Design showcase							NO
Final Report for Semester Data Transfer							NO
Launch							NO
Previous Semester Report Review	Fall 2028						NO
Mid Semester Progress Review							NO
Stakeholder/Budget Review							NO
Mission Advisory Board Progress Review							NO
Licensing Evaluation							NO
Schedule Review							NO
Risk Reevaluation	Fall 2028						NO
Pending Launch Manifest							NO
Final Compiled Mission Report							NO

SolidWorks Renderings

NuKat

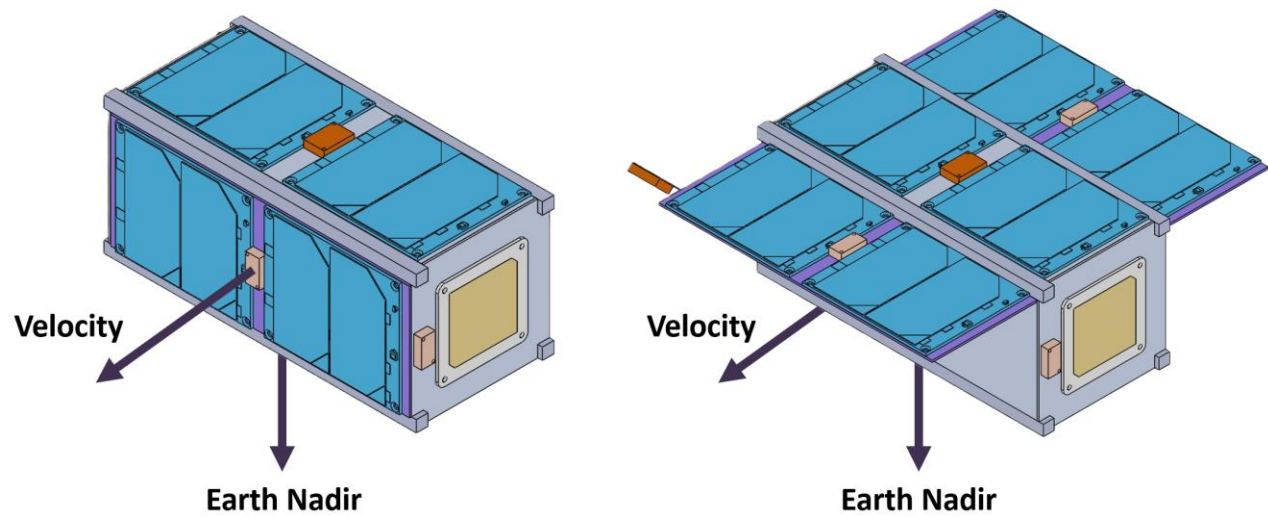


Figure 4: NuKAT Undeployed and Deployed Configurations

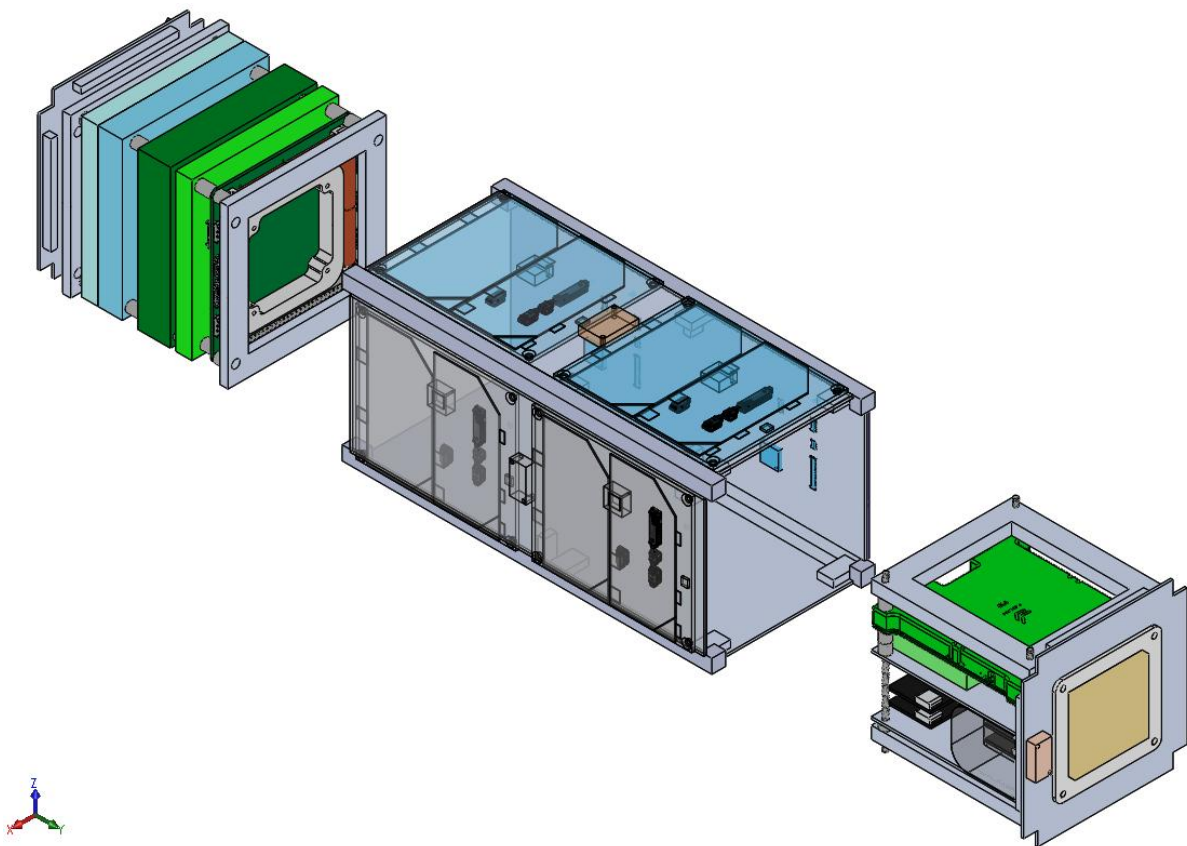


Figure 21: NuKAT E-Stack Exploded View

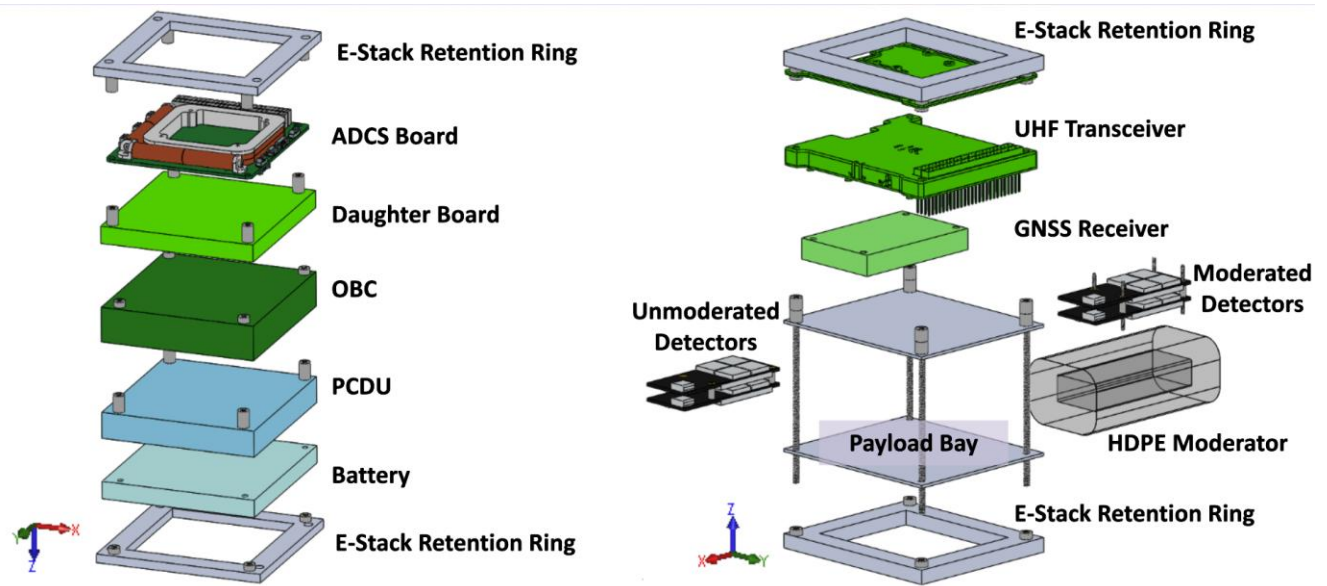


Figure 22: NuKat Avionic E-Stack and Payload E-Stack Exploded View

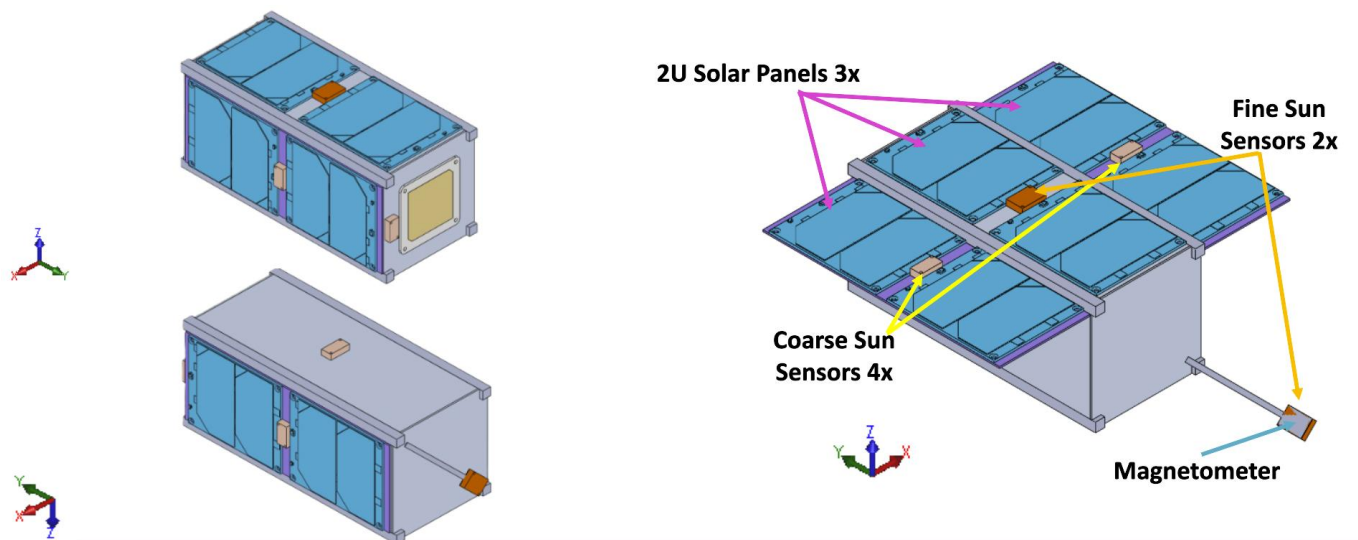


Figure 23: NuKat External Components

PHAT-Sat

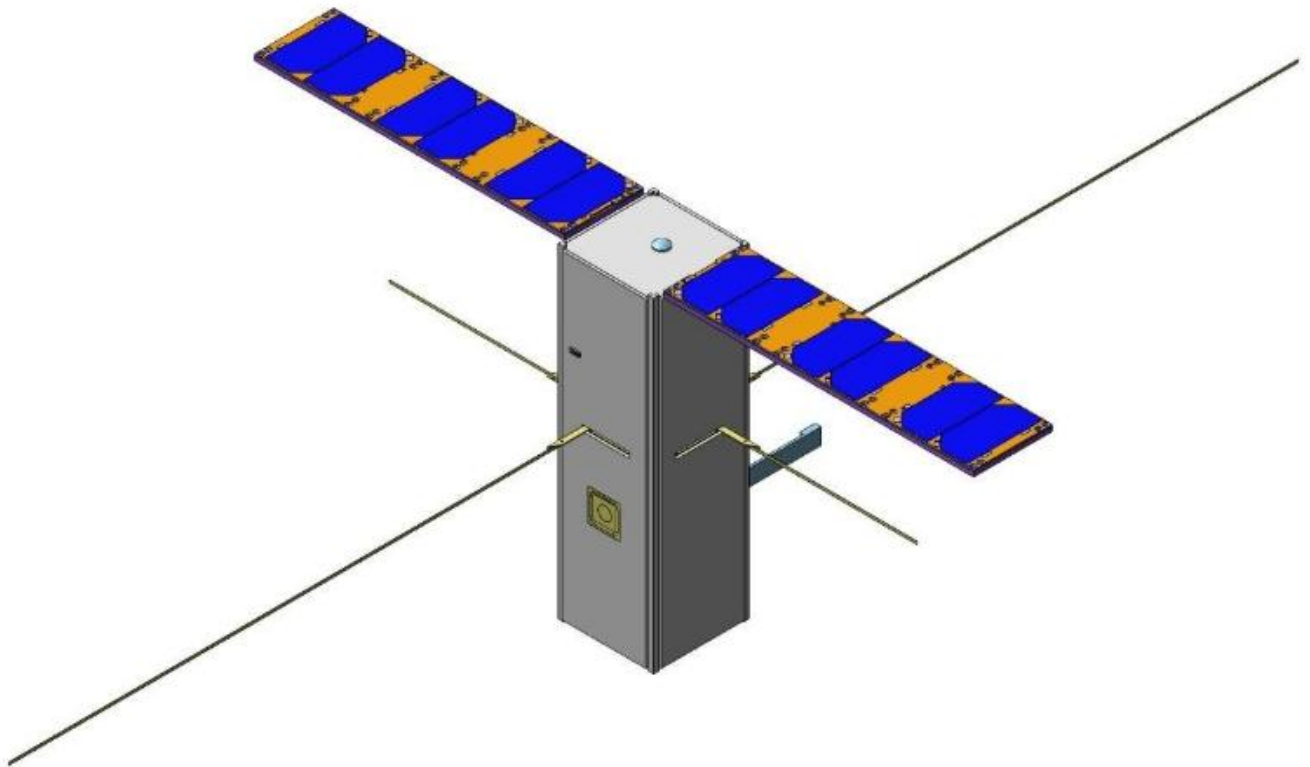


Figure 8: PHAT-Sat Isometric View (Deployed)

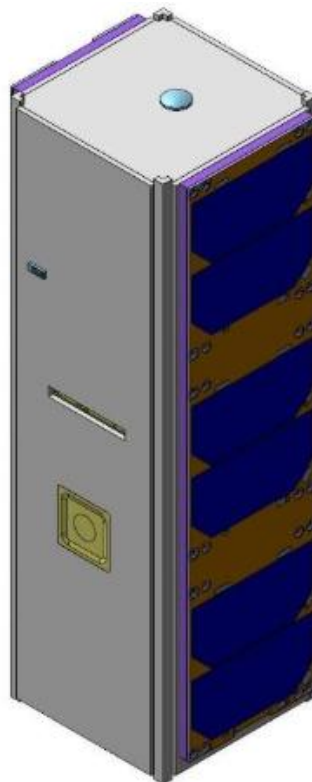


Figure 9: PHAT-Sat Isometric View (Stowed)

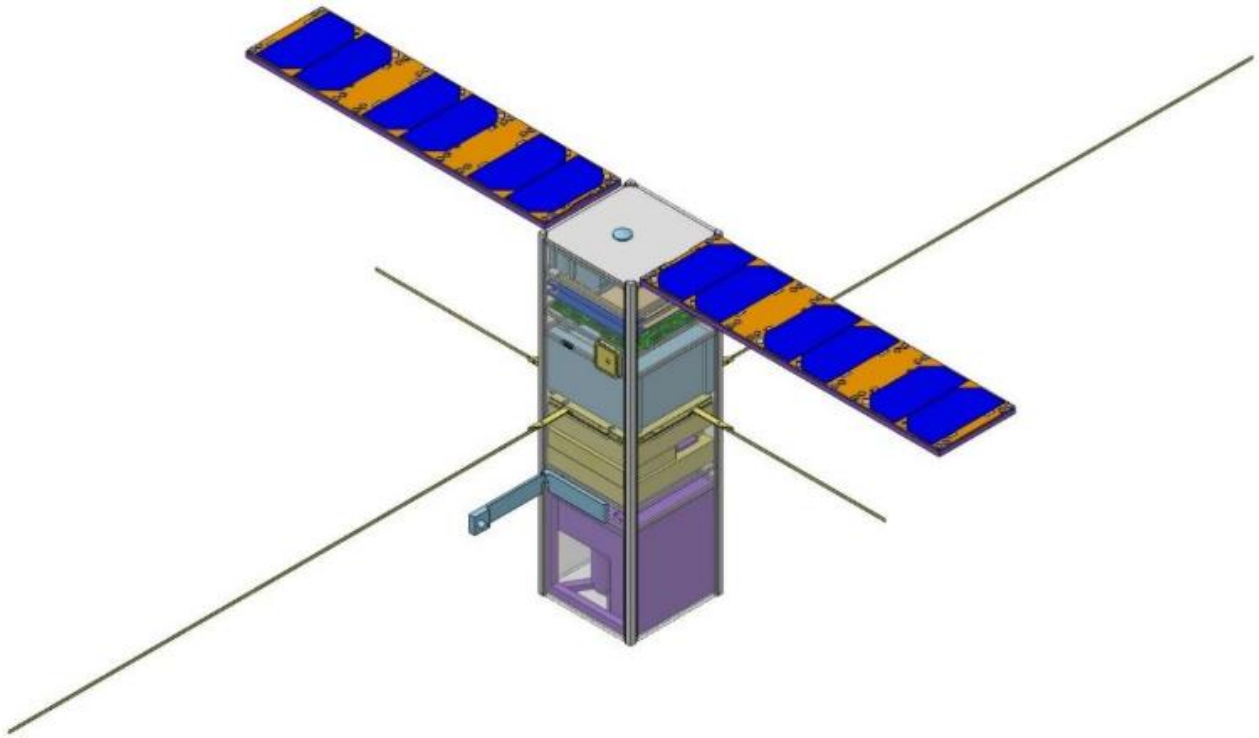


Figure 25: PHAT-Sat Isometric View (Deployed, Transparent Paneling)

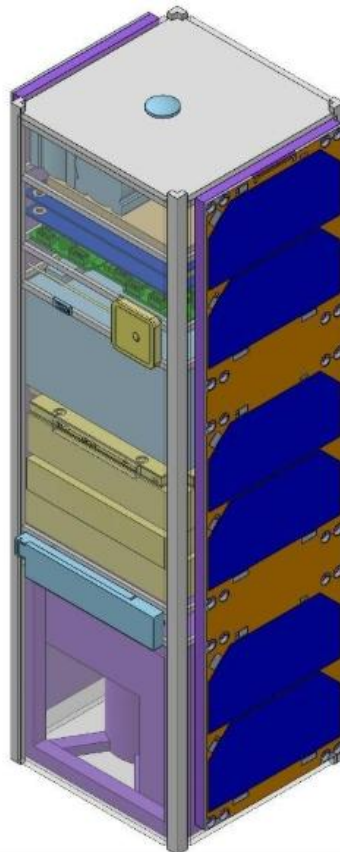


Figure 26: PHAT-Sat Isometric View (Stowed, Transparent Paneling)

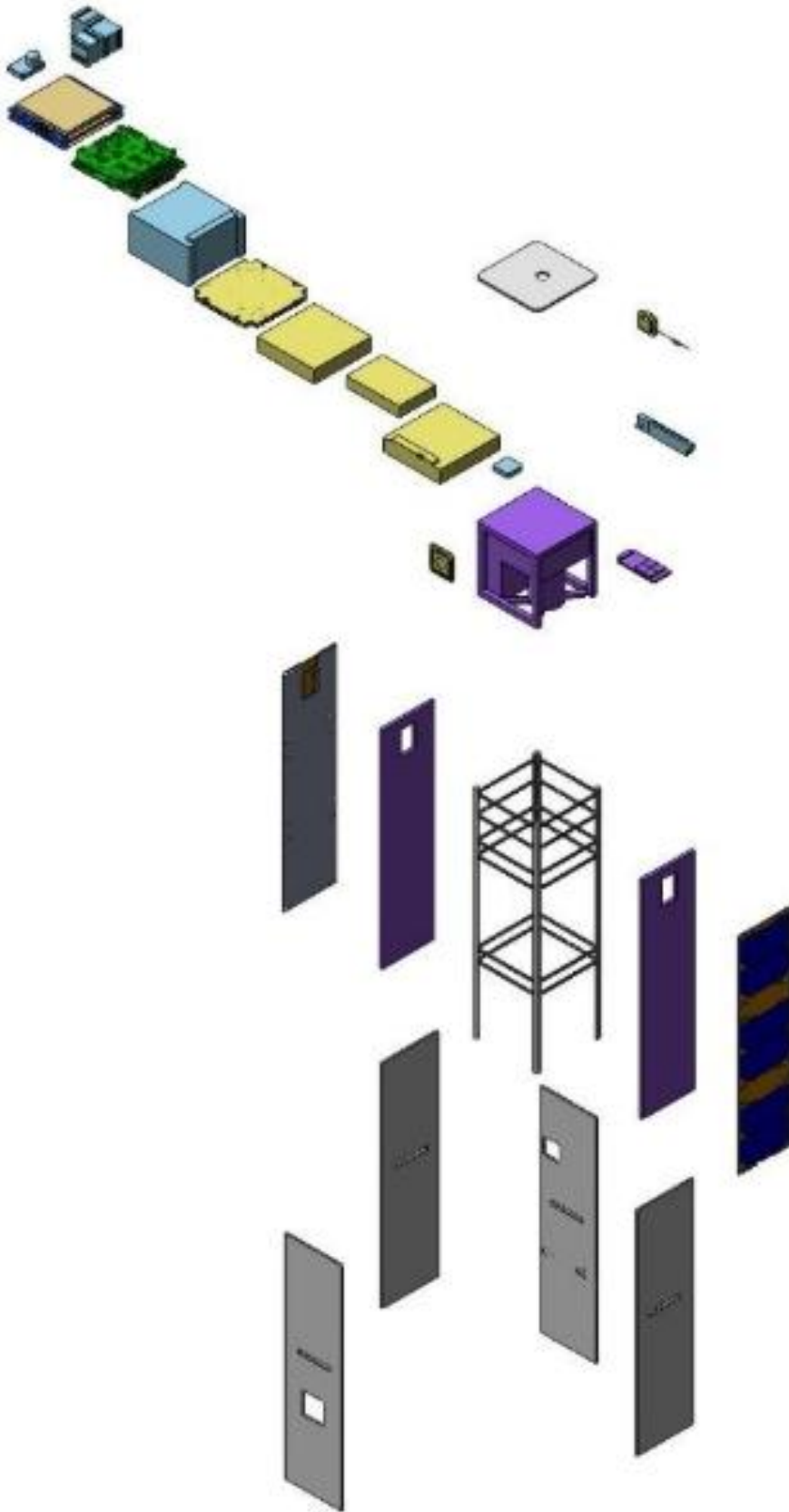


Figure 27: PHAT-Sat Component Exploded Isometric View

Orbit Modeling results

Integration and Test Facility Definition requirements and diagram

This document defines the facility and environment requirements for the Kansas State University CubeSat Integration and Test Laboratory. The lab will serve as the primary location for the assembly and testing of hardware for all university CubeSat missions.

This space will need used for but not limited to:

- Hold a maximum of 12 people, normal operating conditions of approximately 6 people.
- Assembly of CubeSat components and subsystems.
- Electrical integration and testing.
- Software and hardware functional verification.
- Subsystem functional tests.
- Communication and RF testing.
- Storage of tools and sensitive materials in safe conditions.

General Space Requirements:

- Required, the space must be fully enclosed and only accessible through doors that can be securely locked and accessed by team personnel and other authorized people.
- Required, the space should provide a normal, unobstructed layout, rectangular or similar, allowing multiple people to move safely and freely around workbenches and test equipment.
- Required, the space must not be less than 400 square feet.
- Required, the space cannot be in the direct vicinity of another area that will produce intense acoustic vibration. Example: being above a machining shop.
- Required, the space cannot be in the direct vicinity of strong magnetic sources such as large motors, transformers, etc.
- Required, the space must have ceilings no less than 10 feet tall.
- The space must provide a compressed air source, either through an existing facility air line or alternatively we can bring in a portable compressor.
- Recommended: one internal facing window

Power Requirements

- Required, at least twelve(12) standard 120V outlets in accessible locations throughout the room
- Ability to have a proper power strip installed near an area that will hold electrical testing equipment.
- All outlets must be 3 pronged outlets, having a ground socket.
- Proper ground connections.
- 240V supply would be nice if readily available for future expansion, not required.

Door Way

- Required, minimum of one door, dimensions no less than 36" x 80".
- Required, doors must be able to have restricted access to authorized personnel.
- At least one entrance must have a flush door threshold to prevent rolling carts carrying sensitive materials from being jarred when entering the room. (No lip on the ground of the door frame)

Lighting

- Required, the space needs either proper ceiling mounted lighting, enough to adequately illuminate all surfaces, or alternatively has the proper power delivery and mounting surfaces to incorporate lighting solutions.
- Required, emergency lighting for power outages, can likely be incorporated separately.

Climate Control

- Required, the space requires adequate climate control, fit for human comfort.
- Required, the space requires normal ventilations adequate for human occupation.
- Required, the space requires enough power delivery for additional climate adjustments to be brought in, not limited to, dehumidifiers/humidifiers.

Network

- Required, the space must have reliable access to campus WiFi networks.
- If campus desktops normally use ethernet then ethernet hookups are needed.
- A space that receives cellular network connection
- A dedicated landline phone.

Safety

- The space cannot have sprinkler head fire suppression, instead we'll have fire extinguishers in place.
- The space requires close access to an eye wash station, and first aid.
- Emergency lighting for power outages.

Additional questions

- Do you know of any space on campus with one or more of the following?
 - Shock and vibration test rig
 - Lateral and axial force test
 - Thermo-vacuum chamber
 - Radio frequency testing equipment

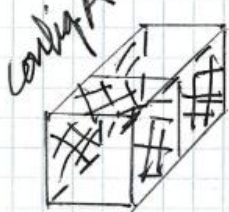


Solar Array Configuration Analysis: NuKAT

2 NuKAT

~~Function example 1, fig. 2.3~~ Fixed 1Us

config A

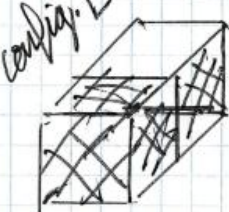


If $\Theta_z = 45^\circ = \pi/4$,
 then $P_{\text{array}} = P_{\text{max}} \cdot \cos \pi/4$

For config A: $3 \cdot P_{\text{max}} \cdot \cos \pi/4 = P_{\text{array}}$
 $\approx 2.121 \cdot P_{\text{max}}$

- EnduroSat: $P_A \approx 5.112 \text{ W}$
- ISSpace: $P_A \approx 6.364 \text{ W}$
- DHV: $P_A \approx 4.879 \text{ W}$
- 2NDspace: $P_A \approx 5.197$

config B



If $\Theta_z = 54.7^\circ$, $P_{\text{max}} = P \cdot \cos 54.7^\circ$
 $\approx P \cdot 0.578$

then for config B
 $P_B = 4 \cdot P \cdot \cos 54.7^\circ$
 $\approx 2.3114 \cdot P$

- EnduroSat: $P_B \approx 5.547 \text{ W}$
- ISSpace: $P_B \approx 6.934 \text{ W}$
- DHV: $P_B \approx 5.3163 \text{ W}$
- 2NDspace: $P_B \approx 5.663$

Generally $P_T = \sum P_k$;
 kth solar panel

Solar Array Configuration Analysis: PHAT-Sat

Solar Panel Comparison 1

pumkin paper example 2

example shows 6 cells on each flap.

In General:

One XTJ cell has $V_{oc} = 2.5 \text{ volts}$,
 $I_{sc} = 0.4 \text{ A}$, & $P_{Bot} = 1 \text{ W}$.

One flap $\rightarrow$ 6S1P or 3S2P;

Check 6S1P flap: $V_{oc} = 15 \text{ volts}$
 $\frac{1}{1} \text{ 6S1P}$ $I_{sc} = 0.4 \text{ A}$
 $P_{Bot} = 6 \text{ W}$

Check 3S2P flap: $V_{oc} = 7.5 \text{ volts}$
 $\frac{1}{1} \text{ 3S2P}$ $I_{sc} = 0.8 \text{ A}$
 $P_{Bot} = 6 \text{ W}$

For 6 flaps at 6S1P in all parallel

$\frac{1}{1} \text{ 6S1P } \dots \times 6$

$V_{oc} = 15 \text{ volts}$
 $I_{sc} = 0.4 \text{ A} \cdot 6 = 2.4 \text{ A}$

$P_{Bot} = 36 \text{ W}$

For 6 flaps at 352P with 2 flaps in parallel:



$$V_{oc} = 22.5 \text{ volts}$$

$$\downarrow \Sigma I_c = 0.800A \cdot 2 = 1.6A$$

$$P_{Bot} = 36W$$

Interface Diagrams

NuKat

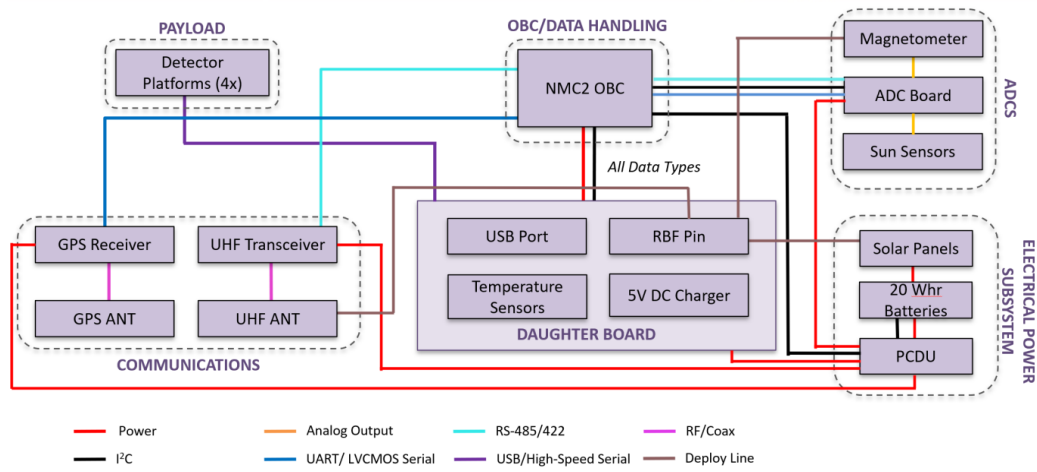


Figure 6: NuKAT Satellite System Block Diagram

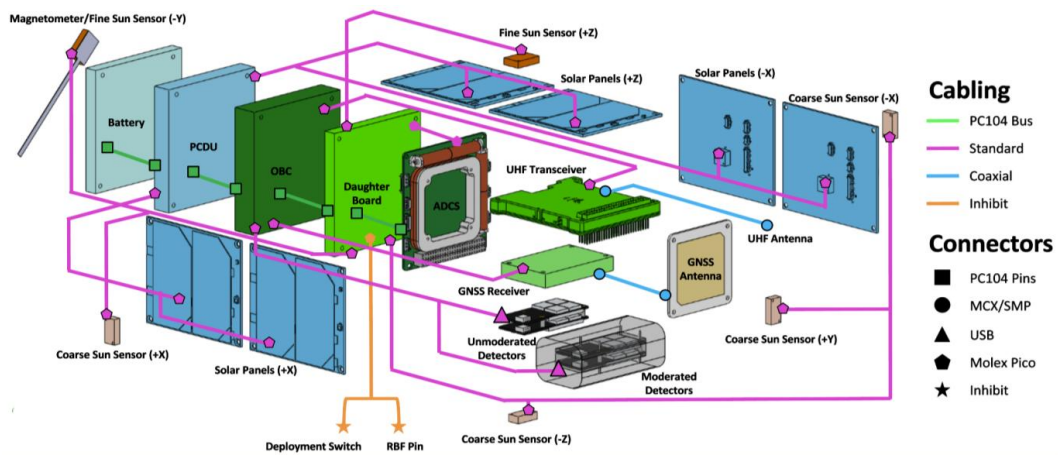


Figure 24: NuKat Cabling and Connection Diagram

PHAT-Sat

Figure 24: NuKat Cabling and Connection Diagram

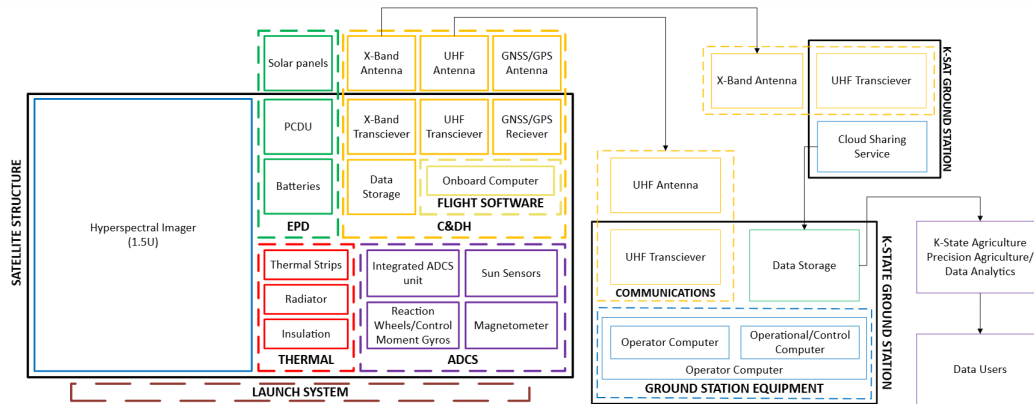


Figure 10: PHAT-Sat System Block Diagram

Mass Budget

NuKat

Table 8: NuKat Mass Budget

Component	Model	Mass (g)	Volume (cm ³)	Critical Dimension (cm)
Detector Platform (4x)	RTD DSC Domino	9.50 (4)	8.30 (4)	0.6 (Z)
HDPE Moderator	Custom	212.15	222.84	3.4 (Z)
ADC Board	Tensor Tech MTQ-200M Board	150	155.66	2.3 (Z)
Coarse Sun Sensor (4x)	Tensor Tech CSS-10	0.50 (4)	1.40 (6)	0.6
Fine Sun Sensor (2x)	Tensor Tech FSS-15	3 (2)	1.721 (2)	2.0 (Y)
OBC	NextAge NMC2	85	138.24	1.5 (Y)
UHF Transceiver	EnduroSat UHF Transceiver	90.22	194.61	2.3 (Y)
UHF Antenna	TBD	TBD	TBD	TBD
GNSS Receiver	NovAtel OEM719	23	38.96	1.1 (Z)
GNSS Antenna	EnduroSat L1/E1 GNSS Antenna	17.1	21.84	0.2 (Y)
PCDU	DHV PicoEPS Power Module	210	129.6	1.5 (Y)
Battery	DHV PicoEPS Battery Module	80	86.4	1.0 (Y)
Solar Panels	EnduroSat 1U	38 (6)	21.61 (6)	1.0 (Y)
Chassis Assembly	Custom	764.56	283.16	2.0 (Y)
Deployable Magnetometer	Custom	3.92	2.05	1.0 (Z-X)
Radiators/Insulators	TBD	TBD	TBD	TBD

Glen Fountain NuKat MDR Comments

1. General Comments

- a. The presentations provide a good start on the development process. As discussed at the time of the review (and discussed below) more quantitative detail is needed prior to the PDR. This is the primary missing content in the presentation.
- b. I would recommend that team consider a series of reviews (or find a longer time for major reviews) to be able to fully present the details necessary to evaluate the concept and design. These reviews should be identified in the schedule. Typically, there are a series of reviews for each subsystem that leads up to the mission PDR/CDR.

- c. Trade studies may be needed once the requirements are better defined and the performance margins determined. Based on the performance margins (or lack thereof) some trade studies may be needed.
- 2. Charts 11- 13
 - a. What is the energy spectrum the detector is expected to sense?
 - b. What is the sensitivity of the spectrum the detector is expected to achieve?
 - c. What is the expected environmental flux the detector is expected to see and what output is the detector expected to produce?
 - d. What is the instrument noise level relative to the expected observations?
 - d. What are the quantitative figures of merit of the DSC Domino detector (relative mass, power, volume, reliability) relative to current technology?
- 3. Chart 16
 - a. What data is to be transmitted? What spacecraft “housekeeping” data and what instrument data. How much of each kind is to be collected between passes for onboard storage to be transmitted to the ground?
 - b. What is the data downlink rate given the pass time and time between passes?
- 4. Charts 21 – 22
 - a. The team has identified the various environmental drivers. For the spacecraft design to close these environments need to be quantified/enveloped. Recommend the team defines the environmental requirements by looking at the: orbit for radiation/thermal inputs; typical launch environments from earlier NASA Space Launch missions, etc.
- 5. Charts 23 – 29
 - a. Each subsystem requirements and performance need to be quantitatively defined. Also, each subsystem performance margins need to be defined.
- 6. Charts 30 – 33
 - a. In general, the comments provided at the review are appropriate with respect to link margins with the system antenna gain on the ground and not on the spacecraft.
 - b. What is the form of the communications to the end users?
- 7. Chart 34
 - a. The chart implies that a verification matrix is being developed. A review of the matrix should be provided at a later review.
- 8. Chart 36
 - a. Greater budget granularity should be developed.
 - b. In addition to the spacecraft a budget; the ground system cost and the source/costs of assembly and test equipment should be identified.

- c. An estimate of the time put in by the team should be made. The time should be identified by the various scheduled tasks (see below) and then tracked to see how well the estimate correlates with the actual time expended.
- 9. Chart 38
 - a. Consider developing a PERT chart schedule. The schedule should be broken down into the detailed tasks need to design, fabricate, assemble and test each subsystem and then the assembly and testing of the spacecraft. The PERT charts will also assure that you understand the dependency of subsequent tasks on their predecessors.